

Noise-Resolved Information Decomposition

A Working Theory of Interaction Order Under Source Degradation

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Abstract

Classical information theory quantifies uncertainty, communication, and dependence, while partial information decomposition aims to separate multivariate information into redundant, unique, and synergistic parts. This note develops a complementary direction: instead of decomposing information only at full source resolution, decompose information by its response to controlled independent degradation of the sources. For independent uniform Boolean sources, define subset coefficients by the squared Fourier-Walsh energy of the centered target likelihoods, weighted by inverse target probability. The main identity shows that the expected chi-square divergence between the posterior and prior target distributions is an exact multi-affine polynomial in the squared source reliabilities. The coefficient of a monomial is the information energy carried by the corresponding source subset; grouping by subset size gives an order spectrum. This yields an operational fragility law: if the first nonzero order is d , then weak-signal Shannon mutual information decays as ρ^{2d} under symmetric source reliability ρ . Exact enumeration experiments verify the identity for deterministic and stochastic targets, including copy, XOR, majority, AND, mixed copy/XOR channels, and random stochastic maps. The note also records a reversible Markov-operator generalization, estimation procedures, limitations, and open problems. A second-pass extension adds response-curve cumulants, a local-observability theorem for arbitrary product observations, exact subset tomography by Möbius inversion, a multiplexer spectrum law, random-target spectra, and a chi-square most-informative bound. A third-pass extension adds a finite-alphabet singular-mode theorem, coefficientwise target data processing, composition laws, sensor-fusion observability, flat count spectra, and modulo-count spectral hiddenness. A fourth-pass extension adds a Shannon collision expansion, a same-NRID-spectrum counterexample for Shannon mutual information, a linear-code weight-enumerator theorem, and an exact binomial spectrum law for bent and inner-product targets. A fifth-pass extension adds fixed-budget subset spectroscopy, a conditioning barrier for spectrum inversion, adversarial leakage geometry, sparse hypergraph recovery, and phase-sensitive cubic posterior spectroscopy. A sixth-pass extension adds reliability-field thermodynamics, source complementarity Hessians, reliability-state observability, availability-correlation laws, and active sensing budget laws for high-order information. A seventh-pass extension adds binary utility envelopes, posterior-amplitude shape, matched-channel counterexamples, leading-layer decision exponents, decision flatness, and replicated-observation formulas. An eighth-pass extension adds posterior-score moment-closure spectroscopy, phase-gauge invariants, odd-moment silence, low-order closure-rank experiments, and finite moment completion of operational utility. A ninth-pass extension adds geometry-resolved Markov semigroup NRID, information heat spectra, spectral thermodynamics, bottleneck persistence, eigenspace orientation blindness, and heat-spectrum tomography.

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1 Status and framing

This document records a candidate framework called *Noise-Resolved Information Decomposition* (NRID). The core mathematical identity is exact under the assumptions stated below. The proposed contribution is the organization and interpretation of that identity as a source-degradation information spectrum. The proof uses standard diagonalization of independent noise in the Fourier-Walsh basis, so priority should not be claimed without a deeper literature search across information theory, noise stability, ANOVA decompositions, Sobol indices, and chi-square mutual information.

The guiding hypothesis is:

Information is not only a scalar amount. It has a decay spectrum under admissible noise, and this spectrum exposes the interaction order required for the information to remain observable.

The immediate blind spot is that many multivariate information decompositions ask how sources provide information about a target at a single operating point. NRID instead asks what information survives a controlled perturbation experiment. This makes interaction order measurable by a response curve.

2 Background and motivation

Shannon’s theory introduced entropy, mutual information, and channel capacity as central quantities for communication and uncertainty [6]. Later expositions such as Cover and Thomas develop these measures and their operational roles in coding and inference [7]. Partial information decomposition (PID), introduced by Williams and Beer, seeks to decompose the information that multiple sources provide about a target into redundant, unique, and synergistic parts [9]. Subsequent work has explored redundancy measures, unique information, pointwise formulations, and alternative synergy-first constructions [10, 11, 12, 13, 14, 15].

A separate mathematical tradition studies functions on product spaces through orthogonal expansions. On the Boolean cube, the Fourier-Walsh basis diagonalizes the standard independent noise operator: a Fourier component supported on a set S is attenuated by $\rho^{|S|}$ under reliability ρ [16, 17]. This degree-dependent attenuation is the source of the decomposition developed here.

3 Boolean source model

Let

$$X = (X_1, \dots, X_n) \in \{-1, +1\}^n$$

be independent fair bits. Let T be a finite target generated from X , possibly stochastically. Write the finite target alphabet as $\mathcal{A}$ and define

$$q_t(x) = \mathbb{P}(T = t \mid X = x), \quad p_t = \mathbb{P}(T = t) = 2^{-n} \sum_x q_t(x),$$

for each $t \in \mathcal{A}$. Assume $p_t > 0$ for every target state retained in the alphabet. Define the centered target likelihood

$$m_t(x) = q_t(x) - p_t.$$

For each subset $S \subseteq [n]$, let

$$\chi_S(x) = \prod_{i \in S} x_i, \quad \chi_\emptyset(x) = 1.$$

The Fourier-Walsh expansion of m_t is

$$m_t(x) = \sum_{S \subseteq [n]} \hat{m}_t(S) \chi_S(x), \quad \hat{m}_t(S) = 2^{-n} \sum_x m_t(x) \chi_S(x).$$

Because m_t is centered, $\hat{m}_t(\emptyset) = 0$.

3.1 Independent degradation

For a reliability vector $\boldsymbol{\rho} = (\rho_1, \dots, \rho_n)$ with $0 \leq \rho_i \leq 1$, let $Y_{\boldsymbol{\rho}}$ be a noisy observation of X defined coordinatewise by

$$\mathbb{P}(Y_i = X_i) = \frac{1 + \rho_i}{2}, \quad \mathbb{P}(Y_i = -X_i) = \frac{1 - \rho_i}{2},$$

independently across i . The symmetric reliability case is $\rho_i = \rho$ for all i .

The Boolean noise operator $T_{\boldsymbol{\rho}}$ satisfies

$$(T_{\boldsymbol{\rho}}f)(y) = \mathbb{E}[f(X) \mid Y_{\boldsymbol{\rho}} = y].$$

For the Fourier-Walsh characters,

$$T_{\boldsymbol{\rho}}\chi_S = \left(\prod_{i \in S} \rho_i \right) \chi_S.$$

4 Definitions

Definition 1 (Posterior chi-square information). *The posterior chi-square information between T and the degraded observation $Y_{\boldsymbol{\rho}}$ is*

$$\mathcal{J}_{\boldsymbol{\rho}}(T; Y) = \mathbb{E}_{Y_{\boldsymbol{\rho}}} [\chi^2(P(T \mid Y_{\boldsymbol{\rho}}) \parallel P(T))],$$

where

$$\chi^2(P \parallel Q) = \sum_t \frac{(P(t) - Q(t))^2}{Q(t)}.$$

Equivalently,

$$\mathcal{J}_{\boldsymbol{\rho}}(T; Y) = \mathbb{E}_{Y_{\boldsymbol{\rho}}} \left[\sum_t \frac{(P(T = t \mid Y_{\boldsymbol{\rho}}) - p_t)^2}{p_t} \right].$$

Definition 2 (Subset NRID coefficient). *For each nonempty subset $S \subseteq [n]$, define*

$$\omega_S(T; X) = \sum_t \frac{\hat{m}_t(S)^2}{p_t}.$$

This coefficient measures the target-likelihood energy that sits exactly on the interaction character χ_S .

Definition 3 (Order NRID coefficient). *The order- r coefficient is*

$$\Omega_r(T; X) = \sum_{|S|=r} \omega_S(T; X) = \sum_t \frac{1}{p_t} \sum_{|S|=r} \hat{m}_t(S)^2.$$

The coefficients are nonnegative by construction. They are not PID atoms. They form an orthogonal energy decomposition of the target-posterior response under source noise. In this sense, NRID is closer to a noise-resolved ANOVA spectrum than to a redundancy lattice.

5 Main identity

Theorem 1 (Subset-resolved noise law). *For independent fair Boolean sources and independent binary symmetric degradation with reliability vector ρ ,*

$$\mathcal{J}_\rho(T; Y) = \sum_{\emptyset \neq S \subseteq [n]} \omega_S(T; X) \prod_{i \in S} \rho_i^2.$$

In the symmetric case $\rho_i = \rho$,

$$\mathcal{J}_\rho(T; Y) = \sum_{r=1}^n \Omega_r(T; X) \rho^{2r}.$$

Proof. For a fixed target state t ,

$$P(T = t \mid Y_\rho = y) - p_t = \mathbb{E}[q_t(X) - p_t \mid Y_\rho = y] = (T_\rho m_t)(y).$$

Expanding m_t in the Fourier-Walsh basis and using diagonalization of the noise operator gives

$$T_\rho m_t = \sum_{S \neq \emptyset} \hat{m}_t(S) \left(\prod_{i \in S} \rho_i \right) \chi_S.$$

Since Y_ρ is uniform on the Boolean cube and the characters are orthonormal,

$$\mathbb{E}_Y [(T_\rho m_t)(Y)^2] = \sum_{S \neq \emptyset} \hat{m}_t(S)^2 \prod_{i \in S} \rho_i^2.$$

Dividing by p_t , summing over t , and exchanging finite sums yields

$$\mathcal{J}_\rho = \sum_{S \neq \emptyset} \left(\sum_t \frac{\hat{m}_t(S)^2}{p_t} \right) \prod_{i \in S} \rho_i^2.$$

The expression in parentheses is ω_S . Grouping terms with $|S| = r$ proves the symmetric identity. $\square$

6 Consequences

6.1 Information order generating function

Let $z = \rho^2$ and define

$$G(z) = \mathcal{J}_{\sqrt{z}}(T; Y) = \sum_{r=1}^n \Omega_r z^r.$$

If $\mathcal{J}_1 > 0$, then

$$\pi_r = \frac{\Omega_r}{\mathcal{J}_1}$$

defines a probability distribution over interaction orders. Therefore,

$$\frac{G(z)}{G(1)} = \sum_{r=1}^n \pi_r z^r$$

is the probability generating function of an *information order random variable* R .

This gives exact summary statistics:

$$\mathbb{E}[R] = \frac{G'(1)}{G(1)},$$

$$\text{Var}(R) = \frac{G''(1)}{G(1)} + \frac{G'(1)}{G(1)} - \left(\frac{G'(1)}{G(1)} \right)^2.$$

Thus the degradation curve itself contains the moments of the information-order distribution.

6.2 Coefficient recovery by derivatives

For the symmetric curve,

$$\Omega_r = \frac{1}{r!} \left. \frac{d^r}{dz^r} G(z) \right|_{z=0}.$$

For the anisotropic curve, write $z_i = \rho_i^2$ and

$$H(z_1, \dots, z_n) = \mathcal{J}_{(\sqrt{z_1}, \dots, \sqrt{z_n})}.$$

Then H is multi-affine and

$$\omega_S = \left(\prod_{i \in S} \frac{\partial}{\partial z_i} \right) H(z_1, \dots, z_n) \Big|_{z_1 = \dots = z_n = 0}.$$

This derivative formula is an operational route from degradation experiments to subset-specific information energies.

6.3 Total quadratic information conservation

At full reliability, $Y = X$, so

$$\mathcal{J}_1(T; X) = \sum_{r=1}^n \Omega_r = \sum_{S \neq \emptyset} \omega_S.$$

The total posterior chi-square information is conserved across orders. Noise does not change the coefficients; it only reweights them by powers of reliability.

6.4 Weak-signal Shannon limit

The Shannon mutual information in bits is

$$I(T; Y_\rho) = \mathbb{E}_Y \text{D}_{\text{KL}}(P(T | Y_\rho) \| P(T)).$$

For small posterior deviations $a_t(y) = P(T = t | Y = y) - p_t$,

$$\text{D}_{\text{KL}}(P(T | Y = y) \| P(T)) = \frac{1}{2 \ln 2} \sum_t \frac{a_t(y)^2}{p_t} + O(\|a(y)\|^3).$$

Consequently,

$$I(T; Y_\rho) = \frac{1}{2 \ln 2} \sum_{r=1}^n \Omega_r \rho^{2r} + O(\rho^{3d}),$$

where d is the first nonzero NRID order. In particular,

$$I(T; Y_\rho) \sim \frac{\Omega_d}{2 \ln 2} \rho^{2d} \quad (\rho \downarrow 0).$$

Corollary 1 (Correlation-order fragility law). *If $d = \min\{r : \Omega_r > 0\}$, then*

$$d = \frac{1}{2} \lim_{\rho \downarrow 0} \frac{d \log \mathcal{J}_\rho}{d \log \rho}.$$

Under the weak-signal Shannon expansion, the same exponent is obtained from $I(T; Y_\rho)$:

$$d = \frac{1}{2} \lim_{\rho \downarrow 0} \frac{d \log I(T; Y_\rho)}{d \log \rho}.$$

6.5 Interpretation

A one-bit copy and a k -bit XOR can both contain one bit of information at full reliability. Their near-blindness behavior is different:

$$I_{\text{copy}}(\rho) \sim \frac{\rho^2}{2 \ln 2}, \quad I_{\text{XOR}_k}(\rho) \sim \frac{\rho^{2k}}{2 \ln 2}.$$

Thus high-order information is not merely distributed differently; it is more fragile under independent source noise. NRID makes this fragility directly measurable.

7 Examples

7.1 Pure copy

Let $T = X_1$. Then $\Omega_1 = 1$ and

$$\mathcal{J}_\rho = \rho^2.$$

The weak-signal Shannon exponent is 2.

7.2 Pure XOR

Let

$$T = X_1 X_2 \cdots X_k.$$

Then $\Omega_k = 1$ and all other orders vanish, so

$$\mathcal{J}_\rho = \rho^{2k}.$$

The information is invisible to all lower-order degradation derivatives.

7.3 Majority of three bits

For $T = \text{majority}(X_1, X_2, X_3)$ with outputs in $\{-1, +1\}$,

$$\Omega_1 = 0.75, \quad \Omega_3 = 0.25,$$

so

$$\mathcal{J}_\rho = 0.75\rho^2 + 0.25\rho^6.$$

The order distribution has mass 0.75 at order 1 and mass 0.25 at order 3, with mean order 1.5.

7.4 AND of two bits

Let $T = \mathbf{1}[X_1 = +1, X_2 = +1]$. Then

$$\Omega_1 = \frac{2}{3}, \quad \Omega_2 = \frac{1}{3},$$

and

$$\mathcal{J}_\rho = \frac{2}{3}\rho^2 + \frac{1}{3}\rho^4.$$

This biased deterministic target has both individual and pairwise information energy.

7.5 Mixed copy/XOR stochastic channel

Define a binary stochastic target by

$$\mathbb{P}(T = +1 \mid X) = \alpha \mathbf{1}[X_1 = +1] + (1 - \alpha) \mathbf{1}[X_1 X_2 = +1].$$

Then

$$\Omega_1 = \alpha^2, \quad \Omega_2 = (1 - \alpha)^2,$$

and

$$\mathcal{J}_\rho = \alpha^2 \rho^2 + (1 - \alpha)^2 \rho^4.$$

This is a compact design equation: direct copy information improves low-reliability robustness, while XOR-like information increases joint-source dependence.

8 Exact enumeration experiments

All experiments in this section were computed by exact enumeration over the Boolean cube, not by Monte Carlo sampling. For a target likelihood table $q_t(x)$, the script computes Fourier-Walsh coefficients directly, evaluates $\mathcal{J}_\rho$ from the posterior distribution, and compares it with the NRID polynomial.

Table 1: Named deterministic and stochastic examples. The fitted Shannon slope is the slope of $\log I(T; Y_\rho)$ versus $\log \rho$ over small ρ .

System	NRID spectrum	Exact $\mathcal{J}_\rho$	Fitted slope	$I(T; Y_{0.1})$ bits
$T = X_1$	$\Omega_1 = 1$	ρ^2	2.0017	7.23×10^{-3}
$T = X_1 X_2$	$\Omega_2 = 1$	ρ^4	4.0000	7.21×10^{-5}
$T = X_1 X_2 X_3$	$\Omega_3 = 1$	ρ^6	6.0000	7.21×10^{-7}
Majority-3	$\Omega_1 = 0.75, \Omega_3 = 0.25$	$0.75\rho^2 + 0.25\rho^6$	2.0030	5.43×10^{-3}
AND-2	$\Omega_1 = 2/3, \Omega_2 = 1/3$	$\frac{2}{3}\rho^2 + \frac{1}{3}\rho^4$	2.0036	4.83×10^{-3}
Mixed, $\alpha = 0.35$	$\Omega_1 = 0.1225, \Omega_2 = 0.4225$	$0.1225\rho^2 + 0.4225\rho^4$	2.0344	9.14×10^{-4}

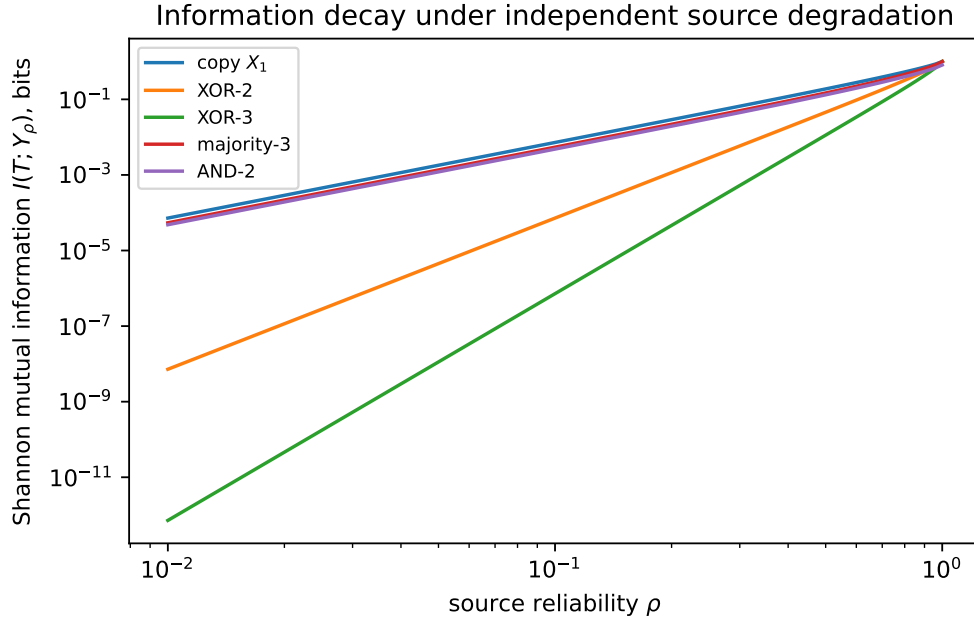


Figure 1: Shannon mutual information decay under independent degradation. The low-reliability slopes reveal the first active NRID order: copy has slope near 2, XOR-2 near 4, and XOR-3 near 6.

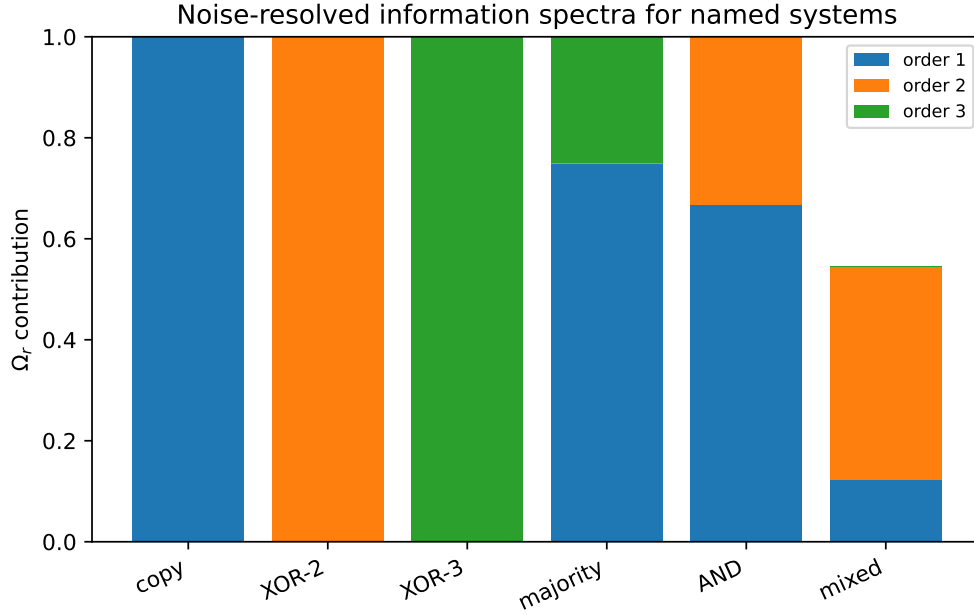


Figure 2: Order spectra for named systems. Each bar decomposes posterior chi-square information at full reliability into order contributions.

8.1 Random stochastic targets

Random stochastic target maps were generated by drawing Gaussian logits for $q_t(x)$ and applying a softmax over three target states. The table reports the maximum relative error between exact posterior chi-square information and the NRID polynomial over $\rho \in \{0.15, 0.42, 0.83\}$. Errors are at floating-point precision.

Table 2: Random stochastic target checks.

n	$ \mathcal{A} $	Seed	Nonzero order coefficients	Max. relative error
3	3	7301	$\Omega_1 = 0.427862658954, \Omega_2 = 0.157131666514, \Omega_3 = 0.006174197214$	8.94×10^{-16}
4	3	7302	$\Omega_1 = 0.150487110305, \Omega_2 = 0.210152994301, \Omega_3 = 0.124589792529, \Omega_4 = 0.005630352265$	2.48×10^{-16}
5	3	7303	$\Omega_1 = 0.058690529870, \Omega_2 = 0.132939714318, \Omega_3 = 0.168998923093, \Omega_4 = 0.058714271649, \Omega_5 = 0.001799353307$	1.56×10^{-15}

9 General reversible-kernel extension

The Boolean result is a special case of a more general spectral identity for reversible Markov degradation.

Proposition 1 (Spectral NRID for reversible degradation). *Let X take values in a finite space $\mathcal{X}$ with distribution μ . Let K be a Markov kernel on $\mathcal{X}$ that is reversible with respect to μ . Generate Y from X using K . Let $\{\phi_a\}_{a \in A}$ be an orthonormal eigenbasis of $L^2(\mu)$ with $\phi_0 \equiv 1$ and*

$$K\phi_a = \lambda_a\phi_a.$$

For $m_t(x) = P(T = t \mid X = x) - p_t$, write

$$m_t = \sum_{a \neq 0} \hat{m}_t(a) \phi_a.$$

Then

$$\mathcal{J}_K(T; Y) = \sum_{a \neq 0} \left(\sum_t \frac{\hat{m}_t(a)^2}{p_t} \right) \lambda_a^2.$$

Proof. Reversibility implies that the posterior operator $f \mapsto \mathbb{E}[f(X) \mid Y]$ is the same self-adjoint operator K acting on functions of Y . Hence

$$P(T = t \mid Y) - p_t = (Km_t)(Y).$$

Expanding m_t in the eigenbasis gives

$$Km_t = \sum_{a \neq 0} \lambda_a \hat{m}_t(a) \phi_a.$$

Taking the $L^2(\mu)$ norm, dividing by p_t , and summing over t yields the identity. $\square$

For product spaces, if $K = K_1 \otimes \cdots \otimes K_n$, the eigenfunctions are products and the eigenvalues multiply. If each coordinate kernel has one constant mode and all nonconstant local modes share eigenvalue ρ_i , the Boolean subset law is recovered. If local spectra contain multiple nonconstant eigenvalues, the decomposition resolves not only order but also the local geometry of the degradation.

9.1 Gaussian semigroup version

For a standard Gaussian source and Ornstein-Uhlenbeck degradation

$$Y = e^{-t}X + \sqrt{1 - e^{-2t}}Z,$$

Hermite polynomials diagonalize the semigroup. Formally, if the centered target likelihoods are square-integrable, then

$$\mathcal{J}_t = \sum_{k \geq 1} \Omega_k e^{-2kt},$$

where Ω_k is the weighted Hermite energy at total degree k . This is the continuous analogue of the Boolean law $\mathcal{J}_\rho = \sum_k \Omega_k \rho^{2k}$ with $\rho = e^{-t}$.

10 Relation to PID and synergy

NRID and PID answer different questions. PID asks which information is redundant, unique, or synergistic with respect to collections of sources. NRID asks how target information is distributed across eigenspaces of a chosen source-degradation operator.

For independent uniform Boolean sources, high NRID order often coincides with intuitive synergy in canonical examples: XOR- k is pure order k , while a copy is pure order 1. However, NRID should not be equated with a full PID. It does not by itself distinguish all redundancy structures within the same order, and it depends on the chosen noise model. Its advantage is that the coefficients are nonnegative, experimentally addressable through degradation curves, and exactly tied to posterior chi-square information.

11 Estimation protocol

A practical estimator can be organized as follows.

1. Collect paired samples (X, T) or a model for $P(T | X)$.
2. Choose a grid of reliabilities ρ_j or anisotropic reliabilities $\boldsymbol{\rho}_j$.
3. Generate degraded samples Y_{ρ_j} by independent source corruption.
4. Estimate $P(T | Y_{\rho_j})$ and compute either posterior chi-square information $\mathcal{J}_{\rho_j}$ or Shannon mutual information $I(T; Y_{\rho_j})$.
5. Fit a nonnegative polynomial in $z = \rho^2$:

$$\mathcal{J}_{\sqrt{z}} \approx \sum_{r=1}^R \Omega_r z^r.$$

6. Infer the first active order from the low- ρ log-log slope or from the lowest nonzero fitted coefficient.

The chi-square version has an exact finite polynomial in the Boolean setting. The Shannon version gives the same leading exponent near $\rho = 0$ but need not be an exact finite polynomial at all reliabilities.

12 Candidate applications

Sensor fusion. A system whose information lies mostly at order 1 should degrade gracefully when sensors become noisy. A system whose information lies mostly at high order may perform well only when many sensors remain simultaneously reliable.

Privacy and secret sharing. Encoding information into high-order interactions makes it vanish rapidly under independent observation noise. XOR-like encodings are the simplest example.

Neural population codes. NRID offers a controlled way to distinguish information carried by individual neural responses from information that appears only in higher-order population interactions.

Representation learning. Feature sets can be probed by degrading features and measuring the information response. The fitted order spectrum may reveal whether a representation stores target information linearly, pairwise, or in higher-order combinations.

Scientific model comparison. Two models can have the same full-resolution mutual information but different NRID spectra. The spectra may separate robust low-order mechanisms from brittle high-order mechanisms.

13 Limitations

1. The Boolean theorem assumes independent uniform sources. Correlated sources require a different orthogonalization and usually lose the simple subset interpretation.
2. The exact polynomial identity is for posterior chi-square information, not for Shannon mutual information. Shannon information shares the leading weak-signal exponent but has higher-order corrections.
3. NRID depends on the chosen degradation operator. This is a feature for experimental design but a limitation for universal interpretation.
4. High order in the Fourier or eigenspace sense is not automatically equivalent to all notions of synergy used in PID.
5. Estimation of high-order coefficients from finite data may be statistically difficult because high-order terms are small near low reliability and numerous at large n .

14 Open problems and conjectural directions

14.1 KL-spectrum beyond the weak-signal limit

Conjecture 1 (Structured KL expansion). *For finite Boolean sources, $I(T; Y_\rho)$ admits an analytic expansion around $\rho = 0$ whose first nonzero exponent is $2d$, where d is the first nonzero NRID order. Higher coefficients are polynomial functionals of the same Fourier likelihood coefficients, including cross-order interactions.*

The first exponent follows from the weak-signal expansion. The deeper problem is to characterize all higher coefficients and determine whether any useful nonnegative decomposition survives beyond chi-square information.

14.2 Correlated-source NRID

For correlated sources, one can replace Fourier-Walsh characters with orthogonal functions under P_X . The open problem is to identify degradation operators whose posterior action is diagonal or nearly diagonal in that basis. A successful theory would separate interaction order from source correlation structure.

14.3 Causal degradation

Independent synthetic noise is only one intervention. A causal NRID would define degradation by interventions on causal mechanisms rather than observed variables. The candidate object is an information spectrum over causal cuts.

14.4 Continuous and geometric NRID

For manifolds, graphs, diffusion processes, and language-like state spaces, the reversible-kernel proposition suggests that information decomposes by Laplacian or Markov eigenmodes. In such settings, “order” may be replaced by frequency, scale, graph cut, or semantic direction.

14.5 Operational coding meaning

The posterior chi-square information has hypothesis-testing and divergence interpretations, but the operational meaning of individual coefficients Ω_r deserves a coding theorem. A possible direction is to relate Ω_r to the low-signal distinguishability of target states when only r -way reliability perturbations are observable.

15 Research ledger

The following items are preserved for later development.

1. **Subset law.** The strongest Boolean identity is anisotropic and subset-specific:

$$\mathcal{J}_\rho = \sum_{S \neq \emptyset} \omega_S \prod_{i \in S} \rho_i^2.$$

This should be the base theorem for future work.

2. **Order distribution.** Normalized coefficients $\pi_r = \Omega_r / \mathcal{J}_1$ define a genuine probability distribution over information order. The degradation curve is its probability generating function.
3. **Fragility exponent.** The leading low-reliability exponent is twice the first active interaction order.
4. **Experimental estimator.** Fit $\mathcal{J}$ as a nonnegative polynomial in $z = \rho^2$. Use Shannon mutual information only for leading-order diagnosis unless higher-order corrections are modeled.

5. **General spectral form.** Reversible degradation gives

$$\mathcal{J}_K = \sum_{a \neq 0} A_a \lambda_a^2, \quad A_a = \sum_t \hat{m}_t(a)^2 / p_t.$$

This may be the correct general object: an information-energy measure over the spectrum of a degradation semigroup.

6. **Design principle.** Low-order information is robust. High-order information is selective and brittle. This tradeoff may be formalized as an information-theoretic robustness/secret-sharing principle.

16 Second research pass: spectroscopy, tomography, and extremal laws

The first pass established the NRID polynomial for independent Boolean source degradation. The second pass tested whether the polynomial is only a verification identity or whether it can support additional laws. Four useful structures emerged: a response-curve cumulant theorem, a local-observability generalization beyond bit-flip noise, exact subset tomography by inclusion-exclusion, and a chi-square analogue of the most-informative Boolean-function principle.

16.1 Response-curve spectroscopy

Let

$$z = \rho^2, \quad G(z) = \mathcal{J}_{\sqrt{z}} = \sum_{r=1}^n \Omega_r z^r.$$

For $0 < z \leq 1$, define the reliability-tilted order weights

$$w_r(z) = \frac{\Omega_r z^r}{G(z)}.$$

These weights form a probability distribution over active orders. Define the *effective surviving order*

$$D(z) = \sum_{r=1}^n r w_r(z).$$

Theorem 2 (Response-curve cumulant law). *Let $\theta = \log z$ and $K(\theta) = \log G(e^\theta)$. Then*

$$K'(\theta) = D(e^\theta), \quad K''(\theta) = \text{Var}_{w(\theta)}(R) \geq 0.$$

More generally, the derivatives of K with respect to θ are the cumulants of the reliability-tilted order variable R . Equivalently,

$$D(\rho) = \frac{1}{2} \frac{d \log \mathcal{J}_\rho}{d \log \rho}$$

and

$$\text{Var}_{w_\rho}(R) = \frac{1}{4} \frac{d^2 \log \mathcal{J}_\rho}{d(\log \rho)^2}.$$

Proof. Since

$$G(e^\theta) = \sum_r \Omega_r e^{r\theta},$$

$K(\theta)$ is the log moment-generating function of the finite measure with masses Ω_r . Normalizing by $G(e^\theta)$ gives $w_r(\theta)$. Differentiating once gives the tilted mean of R ; differentiating twice gives the tilted variance. The conversion from θ to $\log \rho$ uses $\theta = \log z = 2 \log \rho$. $\square$

Corollary 2 (Monotone visible order). *The effective surviving order is nondecreasing as reliability increases:*

$$\frac{dD}{d \log z} = \text{Var}_{w(z)}(R) \geq 0.$$

It is constant if and only if all nonzero NRID mass lies at one order.

This converts the response curve into an information spectrometer. The log-log slope gives the mean order of the information that remains visible at that reliability; the curvature gives the variance of the visible orders. The asymptotic fragility exponent is only the low-reliability endpoint of this richer curve.

16.2 Experimental spectroscopy

Table 3 compares systems with similar full-resolution quadratic information but different reliability response. The random balanced target uses the exact expectation described in Proposition 3. The leaky parity target is stochastic with

$$P(T = +1 \mid X) = \frac{1}{2} (1 + 0.05X_1 + 0.90X_1X_2X_3X_4X_5).$$

It has a tiny first-order leak and a large fifth-order component. Thus its limiting low-reliability exponent is order 1, but at moderate reliability its effective order is close to 5. This exposes an “asymptotic leakage trap”: the first nonzero order can be technically low while almost all practically visible information is high-order.

Table 3: Response spectroscopy for selected systems. D_ρ is the effective surviving order at reliability ρ .

System	first	mean R	$D_{0.5}$	$\mathcal{J}_{0.5}$	$D_{0.9}$	$\mathcal{J}_{0.9}$
Copy/dictator-10	1	1.0000	1.0000	2.5000×10^{-1}	1.0000	8.1000×10^{-1}
Parity-10	10	10.0000	10.0000	9.5367×10^{-7}	10.0000	1.2158×10^{-1}
Majority-9	1	2.4609	1.0252	1.7030×10^{-1}	1.6019	6.6110×10^{-1}
Address/mux, $k = 8$	1	5.0000	2.6000	5.8212×10^{-3}	4.5801	3.6448×10^{-1}
Random balanced, $n = 10$	1	5.0049	2.2406	8.1255×10^{-3}	4.4870	3.6792×10^{-1}
Leaky parity-5	1	4.9877	3.2345	1.4160×10^{-3}	4.9715	2.8446×10^{-1}

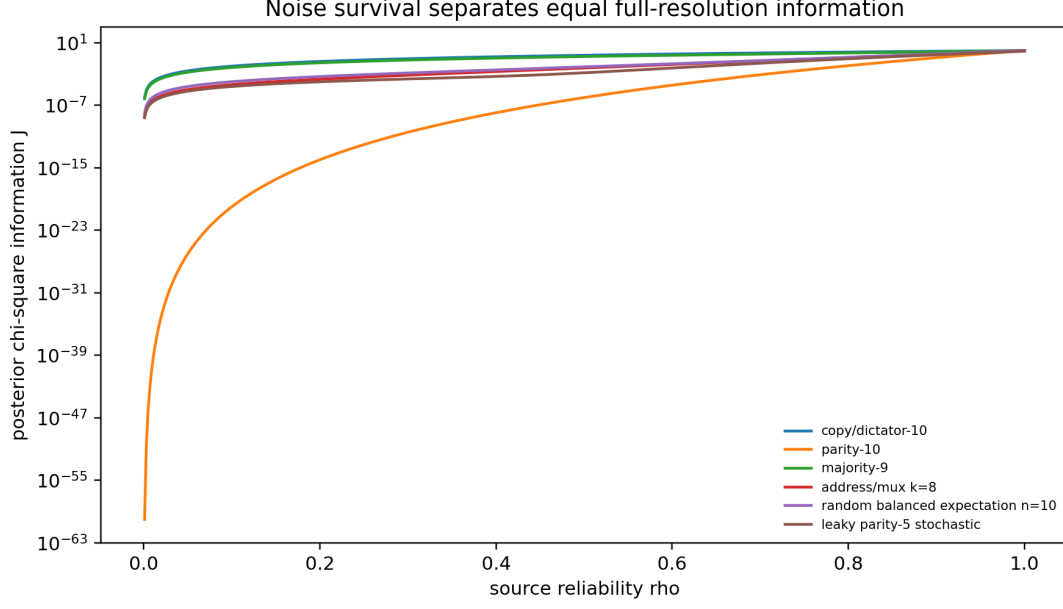


Figure 3: Posterior chi-square survival curves. All deterministic binary examples have $\mathcal{J}_1 = 1$, but the curves separate sharply under degradation.

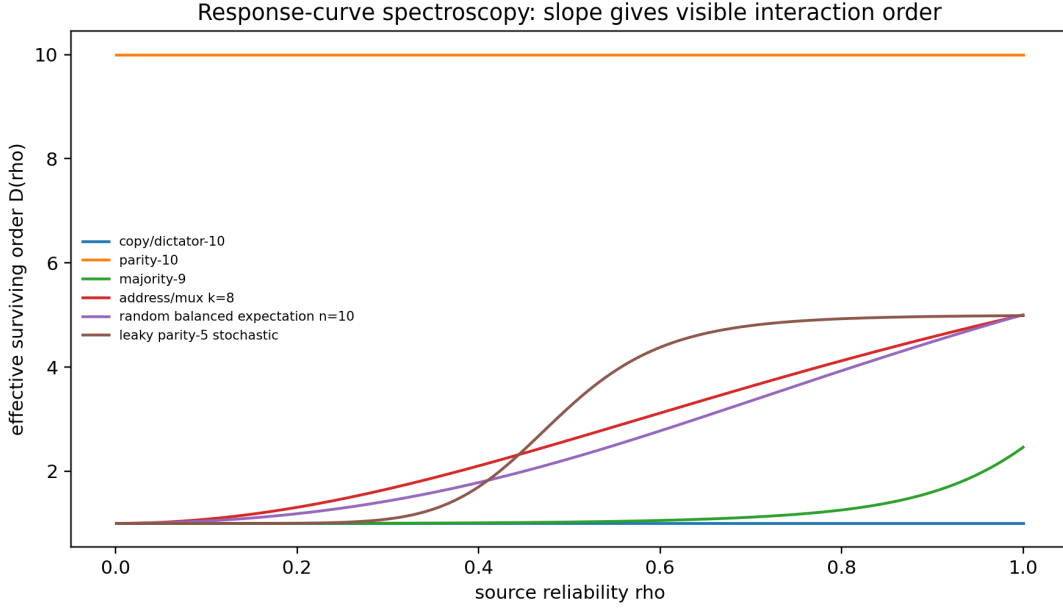


Figure 4: Effective surviving order obtained from the response-curve slope. Curves that are flat correspond to pure-order systems. Mixed systems move from low-order visibility at small ρ toward their full-resolution order distribution as reliability increases.

A finite-difference check of the cumulant theorem across six systems and three reliability values gave maximum absolute error 1.15×10^{-10} for D and 1.78×10^{-5} for the visible-order variance. The larger variance error is due to second-difference numerical conditioning, not a failure of the identity.

16.3 Local observability theorem

The first theorem used binary symmetric channels. The next observation shows that the same polynomial is not tied to bit flips.

Definition 4 (Local observability). *Let $X_i \in \{-1, +1\}$ be uniform and let Y_i be any observation generated from X_i , independently across i . Define the posterior score*

$$a_i(Y_i) = \mathbb{E}[X_i \mid Y_i]$$

and the scalar observability

$$z_i = \mathbb{E}[a_i(Y_i)^2].$$

For a binary symmetric channel with reliability ρ_i , $a_i(Y_i) = \rho_i Y_i$ and $z_i = \rho_i^2$. For an erasure channel that reveals X_i with probability η_i and otherwise outputs an erasure symbol, $a_i(Y_i) = Y_i$ if observed and 0 if erased, hence $z_i = \eta_i$.

Theorem 3 (Product-observation observability law). *For independent fair Boolean sources and arbitrary independent local observation channels $X_i \rightarrow Y_i$,*

$$\mathcal{J}(T; Y) = \sum_{\emptyset \neq S \subseteq [n]} \omega_S(T; X) \prod_{i \in S} z_i.$$

Thus the entire product degradation mechanism enters posterior chi-square information only through the local observabilities $z_i = \mathbb{E}[\mathbb{E}[X_i \mid Y_i]^2]$.

Proof. For each Fourier character,

$$\mathbb{E}[\chi_S(X) \mid Y] = \mathbb{E}\left[\prod_{i \in S} X_i \mid Y\right] = \prod_{i \in S} \mathbb{E}[X_i \mid Y_i] = \prod_{i \in S} a_i(Y_i),$$

using conditional independence across coordinates. The functions $\prod_{i \in S} a_i(Y_i)$ are orthogonal across distinct S , because $\mathbb{E}[a_i(Y_i)] = \mathbb{E}[X_i] = 0$. Their squared norm is

$$\mathbb{E}\left[\prod_{i \in S} a_i(Y_i)^2\right] = \prod_{i \in S} z_i.$$

Substituting the Fourier expansion of each centered target likelihood m_t and summing the weighted squared norms gives the result. $\square$

This theorem suggests that z_i is the correct experimental coordinate for Boolean source observability. Different physical perturbations can become equivalent once they induce the same posterior-score variance.

16.4 Experiment: bit-flip and erasure collapse

The test channel was

$$P(T = +1 \mid X) = \frac{1}{2} (1 + 0.45X_1 + 0.35X_2X_3 + 0.20X_1X_4X_5).$$

Its nonzero subset coefficients are

$$\omega_{\{1\}} = 0.2025, \quad \omega_{\{2,3\}} = 0.1225, \quad \omega_{\{1,4,5\}} = 0.0400.$$

Therefore, under symmetric observability z ,

$$\mathcal{J}(z) = 0.2025z + 0.1225z^2 + 0.0400z^3.$$

Exact enumeration confirmed that a BSC with $\rho = \sqrt{z}$, an erasure channel with retention probability z , and the NRID polynomial agree to floating-point precision.

Table 4: Different degradation mechanisms collapse under the same observability coordinate z .

z	BSC, $\rho = \sqrt{z}$	erasure, $\eta = z$	polynomial
0.15	0.03326625	0.03326625	0.03326625
0.40	0.10316000	0.10316000	0.10316000
0.75	0.23765625	0.23765625	0.23765625

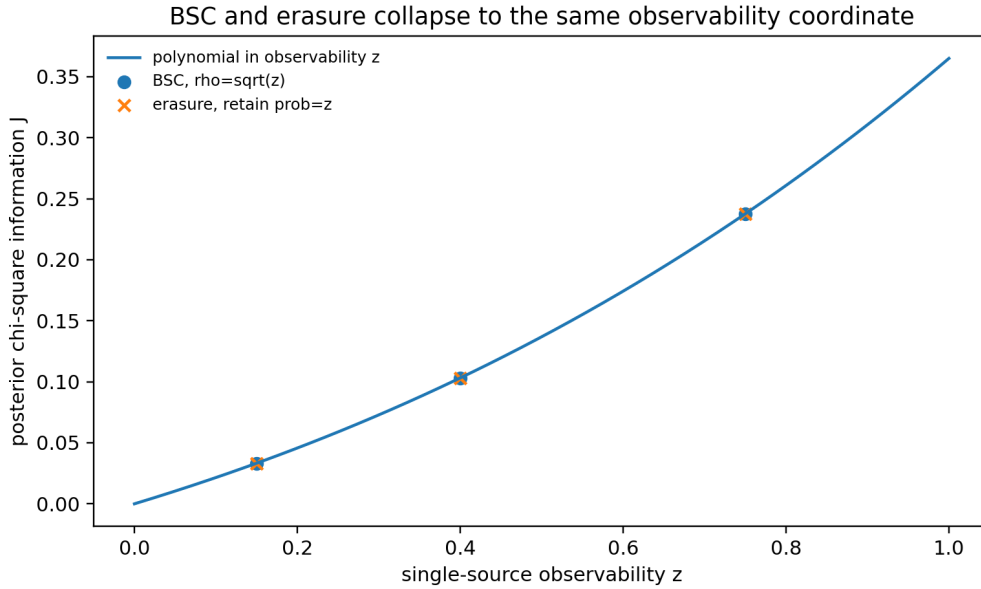


Figure 5: BSC and erasure observations produce the same posterior chi-square information when matched by local observability z .

16.5 Subset tomography by Möbius inversion

Set $z_i = 1$ for coordinates in A and $z_i = 0$ for coordinates outside A . Operationally, this means that coordinates in A are revealed noiselessly and all other coordinates are blinded. Let

$$H(A) = \mathcal{J}(z_i = \mathbf{1}[i \in A]).$$

The subset law gives

$$H(A) = \sum_{S \subseteq A} \omega_S.$$

Therefore, by Möbius inversion on the subset lattice,

$$\omega_S = \sum_{B \subseteq S} (-1)^{|S|-|B|} H(B).$$

This is an exact ablation protocol for recovering the information hypergraph.

For the hypergraph channel above, direct Fourier computation and Möbius recovery give the same atoms:

S	ω_S direct	ω_S recovered
$\{1\}$	0.2025	0.2025
$\{2, 3\}$	0.1225	0.1225
$\{1, 4, 5\}$	0.0400	0.0400

with zero recovery error in exact arithmetic.

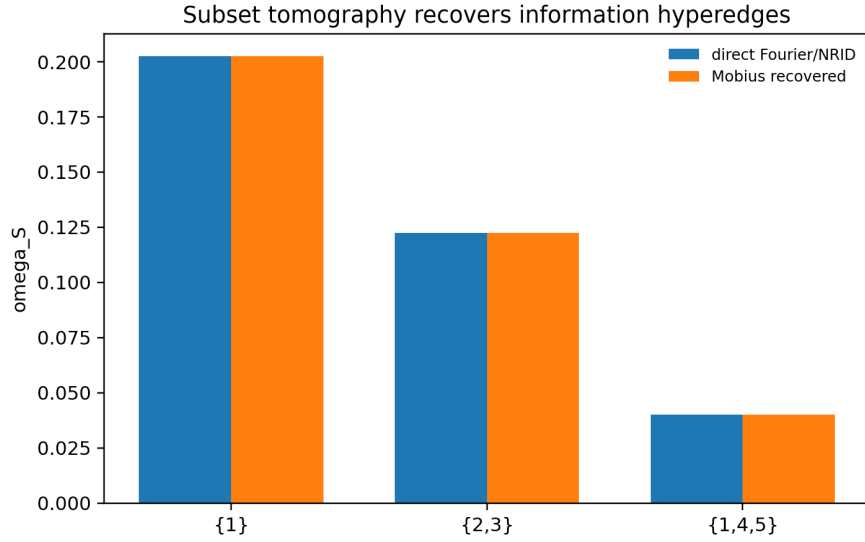


Figure 6: Subset tomography recovers the nonzero information hyperedges from ablation values $H(A)$.

This creates a direct experimental target: design reliability interventions, measure $\mathcal{J}$, and reconstruct the hypergraph of informational interactions. For many variables the full subset lattice is infeasible, but sparse recovery may be possible when only a small number of hyperedges carry most of the information.

16.6 Multiplexer/addressing law

A multiplexer is a useful test case because it routes one of many data lines to the output according to address bits. Let $A = (A_1, \dots, A_k)$ be k address bits and let $\{D_a : a \in \{-1, +1\}^k\}$ be 2^k independent data bits. Define the target

$$T = D_A.$$

Equivalently,

$$T = \sum_{a \in \{-1, +1\}^k} D_a 2^{-k} \prod_{j=1}^k (1 + a_j A_j).$$

Every Fourier term contains one data bit and some subset of the address bits. Hence the order spectrum is exactly a shifted binomial law.

Proposition 2 (Multiplexer law). *For the k -address-bit multiplexer,*

$$\Omega_r = \frac{\binom{k}{r-1}}{2^k}, \quad r = 1, \dots, k+1.$$

Consequently,

$$\mathcal{J}_\rho = \rho^2 \left(\frac{1 + \rho^2}{2} \right)^k$$

and

$$D(\rho) = 1 + \frac{k\rho^2}{1 + \rho^2}.$$

At full reliability, the mean order is $1 + k/2$ and the order variance is $k/4$.

Proof. The expansion above has, for each data line D_a , one term for every subset of the k address bits. A term using s address bits has total order $s + 1$ and coefficient magnitude 2^{-k} . There are $2^k \binom{k}{s}$ such terms, so the Fourier energy at order $r = s + 1$ is

$$2^k \binom{k}{s} 2^{-2k} = \binom{k}{s} / 2^k.$$

The formulas for $\mathcal{J}_\rho$ and $D(\rho)$ follow by summing the binomial generating function. $\square$

Direct enumeration for $k = 1, 2, 3$ matched the formula exactly. For $k = 8$, the full-resolution mean order is 5, but the effective order is only 2.6 at $\rho = 0.5$ and 4.58 at $\rho = 0.9$. Thus address-based routing is neither pure low-order nor pure high-order: it has a binomial ladder of interaction orders.

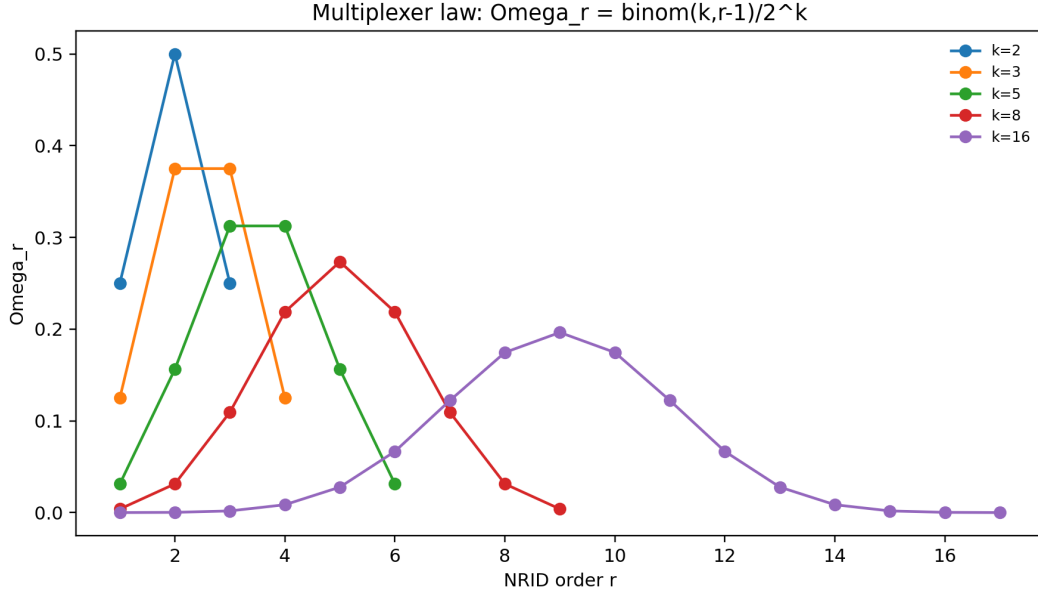


Figure 7: The multiplexer/addressing spectrum is a shifted binomial distribution over NRID orders.

16.7 Random balanced Boolean targets

For a balanced deterministic Boolean target $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$, NRID reduces to the usual Fourier degree weights:

$$\Omega_r = \sum_{|S|=r} \hat{f}(S)^2.$$

This connects NRID to Boolean-function analysis and to functional ANOVA/Sobol-style interaction decompositions [19, 20, 23].

Proposition 3 (Expected spectrum of a random balanced target). *Let f be chosen uniformly from balanced Boolean functions on n bits. Then, for every $r \geq 1$,*

$$\mathbb{E}[\Omega_r] = \frac{\binom{n}{r}}{2^n - 1}.$$

Thus the expected full-reliability mean order is

$$\mathbb{E}[R] = \frac{n2^{n-1}}{2^n - 1} \approx \frac{n}{2}.$$

Proof. For every balanced Boolean function, Parseval gives

$$\sum_{S \neq \emptyset} \hat{f}(S)^2 = 1,$$

because the constant coefficient vanishes. The uniform distribution over balanced truth tables is invariant under permutations of the nonconstant Fourier characters induced by cube symmetries. Hence each of the $2^n - 1$ nonempty characters has the same expected squared coefficient, namely $1/(2^n - 1)$. Summing over the $\binom{n}{r}$ characters of size r gives the formula. $\square$

An average over 400 random balanced truth tables for $n = 10$ gave maximum absolute deviation 7.17×10^{-4} from the formula. The result suggests that arbitrary balanced Boolean targets are typically middle-order objects, even though a small first-order coefficient is usually present and controls the strict $\rho \downarrow 0$ exponent.

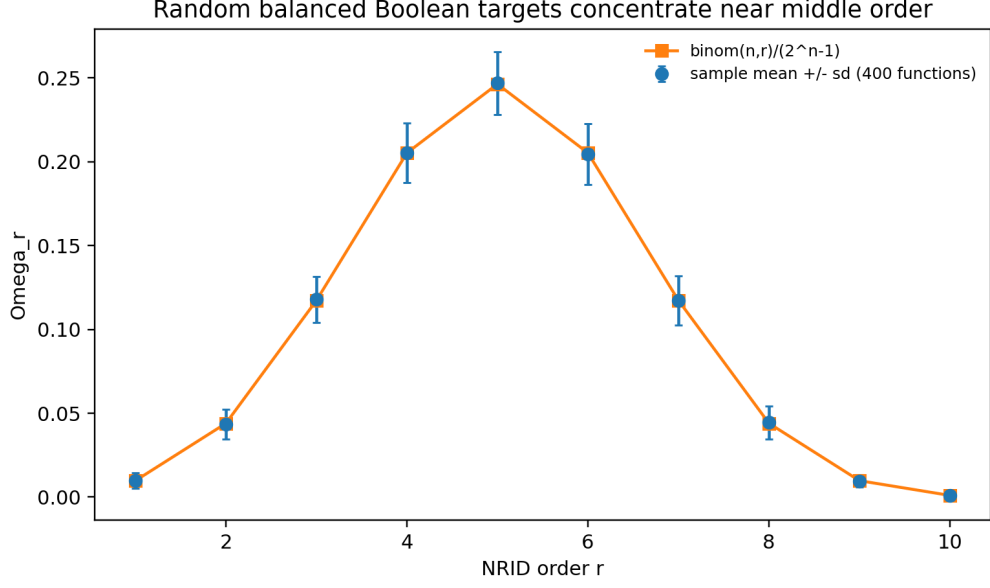


Figure 8: Exact spectra averaged over random balanced Boolean targets on 10 bits. The sample mean matches $\binom{n}{r}/(2^n - 1)$.

16.8 A chi-square most-informative theorem

The Courtade-Kumar conjecture asks, in one form, whether dictator functions maximize Shannon mutual information between a Boolean function and noisy inputs under a binary symmetric channel [21]. Recent work continues to make progress on high-noise and coordinate-wise variants [22]. NRID gives a complete analogue for posterior chi-square information.

Theorem 4 (Universal symmetric chi-square bound). *Let T be any finite target with support size K generated from independent fair Boolean sources, and let Y_ρ be the symmetric BSC-degraded source. Then*

$$\mathcal{J}_\rho(T; Y) \leq (K - 1)\rho^2.$$

For a binary target, $\mathcal{J}_\rho(T; Y) \leq \rho^2$.

Proof. The NRID polynomial gives

$$\mathcal{J}_\rho = \sum_{r=1}^n \Omega_r \rho^{2r} \leq \rho^2 \sum_{r=1}^n \Omega_r = \rho^2 \mathcal{J}_1.$$

At full reliability, write $q_t(X) = P(T = t | X)$ and $p_t = \mathbb{E} q_t(X)$. Then

$$\mathcal{J}_1 = \sum_t \frac{\text{Var}(q_t(X))}{p_t}.$$

Since $0 \leq q_t \leq 1$ and $\mathbb{E} q_t = p_t$,

$$\text{Var}(q_t) \leq p_t(1 - p_t).$$

Therefore

$$\mathcal{J}_1 \leq \sum_t (1 - p_t) = K - 1.$$

□

For binary deterministic targets, equality in the upper bound requires all NRID mass to be at order 1. Among Boolean functions this is attained exactly by dictators and anti-dictators. Exhaustive enumeration of all 12,870 balanced Boolean functions on 4 bits confirmed that the maximum of $\mathcal{J}_\rho$ is ρ^2 at $\rho \in \{0.25, 0.50, 0.75, 0.90\}$, with 8 maximizers, corresponding to the 4 dictators and their complements.

16.9 Applications suggested by the second pass

Information spectroscopy. A system can be probed by injecting controlled noise and estimating $\mathcal{J}_\rho$. The first derivative of $\log \mathcal{J}_\rho$ estimates the visible order, while the second derivative estimates the spread of visible orders. This may be useful when a full PID calculation is infeasible.

Ablation-based hypergraph discovery. The Möbius formula turns subset ablations into information hyperedge estimates. This gives a concrete route to feature-interaction discovery, epistasis mapping, multi-sensor diagnostics, and neural population analysis.

Robustness budgeting. The bound $\mathcal{J}_\rho \leq (K - 1)\rho^2$ separates a universal reliability ceiling from the order spectrum that determines how quickly a particular system falls below that ceiling. Low-order information is robust; high-order information may be accurate at full reliability but brittle under source degradation.

Multiplexers, attention, and conditional computation. The multiplexer law suggests that address-based routing creates a shifted binomial order spectrum. Systems that select information conditionally, including digital multiplexers, lookup tables, sparse mixture-of-expert gates, and attention-like mechanisms, may have predictable reliability profiles governed by the depth and uncertainty of their addressing variables.

Privacy and secret sharing. High-order encodings suppress observability under partial or noisy measurements. XOR is the extreme pure-order case; multiplexer and random balanced targets provide more graded designs. NRID can quantify the tradeoff between recoverability at high reliability and invisibility at low reliability.

16.10 New targets for further study

1. **Finite-sample spectroscopy.** Develop estimators and confidence intervals for $D(\rho)$ and the curvature-derived order variance from empirical estimates of $\mathcal{J}_\rho$.
2. **Sparse information hypergraph recovery.** Replace full subset Möbius inversion with compressed sensing or adaptive experimental design when ω_S is sparse.
3. **Biased and correlated sources.** Generalize local observability using orthonormal local bases, canonical correlations, or Markov operators so that source bias and dependence do not masquerade as interaction order.
4. **KL-order cumulants.** Determine which parts of the response-curve cumulant law survive for Shannon mutual information beyond the weak-signal limit.
5. **Coding theorem for atoms.** Seek an operational interpretation of ω_S in hypothesis testing, prediction under quadratic loss, or low-signal communication.

6. **Architecture spectra.** Compute NRID spectra of common computational primitives: lookup tables, decision trees, sparse gates, error-correcting decoders, and attention heads.

16.11 Research ledger update

The following additions should be preserved as high-value results.

1. The log response curve is a cumulant-generating function for the reliability-tilted order distribution.
2. Local observation mechanisms collapse to $z_i = \mathbb{E}[\mathbb{E}[X_i | Y_i]^2]$ for posterior chi-square information.
3. Subset ablations recover ω_S exactly by Möbius inversion.
4. The k -address-bit multiplexer has spectrum $\Omega_r = \binom{k}{r-1}/2^k$ and response $\mathcal{J}_\rho = \rho^2((1 + \rho^2)/2)^k$.
5. A random balanced Boolean target has expected spectrum $\mathbb{E}[\Omega_r] = \binom{n}{r}/(2^n - 1)$.
6. For any K -state target, $\mathcal{J}_\rho \leq (K - 1)\rho^2$ under symmetric BSC degradation; the binary deterministic maximizers are dictators and anti-dictators.

A Third research pass: modal alphabets, composition algebra, and symmetric summaries

This pass asks whether the Boolean NRID identity is an accident of binary sources or a visible part of a larger spectral principle. The answer appears to be the latter. Exact experiments also reveal three additional structures: target-side data processing contracts every information hyperedge, independent target composition has a simple multiplicative calculus, and symmetric summaries such as exact counts and modular residues have unexpectedly sharp order spectra.

A.1 Finite-alphabet modal NRID

Let the sources $X_1, \dots, X_n$ be independent finite variables with laws μ_i . Each source is observed locally through an arbitrary channel $K_i(y_i | x_i)$, producing Y_i , and the local observation channels are conditionally independent given X . Let ν_i be the marginal law of Y_i . Define the conditional-expectation operator

$$A_i : L^2(\mu_i) \rightarrow L^2(\nu_i), \quad (A_i f)(y_i) = \mathbb{E}[f(X_i) | Y_i = y_i].$$

Constants are singular functions with singular value 1. Choose a singular system

$$A_i \phi_{i,a} = \sigma_{i,a} \psi_{i,a}, \quad a = 0, 1, \dots, d_i - 1,$$

where $\phi_{i,0} = \psi_{i,0} = 1$, $\sigma_{i,0} = 1$, and the nonconstant singular values lie in $[0, 1]$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$ define

$$\phi_\alpha(x) = \prod_i \phi_{i,\alpha_i}(x_i), \quad \psi_\alpha(y) = \prod_i \psi_{i,\alpha_i}(y_i), \quad \sigma_\alpha = \prod_i \sigma_{i,\alpha_i}.$$

For each target state t , expand the centered likelihood

$$m_t(x) = \mathbb{P}(T = t | X = x) - p_t = \sum_{\alpha \neq 0} \hat{m}_t(\alpha) \phi_\alpha(x).$$

Define the modal atom

$$\omega_\alpha = \sum_t \frac{\hat{m}_t(\alpha)^2}{p_t}.$$

Theorem 5 (Finite-alphabet modal noise law). *For arbitrary independent finite sources and arbitrary independent local observation channels,*

$$\mathcal{J}(T; Y) = \sum_{\alpha \neq 0} \omega_\alpha \sigma_\alpha^2.$$

Equivalently, each target-likelihood mode is attenuated by the squared singular value of the observation mode that must carry it.

Proof. The product conditional-expectation operator $A = A_1 \otimes \dots \otimes A_n$ maps m_t to

$$Am_t = \sum_{\alpha \neq 0} \hat{m}_t(\alpha) \sigma_\alpha \psi_\alpha.$$

The product singular functions ψ_α are orthonormal in $L^2(\nu_1 \otimes \cdots \otimes \nu_n)$. Therefore

$$\mathbb{E}_Y[(P(T = t | Y) - p_t)^2] = \sum_{\alpha \neq 0} \hat{m}_t(\alpha)^2 \sigma_\alpha^2.$$

Dividing by p_t and summing over target states gives the identity. $\square$

This theorem exposes a limitation of scalar observability. For binary sources there is only one nonconstant source mode, so a local channel is summarized by a single number $z_i = \sigma_i^2$. For q -ary or otherwise finite sources, one source can have several nonconstant modes. Two sensors can have the same scalar quantity $\sum_a \sigma_{i,a}^2$ yet reveal different target information if they reveal different modes.

Table 5: Mode-selective observability for ternary sources. The two highlighted channels have the same sum of squared nonconstant singular values, 0.36 per coordinate, but different posterior chi-square information because they expose different source modes.

Scenario	scalar energy per coordinate	exact $\mathcal{J}$	modal prediction
balanced modes (0.55, 0.55)	0.605, 0.605	0.04371545	0.04371545
mode 1 only	0.36, 0.36	0.03461780	0.03461780
mode 2 only	0.36, 0.36	0.01068451	0.01068451
cross mode 1 $\rightarrow$ 2	0.36, 0.36	0.04015665	0.04015665
cross mode 2 $\rightarrow$ 1	0.36, 0.36	0.01314622	0.01314622

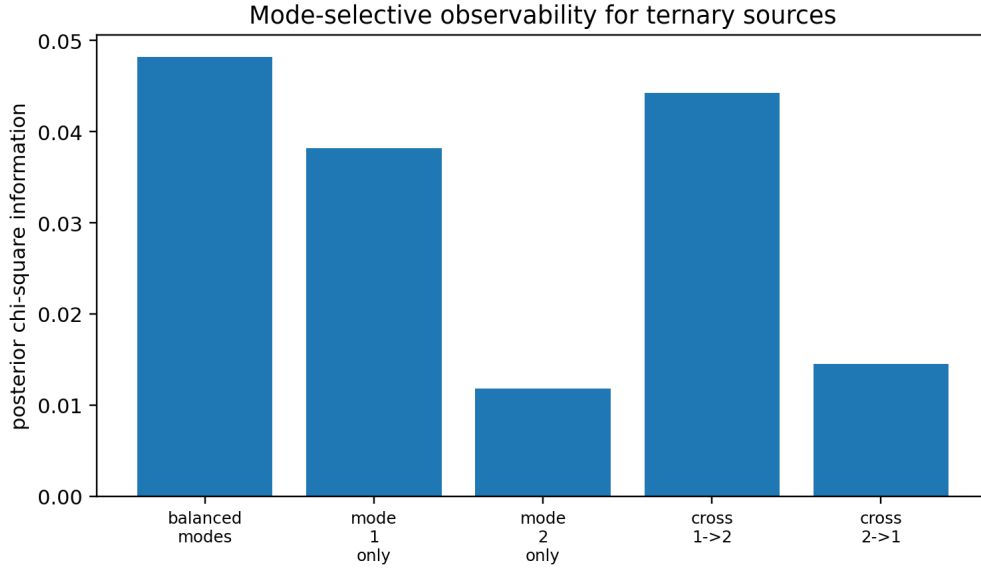


Figure 9: Mode-selective observability. Scalar source reliability is insufficient for nonbinary alphabets; the information response depends on which singular modes of each source are observable.

For a q -ary symmetric channel, all nonconstant modes share the same eigenvalue λ . The modal law collapses to an order law again:

$$\mathcal{J}_\lambda = \sum_{r=1}^n \Omega_r^{(q)} \lambda^{2r},$$

where r is the number of nonconstant source modes in the product basis. Exact enumeration over ternary sources gave maximum absolute error 1.53×10^{-16} between direct posterior chi-square information and the polynomial prediction. In a four-source ternary test, the spectra were

$$T = X_1 : \quad \mathcal{J}_\lambda = 2\lambda^2,$$

$$T = \sum_i X_i \bmod 3 : \quad \mathcal{J}_\lambda = 2\lambda^8,$$

while a random stochastic target had

$$\mathcal{J}_\lambda = 0.303631\lambda^2 + 0.319365\lambda^4 + 0.244079\lambda^6 + 0.0676803\lambda^8.$$

Thus the Boolean order law is the one-mode-per-source special case of a general singular-mode law.

A.2 Target-side data processing contracts individual hyperedges

NRID also has a target-side data-processing law. Suppose U is produced from T by a channel $R(u | t)$, with

$$p'_u = \sum_t R(u | t)p_t, \quad \hat{m}'_u(S) = \sum_t R(u | t)\hat{m}_t(S).$$

Then the subset atom for the postprocessed target is

$$\omega'_S = \sum_u \frac{\hat{m}'_u(S)^2}{p'_u}.$$

Theorem 6 (Coefficientwise target data processing). *For every source subset S ,*

$$\boxed{\omega'_S \leq \omega_S}.$$

Consequently, every order coefficient and every response curve is contracted:

$$\Omega'_r \leq \Omega_r, \quad \mathcal{J}'_\rho \leq \mathcal{J}_\rho \quad \text{for all } \rho \in [0, 1].$$

Proof. For fixed S , write $a_t = \hat{m}_t(S)$. Since $\sum_t a_t = 0$, this is a signed perturbation of the target prior. For each u , Cauchy–Schwarz gives

$$\frac{(\sum_t R(u | t)a_t)^2}{\sum_t R(u | t)p_t} \leq \sum_t R(u | t) \frac{a_t^2}{p_t}.$$

Summing over u and using $\sum_u R(u | t) = 1$ yields $\omega'_S \leq \omega_S$. □

Exact random-target tests supported the theorem numerically. A random five-source, three-state target had total full-resolution posterior chi-square information 0.661054 before target garbling and 0.0510625 after garbling. No subset atom increased; the largest after-minus-before atom difference was -1.75×10^{-3} .

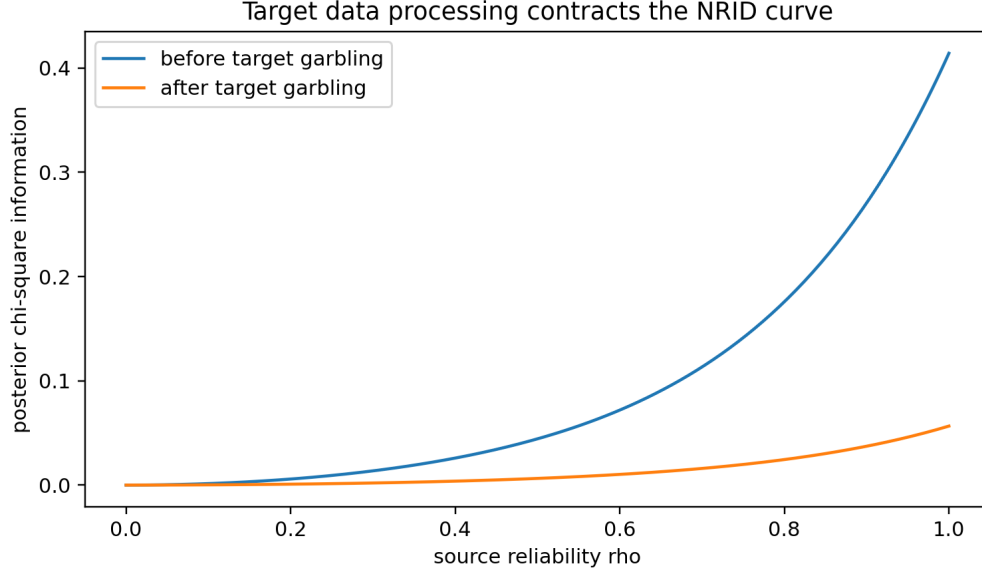


Figure 10: Target data processing contracts the whole NRID response curve. The coefficientwise theorem is stronger: each individual source hyperedge is contracted.

A.3 Composition algebra: collection is not fusion

Let A and B be independent target variables carried by disjoint source blocks. Write their NRID generating functions as

$$G_A(z) = \mathcal{J}_A(z), \quad G_B(z) = \mathcal{J}_B(z).$$

There are two natural ways to combine them, and they have different algebra.

First, if the new target is the collected pair $T = (A, B)$, then likelihood ratios multiply and

$$1 + G_{(A,B)}(z) = (1 + G_A(z))(1 + G_B(z)).$$

The collection retains both component spectra and adds cross-union atoms.

Second, if A and B are balanced deterministic $\{\pm 1\}$ targets and the new target is their fused product $C = AB$, then the centered target function is the product of the two centered component functions. Since the source blocks are disjoint,

$$G_C(z) = G_A(z)G_B(z).$$

Fusion therefore raises interaction order; collection preserves low-order access.

Table 6: Exact composition laws. Here $z = \rho^2$. The collected pair (X_1, X_2X_3) has order 1, 2, and 3 atoms, but fusing them by multiplication creates pure parity $X_1X_2X_3$ at order 3.

System	empirical spectrum	formula	$\mathcal{J}_1$	half-life ρ
majority ₃ fused with majority ₃	$0.5625z^2 + 0.375z^4 + 0.0625z^6$	G_{maj}^2	1	0.884800
(majority ₃ , majority ₃)	$1.5z + 0.5625z^2 + 0.5z^3 + 0.375z^4 + 0.0625z^6$	$(1 + G_{\text{maj}})^2 - 1$	3	0.820468
(X_1, X_2X_3)	$z + z^2 + z^3$	$(1 + z)(1 + z^2) - 1$	3	0.831513
$X_1X_2X_3$	z^3	z^3	1	0.890899
identity target $T = X$ on 5 bits	$5z + 10z^2 + 10z^3 + 5z^4 + z^5$	$(1 + z)^5 - 1$	31	0.867093

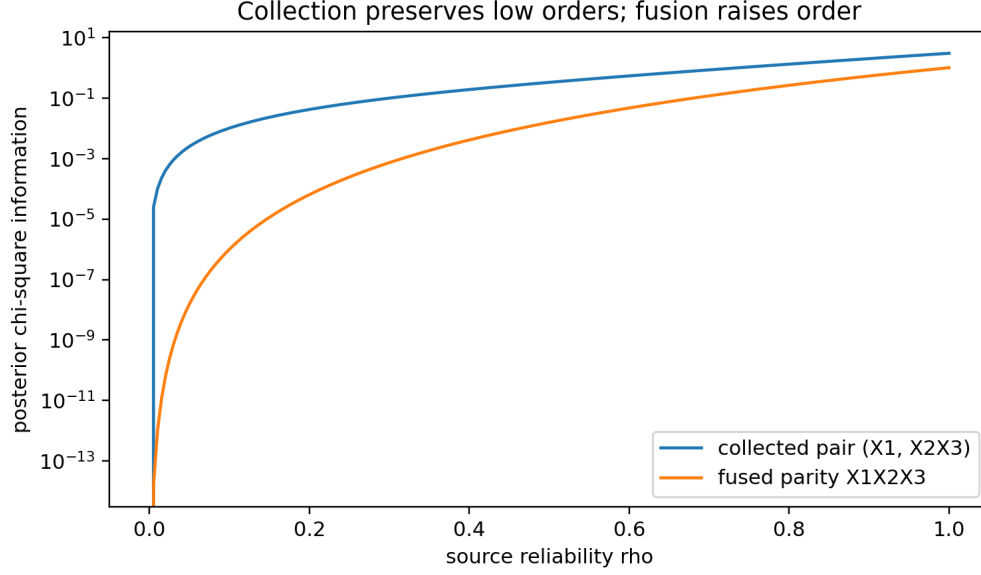


Figure 11: Collection versus fusion. The pair target keeps low-order information visible, while product fusion converts the same ingredients into a pure third-order parity.

A.4 Information hypergraph calculus

The subset polynomial

$$\mathcal{J}(z_1, \dots, z_n) = \sum_{S \neq \emptyset} \omega_S \prod_{i \in S} z_i$$

can be read as a weighted information hypergraph. Because every coefficient is nonnegative, it is multi-affine and supermodular on $[0, 1]^n$. In particular,

$$\frac{\partial \mathcal{J}}{\partial z_i} = \sum_{S \ni i} \omega_S \prod_{j \in S \setminus \{i\}} z_j,$$

and

$$\frac{\partial^2 \mathcal{J}}{\partial z_i \partial z_j} = \sum_{S \supseteq \{i, j\}} \omega_S \prod_{k \in S \setminus \{i, j\}} z_k \geq 0.$$

Thus the marginal value of observing one source can increase when another source becomes observable. This gives a calculus for synergy: positive mixed partial derivatives are exactly the surviving higher-order information hyperedges.

The same polynomial supports a simple notion of an information half-life. For pure order d ,

$$\mathcal{J}_\rho = \rho^{2^d}, \quad \rho_{1/2} = 2^{-1/(2^d)}.$$

A tenth-order parity requires $\rho = 0.965936$ to retain half its full-resolution chi-square information, whereas a dictator bit requires only $\rho = 0.707107$.

A.5 Repeated sensors and observability amplification

The Boolean local-observability theorem can be used to analyze sensor fusion. If the same hidden bit $X \in \{-1, +1\}$ is measured m independent times through BSC reliability ρ , the effective local

observability is

$$z_m(\rho) = \mathbb{E} [\mathbb{E}[X \mid Y_1, \dots, Y_m]^2] .$$

Writing $a = \text{artanh } \rho$ and $p = (1 + \rho)/2$,

$$z_m(\rho) = \sum_{k=0}^m \binom{m}{k} p^k (1-p)^{m-k} \tanh^2((2k-m)a) .$$

For $m = 1$, $z_1 = \rho^2$; for $m = 2$, $z_2 = 2\rho^2/(1 + \rho^2)$. Thus multiple weak sensors can be collapsed into one effective source reliability before applying the NRID polynomial.

At $\rho = 0.5$, repeated independent observations give

$$z_1 = 0.25, \quad z_2 = 0.4, \quad z_3 = 0.517857, \quad z_6 = 0.731173.$$

This suggests an engineering interpretation: source-side redundancy pushes the system to the right on the same information survival curve, while target-side postprocessing pushes the curve downward.

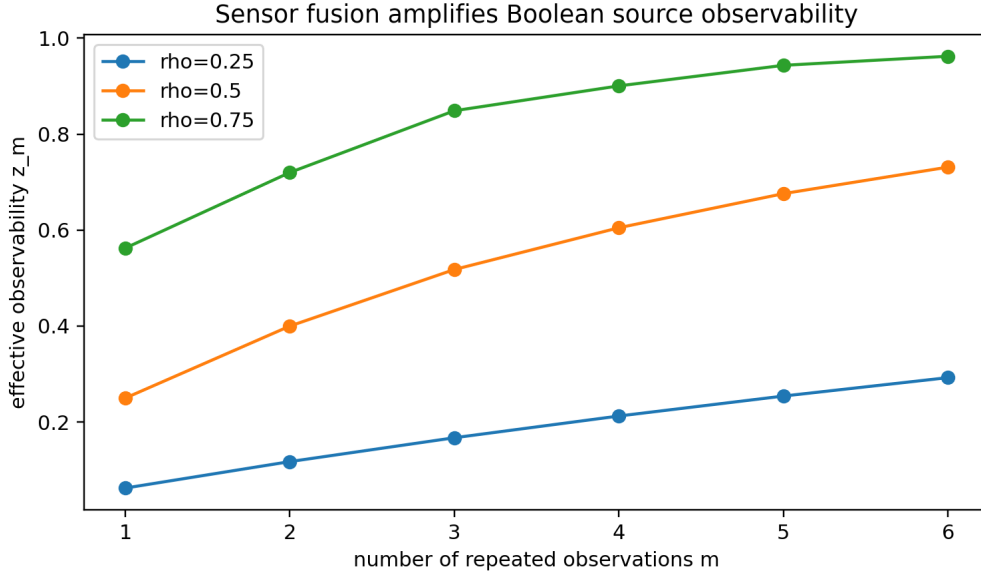


Figure 12: Repeated independent observations of one Boolean source amplify local observability. The resulting scalar $z_m(\rho)$ can be inserted into the subset NRID polynomial.

A.6 Deterministic binary targets and normalized Fourier spectra

For a deterministic binary target $T = f(X) \in \{0, 1\}$ with $p = \mathbb{P}(f = 1)$,

$$\omega_S = \frac{\hat{f}(S)^2}{p(1-p)} .$$

Thus, for binary deterministic targets, NRID is exactly the normalized Fourier energy distribution of the classifier. If f is balanced and encoded as a $\{\pm 1\}$ function, then Ω_r is simply the level- r Fourier weight. This connects the mean NRID order

$$\bar{r} = \frac{\sum_r r \Omega_r}{\sum_r \Omega_r}$$

to average sensitivity and total influence. For majority on odd n , exact enumeration gave the following trend.

Table 7: Majority targets: the mean NRID order equals total influence for balanced $\{\pm 1\}$ functions. The first-order mass decreases slowly while the mean order grows.

n	Ω_1	mean order	$\mathcal{J}_{0.5}$	$\mathcal{J}_{0.9}$
3	0.750000	1.500000	0.191406	0.740360
5	0.703125	1.875000	0.178360	0.701602
7	0.683594	2.187500	0.173121	0.677312
9	0.672913	2.460938	0.170302	0.661098
11	0.666183	2.707031	0.168538	0.649797
13	0.661557	2.932617	0.167330	0.641648

This bridge gives NRID an immediate learning-theoretic interpretation: robust binary labels are low-degree Fourier objects, whereas labels with high spectral order are intrinsically brittle under feature noise. This is adjacent to, but distinct from, the most-informative Boolean function program, which asks which Boolean functions maximize Shannon mutual information through noisy inputs [21, 22].

A.7 Flat spectrum law for exact counts

A particularly surprising result appears for the exact Hamming weight

$$W = \#\{i : X_i = +1\} \in \{0, 1, \dots, n\}.$$

The target has $n + 1$ states, so a deterministic target has total full-resolution posterior chi-square information n . The distribution across orders is exactly flat.

Theorem 7 (Flat count spectrum). *For independent fair Boolean sources and target $W = \#\{i : X_i = +1\}$,*

$$\boxed{\Omega_r(W; X) = 1 \quad \text{for every } r = 1, \dots, n.}$$

Equivalently, each subset S of size r has

$$\omega_S = \binom{n}{r}^{-1}.$$

Therefore

$$\boxed{\mathcal{J}_\rho(W; Y) = \rho^2 + \rho^4 + \dots + \rho^{2n}.$$

Proof. For the state $W = j$, the indicator $q_j(x) = \mathbf{1}[W(x) = j]$ has Fourier coefficient on any subset S of size r

$$\hat{q}_j(S) = 2^{-n} [u^j] (1 + u)^{n-r} (u - 1)^r.$$

The bracketed coefficient is a binary Krawtchouk polynomial $K_r(j)$. Since $p_j = \binom{n}{j} 2^{-n}$,

$$\omega_S = 2^{-n} \sum_{j=0}^n \frac{K_r(j)^2}{\binom{n}{j}}.$$

The dual Krawtchouk orthogonality relation gives

$$\sum_{j=0}^n \frac{K_r(j)^2}{\binom{n}{j}} = \frac{2^n}{\binom{n}{r}},$$

so $\omega_S = \binom{n}{r}^{-1}$ and $\Omega_r = \binom{n}{r} \omega_S = 1$. $\square$

This law is counterintuitive: the exact count is an additive statistic, but exact count information is not purely first-order. It stores one unit of chi-square information in every interaction order. The first-order part gives the noisy linear estimate of the count; higher orders encode increasingly fine corrections needed to identify the exact integer value.

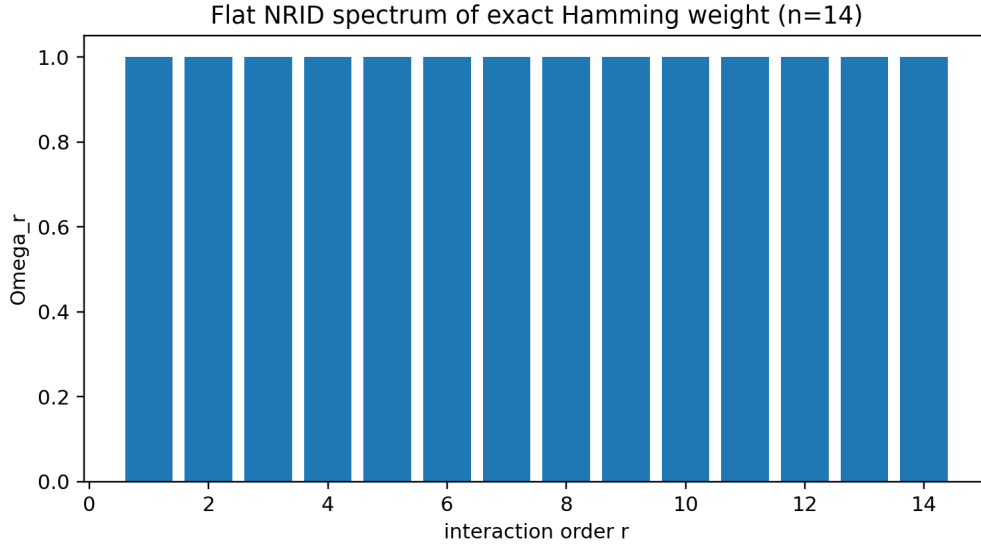


Figure 13: The exact Hamming-weight target has a flat NRID spectrum: one unit of posterior chi-square information at every order. Exact enumeration verified this through $n = 14$ with zero numerical error at displayed precision.

Two consequences follow immediately. First, for any fixed $\rho < 1$,

$$\lim_{n \rightarrow \infty} \mathcal{J}_\rho(W; Y) = \frac{\rho^2}{1 - \rho^2}.$$

Second, to recover a constant fraction of the full-resolution chi-square information $\mathcal{J}_1 = n$, reliability must approach 1 as n grows. Exact threshold calculations gave:

Table 8: Reliability ρ required to recover a fraction α of the full count chi-square information $\mathcal{J}_1 = n$.

n	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.9$
8	0.66695	0.91815	0.98820
16	0.78453	0.95508	0.99372
32	0.87289	0.97638	0.99676
64	0.92999	0.98788	0.99835
128	0.96309	0.99386	0.99917

The flat count spectrum is also a discrete precursor of the Gaussian Mehler spectrum. Let C be the noisy output count. Since the posterior law of W depends on the noisy vector only through C , $I(W; Y) = I(W; C)$. Exact transition-matrix calculations show convergence to the mutual information of correlated Gaussian variables with correlation ρ :

$$I(W; Y) \longrightarrow -\frac{1}{2} \log_2(1 - \rho^2),$$

while the chi-square information converges to $\rho^2/(1 - \rho^2)$. At $n = 128$, the Shannon mutual-information errors were below 10^{-4} bits for $\rho \leq 0.8$.

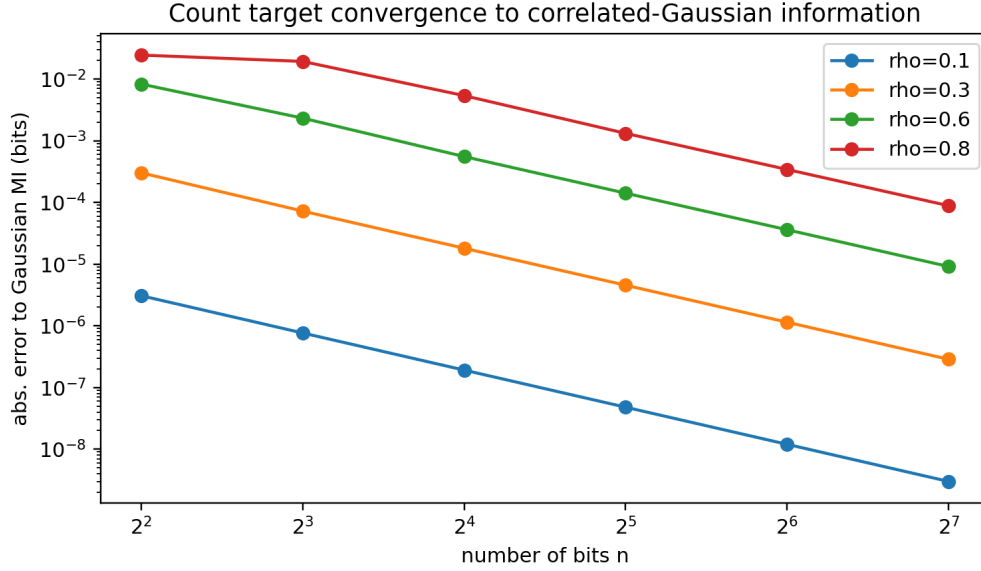


Figure 14: Exact count target: convergence of Shannon mutual information to the correlated-Gaussian limit. The chi-square convergence is already explicit from the flat spectrum formula.

A.8 Modulo-count spectra and spectral hiddenness

The count law suggests studying arithmetic summaries of the count. Let

$$R_m = W \bmod m.$$

For $m = 2$, this is parity and all information is at order n . For $m \geq 3$, low-order coefficients are not generally zero, but most spectral mass can still be concentrated at high order. Using roots of unity, if the residue classes are approximated as exactly uniform, the spectrum is

$$\tilde{\Omega}_r^{(m,n)} = \binom{n}{r} \sum_{\ell=1}^{m-1} \cos^{2(n-r)} \left(\frac{\pi \ell}{m} \right) \sin^{2r} \left(\frac{\pi \ell}{m} \right).$$

For fixed m , the residue probabilities become uniform exponentially fast with n , so the expression is asymptotically exact. It is exact for parity and extremely accurate for small m even at moderate n ; for $m = 3, n = 14$ the relative L^1 spectral error was 6.10×10^{-5} .

Table 9: Modulo-count spectra at $n = 14$. Low-order mass is the normalized mass at orders 1, 2, 3. For mod 3, the formal first nonzero order is 1, but only 3.98×10^{-5} of the mass lies at orders at most 3.

m	total	mean order	sd	mode	first nonzero	mass $r \leq 3$
2	1.000000	14.000	0.000	14	14	0
3	2.000000	10.500	1.620	11	1	3.982×10^{-5}
4	3.000000	9.333	3.636	14	1	1.908×10^{-2}
5	4.000000	8.750	4.174	13	1	1.147×10^{-1}
6	5.000000	8.400	4.404	14	1	2.075×10^{-1}
7	6.000000	8.167	4.507	13	1	2.481×10^{-1}

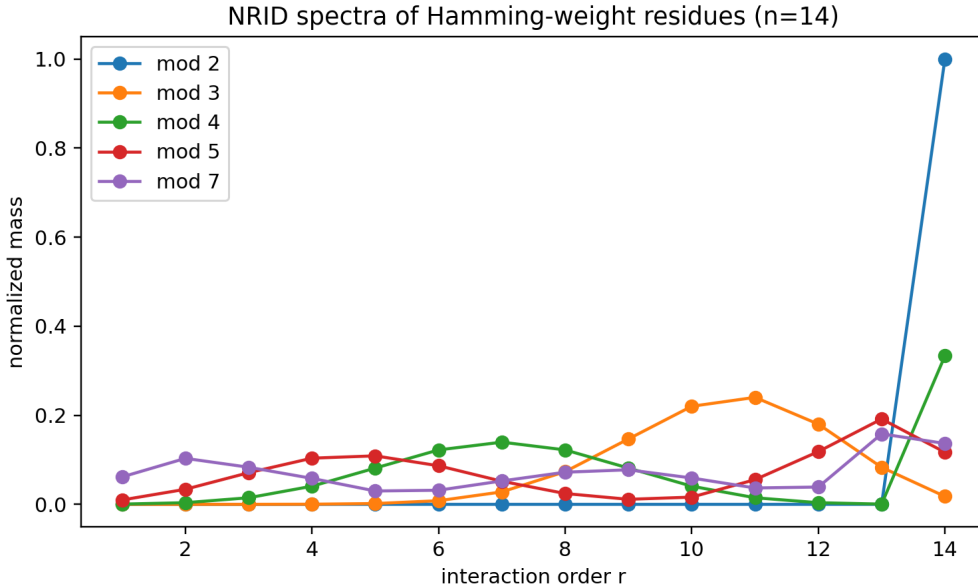


Figure 15: Normalized NRID spectra for Hamming-weight residues at $n = 14$. Parity is strictly order 14; mod 3 is concentrated near order 10.5; larger moduli gradually move mass toward the middle orders.

This motivates a distinction between two forms of hiddenness. A target is *strictly order-hidden* to depth $d - 1$ if all atoms below order d vanish. Parity has this property. A target is *spectrally hidden* if the low-order atoms are nonzero but exponentially small in system size. Modulo-3 count is spectrally hidden: low-order leakage exists, but almost all information is high-order.

A.9 Applications suggested by the third pass

Interpretable machine learning. For deterministic binary labels, NRID is a normalized Fourier spectrum. The low-order mass measures how much of a classifier’s label information can survive independent feature degradation. This is a direct robustness diagnostic and complements functional ANOVA and SHAP-style interaction decompositions [23].

Sensor and experiment design. The modal theorem says that nonbinary sensors must be matched to the source modes that actually carry target information. Sensor specifications based only on scalar accuracy can be misleading.

Privacy and secret sharing. Strict order-hidden targets behave like correlation-immune functions: no low-order subset reveals information. Spectrally hidden targets, such as modular counts, leak small low-order information but place most mass at high order. NRID therefore gives a quantitative leakage spectrum rather than a binary privacy verdict.

Distributed monitoring. Exact count information is spread uniformly across all orders. Low-reliability distributed sensors can recover coarse count information, but the fine integer value requires reliability close to one. This may matter for population counts, fault counts, digital counters, and biological copy-number sensing.

Representation learning. Target postprocessing contracts every hyperedge, whereas source-side sensor fusion increases effective observability. This separates two mechanisms that are often conflated: compressing the target destroys information coefficients; repeated source observations move along the same coefficient polynomial.

A.10 Targets for further study after the third pass

1. Prove sharp finite- n error bounds for the modulo-count approximation and characterize the transition from parity-like spectra to count-like flat spectra as m grows with n .
2. Develop estimators for modal NRID in nonbinary data using singular-vector learning of local observation operators.
3. Determine whether coefficientwise target data processing has a converse: can any coordinatewise contraction of atoms be realized by some target garbling?
4. Study NRID spectra of common model classes: decision trees, threshold circuits, neural network Booleanizations, error-correcting decoders, and cryptographic S-boxes.
5. Extend the count/Gaussian bridge to Poisson, multinomial, and continuous limits, replacing Krawtchouk modes with Hermite, Charlier, or other orthogonal-polynomial chaos expansions.
6. Investigate whether the collection/fusion algebra leads to an information-theoretic “chemistry” of target construction, where operations on targets have predictable spectral reactions.

A.11 Research ledger update

The third pass produced the following items worth preserving:

1. An exact finite-alphabet singular-mode NRID theorem.
2. A counterexample showing that scalar observability fails for nonbinary sources.
3. A coefficientwise target data-processing theorem.
4. Exact composition laws distinguishing collected targets from fused targets.
5. A supermodular information-hypergraph calculus based on mixed partial derivatives of $\mathcal{J}(z)$.

6. A repeated-sensor observability formula for multiple independent views of one Boolean source.
7. The flat count spectrum theorem: exact Hamming weight has $\Omega_r = 1$ for every order.
8. The count/Gaussian bridge: $\mathcal{J} \rightarrow \rho^2/(1 - ho^2)$ and $I \rightarrow -\frac{1}{2} \log_2(1 - ho^2)$.
9. An asymptotic root-of-unity spectrum for modulo-count targets.
10. The distinction between strict order-hiddenness and spectral hiddenness.

B Fourier details for selected examples

B.1 AND-2

For $T = \mathbf{1}[X_1 = +1, X_2 = +1]$,

$$q_1(x) = \frac{(1 + x_1)(1 + x_2)}{4} = \frac{1}{4} + \frac{x_1}{4} + \frac{x_2}{4} + \frac{x_1 x_2}{4}.$$

Thus $p_1 = 1/4$, $p_0 = 3/4$, $m_1 = x_1/4 + x_2/4 + x_1 x_2/4$, and $m_0 = -m_1$. Therefore

$$\begin{aligned}\Omega_1 &= \left(\frac{1}{1/4} + \frac{1}{3/4} \right) \left(2 \cdot \frac{1}{16} \right) = \frac{2}{3}, \\ \Omega_2 &= \left(\frac{1}{1/4} + \frac{1}{3/4} \right) \left(\frac{1}{16} \right) = \frac{1}{3}.\end{aligned}$$

B.2 Mixed copy/XOR

For

$$q_+(x) = \alpha \frac{1 + x_1}{2} + (1 - \alpha) \frac{1 + x_1 x_2}{2},$$

we have $p_+ = 1/2$ and

$$m_+(x) = \frac{\alpha}{2} x_1 + \frac{1 - \alpha}{2} x_1 x_2.$$

Since $m_- = -m_+$ and $p_- = 1/2$,

$$\Omega_1 = \alpha^2, \quad \Omega_2 = (1 - \alpha)^2.$$

C Reproducibility note

The exact enumeration code associated with this draft performs the following operations:

1. enumerate $\{-1, +1\}^n$;
2. build the Fourier-Walsh character matrix;
3. compute $\hat{m}_t(S)$ by uniform averaging;
4. compute Ω_r and ω_S ;
5. compute $P(T | Y_\rho)$ exactly by summing over X ;
6. evaluate $\mathcal{J}_\rho$ and $I(T; Y_\rho)$ exactly for each reliability value.

The random stochastic checks use Gaussian logits followed by a softmax over target states. No Monte Carlo sampling is used in the reported verification table.

D Fourth research pass: Shannon collisions, code spectra, and bent hiddenness

This pass probes two issues left open by the preceding sections. First, it asks whether the NRID order spectrum, which is exact for expected posterior χ^2 information, is also complete for Shannon mutual information. The answer is negative, but the failure is structured: Shannon information is governed by a hierarchy of even collision moments whose first layer is NRID. Second, it searches for constructive families whose spectra can be written in closed form. Linear syndrome maps and bent functions provide such families.

D.1 Shannon collision expansion

For a balanced deterministic binary target $g : \{-1, +1\}^n \rightarrow \{-1, +1\}$, define

$$m_\rho(y) = \mathbb{E}[g(X) \mid Y_\rho = y] = (T_\rho g)(y).$$

Then

$$I(g; Y_\rho) = \mathbb{E}_y \left[1 - h_2 \left(\frac{1 + m_\rho(y)}{2} \right) \right] = \sum_{k \geq 1} \frac{\mathbb{E}[m_\rho(Y)^{2k}]}{2k(2k-1) \ln 2}.$$

If

$$m_\rho(y) = \sum_S \rho^{|S|} a_S \chi_S(y),$$

then orthogonality gives

$$\mathbb{E}[m_\rho(Y)^{2k}] = \sum_{S_1 \oplus \dots \oplus S_{2k} = \emptyset} \prod_{j=1}^{2k} \rho^{|S_j|} a_{S_j}.$$

Thus the $k = 1$ layer is precisely the NRID polynomial,

$$\mathbb{E}[m_\rho^2] = \sum_r \Omega_r \rho^{2r},$$

while higher layers count signed even closures among Fourier hyperedges. The result can be read as a *Shannon-NRID hierarchy*: NRID is the power spectrum, and Shannon information at finite reliability also depends on phase information through four-hyperedge, six-hyperedge, and higher closures.

D.2 Same-spectrum counterexample for Shannon information

An exhaustive enumeration of all balanced functions on four bits found two classifiers with identical NRID spectra but different Shannon mutual information. Their positive truth-table index sets are

$$A_+ = \{0, 1, 2, 3, 4, 5, 6, 15\}, \quad B_+ = \{0, 1, 2, 3, 4, 5, 8, 11\}.$$

Both have

$$\Omega(z) = 0.5625z + 0.1875z^2 + 0.1875z^3 + 0.0625z^4.$$

The search covered 12870 balanced functions and found 37 distinct level spectra. The largest same-spectrum Shannon gap at $\rho = 0.85$ was 0.005650 bits in a spectrum class of size 448.

ρ	shared $\mathcal{J}_\rho$	I_A	I_B	gap	$M_4(A)$	$M_4(B)$
0.50	0.155518	0.115969	0.117960	0.001991	0.028949	0.041309
0.75	0.415360	0.329799	0.335236	0.005437	0.196384	0.211908
0.85	0.592029	0.491318	0.496968	0.005650	0.373297	0.380502
0.95	0.839670	0.754603	0.757882	0.003279	0.711216	0.711755

Table 10: Exact same-spectrum counterexample. The shared χ^2 response does not determine Shannon mutual information once fourth and higher collision moments matter.

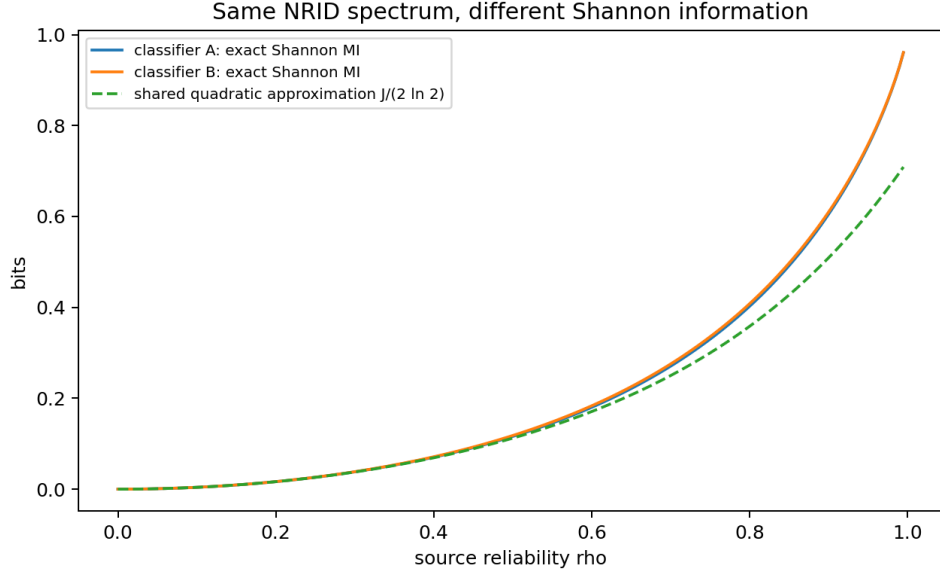


Figure 16: Two four-bit classifiers with identical NRID response but different Shannon information.

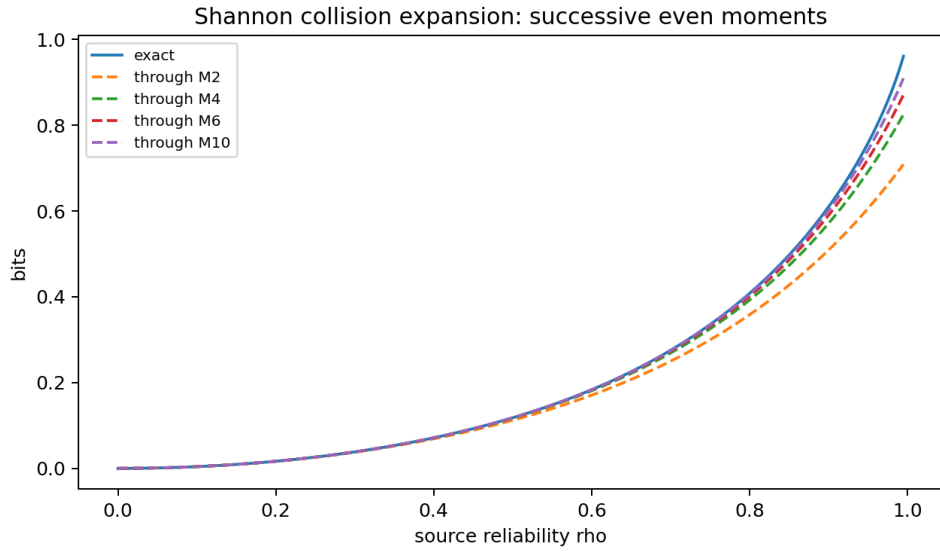


Figure 17: Successive even collision moments approximate exact Shannon information.

D.3 Linear-code theorem

Let $B \in \mathbb{F}_2^n$ be uniform and let $T = HB \bmod 2$, where H is an $m \times n$ binary matrix of rank m . Define the row code

$$C = \{aH : a \in \mathbb{F}_2^m\}.$$

Then

$$\Omega_r(T; B) = |\{c \in C \setminus \{0\} : \text{wt}(c) = r\}|$$

and

$$\mathcal{J}_\rho(T; Y) = \sum_{c \in C \setminus \{0\}} \rho^{2 \text{wt}(c)}.$$

Indeed, each nonzero target character $a \cdot T$ pulls back to the source parity indexed by aH . Since the target is uniform on 2^m states, each nonzero target character contributes one χ^2 atom. Grouping atoms by support size gives the row-code weight enumerator. Hence the first nonzero NRID order is the minimum distance of the row code.

Target	n	rank	first	$\mathcal{J}_1$	$\mathcal{J}(z)$	$\rho_{1/2}$
Hamming syndrome / simplex [7,3,4]	7	3	4	7	$7z^4$	0.917004
Simplex [15,4,8] vector target	15	4	8	15	$15z^8$	0.957603
Random rank-4 length-12 syndrome	12	4	4	15	$1z^4 + 5z^5 + 3z^6 + 2z^7 + 2z^8 + 1z^9 + 1z^{10}$	0.946080

Table 11: Linear syndrome spectra. The simplex examples are monochromatic: all information appears at one interaction order.

For the m -dimensional simplex construction with $n = 2^m - 1$ columns, every nonzero row-code word has weight 2^{m-1} , so

$$\mathcal{J}_\rho = (2^m - 1)\rho^{2^m}.$$

This gives a compact family of multi-state targets whose information is strictly absent below order 2^{m-1} .

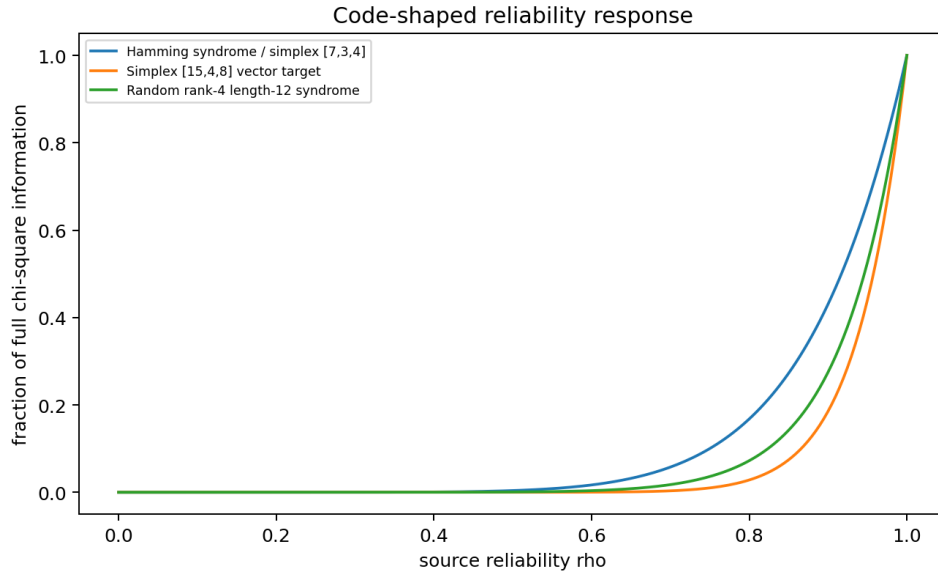


Figure 18: Code-shaped response curves.

D.4 Bent and inner-product spectra

Let g be a bent Boolean function on even n , so $\widehat{g}(S)^2 = 2^{-n}$ for every S . Bent functions have $\nu^2 = (\mathbb{E} g)^2 = 2^{-n}$. For the binary target $T = g(X)$,

$$\omega_S = \frac{\widehat{g}(S)^2}{1 - \nu^2}, \quad S \neq \emptyset.$$

Therefore

$$\Omega_r = \frac{\binom{n}{r}}{2^n - 1}, \quad r = 1, \dots, n,$$

and

$$\mathcal{J}_\rho = \frac{(1 + \rho^2)^n - 1}{2^n - 1}.$$

The inner-product target $g(x, y) = (-1)^{x \cdot y}$ on $n = 2m$ bits realizes this law. This is not strict hiddenness, because first-order atoms exist, but their total mass $n/(2^n - 1)$ is exponentially small. Most information sits near order $n/2$.

m	n	$\mathbb{E} g$	mean order	mode	max error	$\rho_{1/2}$
2	4	0.250000	2.133333	2	0.0e+00	0.841116
4	8	0.062500	4.015686	4	0.0e+00	0.913730
6	12	0.015625	6.001465	6	2.8e-17	0.942225
8	16	0.003906	8.000122	8	0.0e+00	0.956665

Table 12: Inner-product spectra match the binomial bent formula through $n = 16$ to numerical precision.

The reliability threshold for recovering a fraction α of full χ^2 information is explicit for pure-order and bent spectra:

$$\rho_\alpha^{\text{pure}} = \alpha^{1/(2d)}, \quad \rho_\alpha^{\text{bent}} = \sqrt{(1 + \alpha(2^n - 1))^{1/n} - 1}.$$

n	α	parity ρ	bent ρ	count ρ
8	0.5	0.957603	0.913730	0.918151
16	0.5	0.978572	0.956665	0.955078
32	0.5	0.989228	0.978337	0.976383
64	0.5	0.994599	0.989169	0.987879

Table 13: Half-information reliability thresholds.

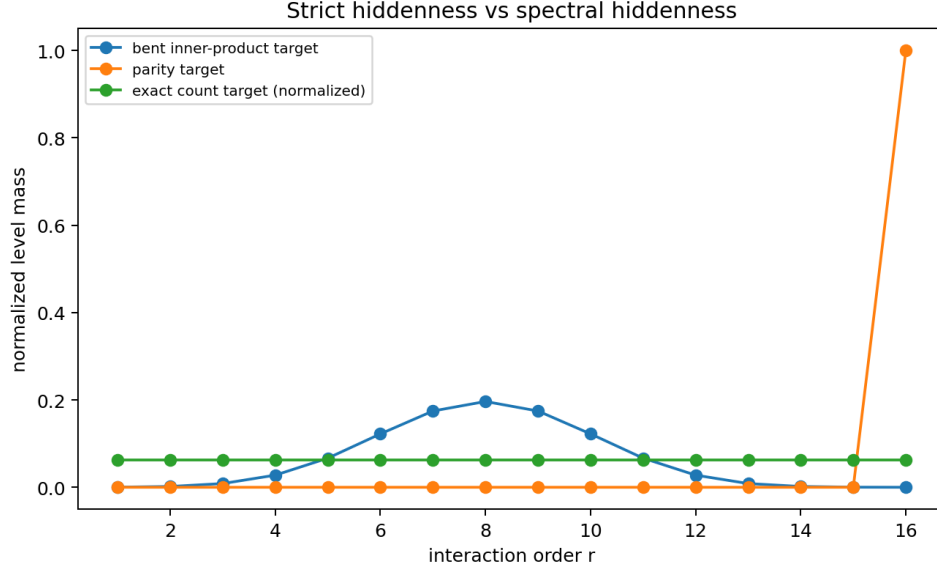


Figure 19: Strict hiddenness, flat spectra, and bent spectral hiddenness at $n = 16$.

D.5 Applications and targets opened by this pass

- **Coding design.** The linear-code theorem turns row-code weight enumerators into information-response curves. Minimum distance is strict information-hiddenness order.
- **Leakage-resilient computation.** Parity, simplex syndromes, and bent inner products represent different leakage profiles: strict hiddenness, monochromatic vector hiddenness, and spectral hiddenness.
- **Classifier interpretation.** Same χ^2 order spectra need not imply same Shannon information at finite reliability; collision moments should be estimated when reliability is not infinitesimal.
- **Cryptographic functions.** NRID spectra may be used to compare nonlinear Boolean functions, S-box coordinate systems, graph indicators, and masking schemes.

New technical targets are: prove truncation bounds for the Shannon collision expansion; classify same-spectrum functions by extremal fourth and sixth collisions; develop MacWilliams-style duality for NRID of linear maps; extend the code theorem to q -ary association schemes; analyze standard S-boxes as vectorial targets; and design estimators for collision moments that avoid full truth-table enumeration.

E Fifth research pass: budgeted observability, hypergraph geometry, and phase-sensitive probes

The preceding sections established NRID as an exact positive information hypergraph for posterior χ^2 information under independent source degradation. This pass studies what is observable when experiments are resource constrained, targeted, or sensitive to signed phase. Five findings emerged. First, observing exactly m sources gives a binomial thinning of the order spectrum. Second, the inverse is exact but severely ill-conditioned at large n . Third, order is not geometry: the isotropic

spectrum does not determine vulnerability to targeted source compromise. Fourth, subset probes can recover sparse information hypergraphs. Fifth, quadratic NRID is phase-blind, while cubic posterior moments detect signed Fourier cycles.

E.1 Fixed-budget subset spectroscopy

Let the Boolean NRID hypergraph have nonnegative atoms ω_S and anisotropic polynomial

$$\mathcal{J}(z_1, \dots, z_n) = \sum_{\emptyset \neq S \subseteq [n]} \omega_S \prod_{i \in S} z_i.$$

If a coordinate subset A is observed noiselessly and all other coordinates are completely blinded, then

$$\mathcal{J}(A) = \sum_{S \subseteq A} \omega_S.$$

Define the fixed-budget curve

$$A_m = \mathbb{E}_{A: |A|=m} \mathcal{J}(A).$$

Theorem 8 (Fixed-budget thinning law). *For $m = 0, 1, \dots, n$,*

$$A_m = \sum_{r=1}^m \Omega_r \frac{\binom{n-r}{m-r}}{\binom{n}{m}} = \sum_{r=1}^m \Omega_r \frac{\binom{m}{r}}{\binom{n}{r}}.$$

Consequently the order spectrum is recovered by the triangular inverse

$$\Omega_m = \binom{n}{m} \left(A_m - \sum_{r=1}^{m-1} \Omega_r \frac{\binom{m}{r}}{\binom{n}{r}} \right).$$

Proof. A hyperedge S of size r contributes to $\mathcal{J}(A)$ iff $S \subseteq A$. For a uniformly sampled m -subset, this event has probability

$$\frac{\binom{n-r}{m-r}}{\binom{n}{m}} = \frac{\binom{m}{r}}{\binom{n}{r}}.$$

Summing over all hyperedges and grouping by $r = |S|$ gives the identity. The inverse follows because the linear map from $(\Omega_1, \dots, \Omega_n)$ to $(A_1, \dots, A_n)$ is lower triangular with nonzero diagonal. $\square$

The independent-reveal curve is a Bernstein mixture of fixed-budget curves. If $M \sim \text{Binomial}(n, p)$, then

$$\mathbb{E}[A_M] = \sum_{r=1}^n \Omega_r p^r.$$

Thus dropout with independent reveal probability p and fixed-cardinality masking are two experimental views of the same spectrum.

Corollary 3 (Budgeted hiddenness bound). *If $\Omega_1 = \dots = \Omega_{d-1} = 0$ and $\mathcal{J}_1 = \sum_r \Omega_r$, then for $m \geq d$,*

$$A_m \leq \mathcal{J}_1 \frac{\binom{m}{d}}{\binom{n}{d}}.$$

Equality is attained by pure d -order information.

The bound gives an operational reading of high-order synergy: it is not only absent from smaller-than- d observations; at budget m it is diluted by the probability that a random m -subset contains the required d -source interaction.

Target	$\mathcal{J}(1)$	first	mean order	$A_1/\mathcal{J}(1)$	$A_4/\mathcal{J}(1)$	max error
copy X_0	1.000	1	1.000	0.125	0.500	0
full parity	1.000	8	8.000	0	0	0
exact Hamming weight	8.000	1	4.500	0.0156	0.100	1.24×10^{-14}
seeded stochastic 3-state	1.03	1	1.81	0.0531	0.308	3.33×10^{-16}

Table 14: Exact fixed-budget spectroscopy for $n = 8$. Direct posterior χ^2 information for every subset agreed with the hypergraph formula to numerical precision.

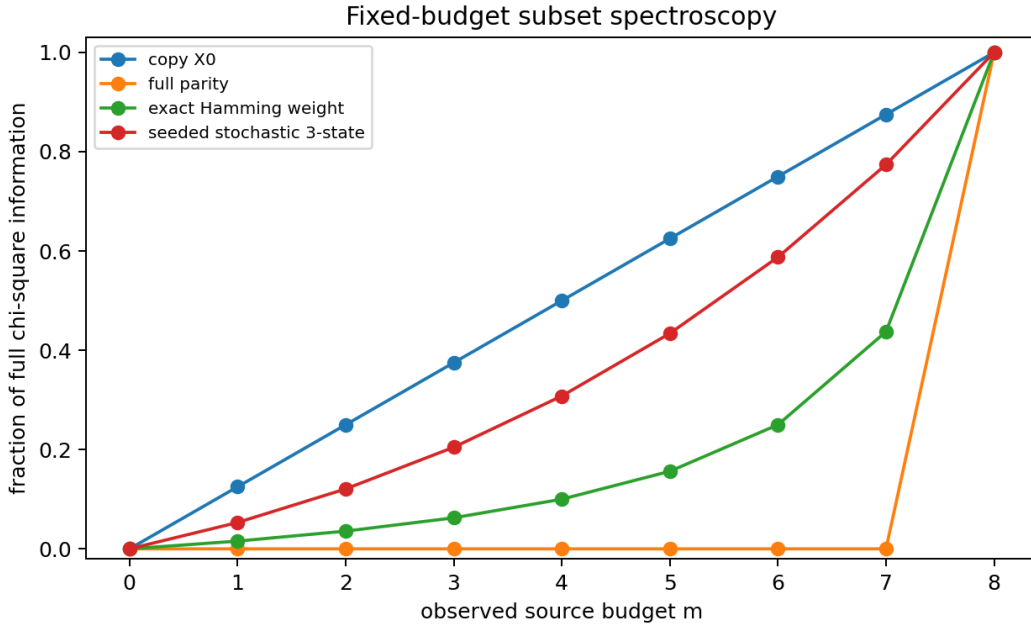


Figure 20: Fixed-budget subset spectroscopy. A one-coordinate copy is linearly discoverable by random subsets; full parity is invisible until all coordinates are observed; exact Hamming weight has large full information but a slowly rising fractional budget curve.

For random balanced Boolean targets, the earlier random-spectrum law gives

$$\mathbb{E}[\Omega_r] = \frac{\binom{n}{r}}{2^n - 1}.$$

Therefore

$$\mathbb{E}[A_m] = \frac{2^m - 1}{2^n - 1}.$$

For $n = 8$, an exact truth-table experiment over 350 random balanced functions gave maximum budget-curve deviation 2.41×10^{-4} from this formula.

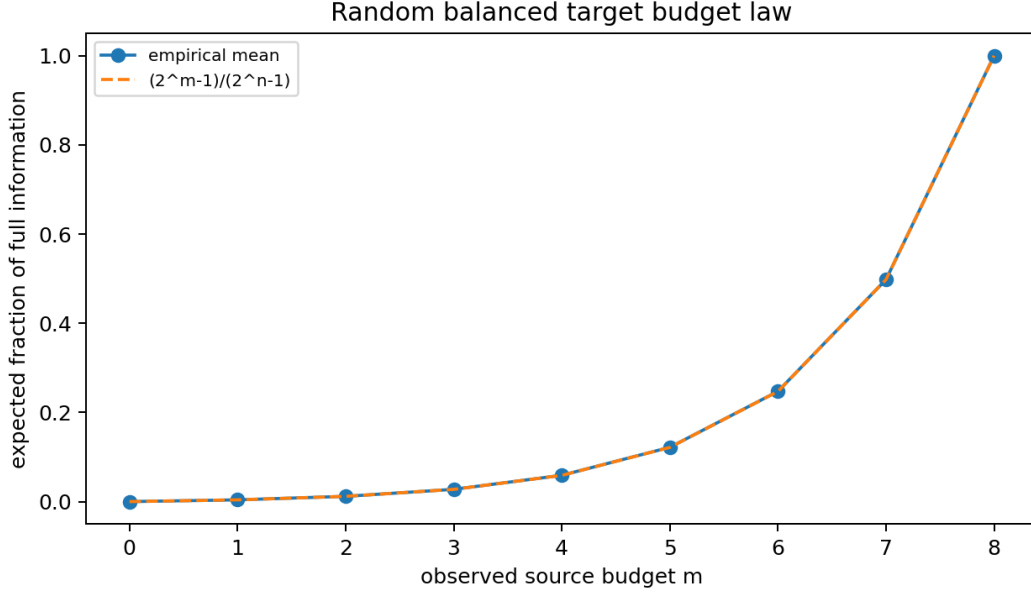


Figure 21: Random balanced target budget law for $n = 8$. The expected revealed fraction follows $(2^m - 1)/(2^n - 1)$.

E.2 Conditioning barrier

The fixed-budget inverse is exact but poorly conditioned. The transform

$$M_{m,r} = \frac{\binom{m}{r}}{\binom{n}{r}}, \quad 1 \leq r \leq m \leq n,$$

has rapidly increasing condition number. Exact numerical values were

$$\kappa(M_{10}) \approx 4.26 \times 10^4, \quad \kappa(M_{20}) \approx 2.68 \times 10^9, \quad \kappa(M_{30}) \approx 1.63 \times 10^{14}.$$

This identifies a statistical barrier: low-order coefficients are accessible from low-budget measurements, but high-order recovery from averaged budget curves requires regularization, nonnegativity constraints, sparse priors, or targeted probes.

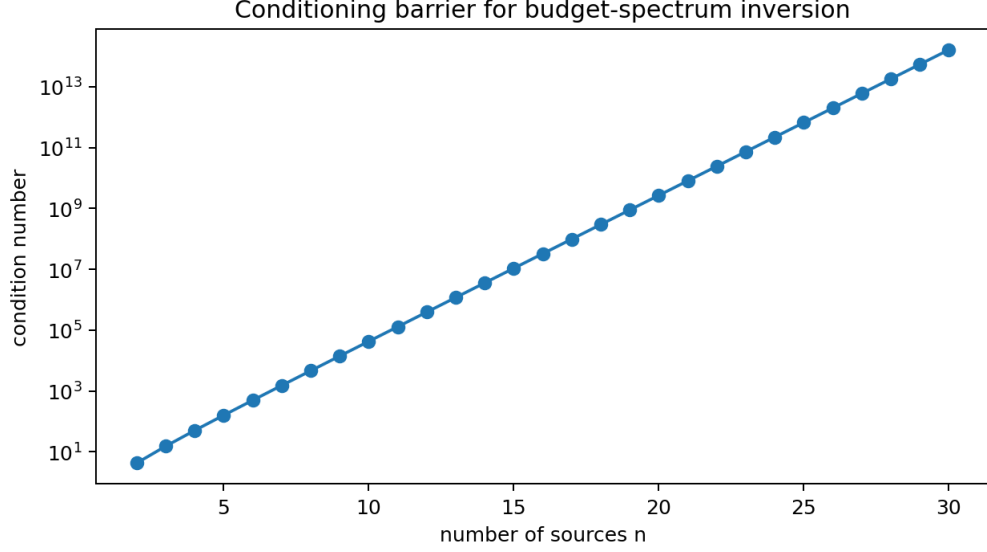


Figure 22: Condition number of the fixed-budget spectrum transform. The exact inverse becomes statistically unstable for large n .

E.3 Order is not geometry: adversarial leakage

The isotropic response $G(z) = \sum_r \Omega_r z^r$ records how much information exists at each order, but it does not record where the hyperedges are. Define adversarial budget leakage by

$$V_m = \max_{|A|=m} \mathcal{J}(A) = \max_{|A|=m} \sum_{S \subseteq A} \omega_S.$$

Two systems can have the same order spectrum and same random fixed-budget curve while having very different V_m .

A sharp construction fixes total mass W at order r . If all mass is placed on one r -hyperedge, then $V_r = W$. If the same mass is uniformly spread over all $\binom{n}{r}$ hyperedges, then $V_r = W/\binom{n}{r}$. Hence the adversarial leakage ratio can be $\binom{n}{r}$ with identical Ω_r .

This was tested with $n = 8$ stochastic binary targets carrying the same pair mass $W = 1/28$. One target placed all mass on a single pair; the other spread it uniformly over all pairs. Their spectra and random budget curves matched to numerical precision, but their adversarial two-source leakage differed by a factor of 28.

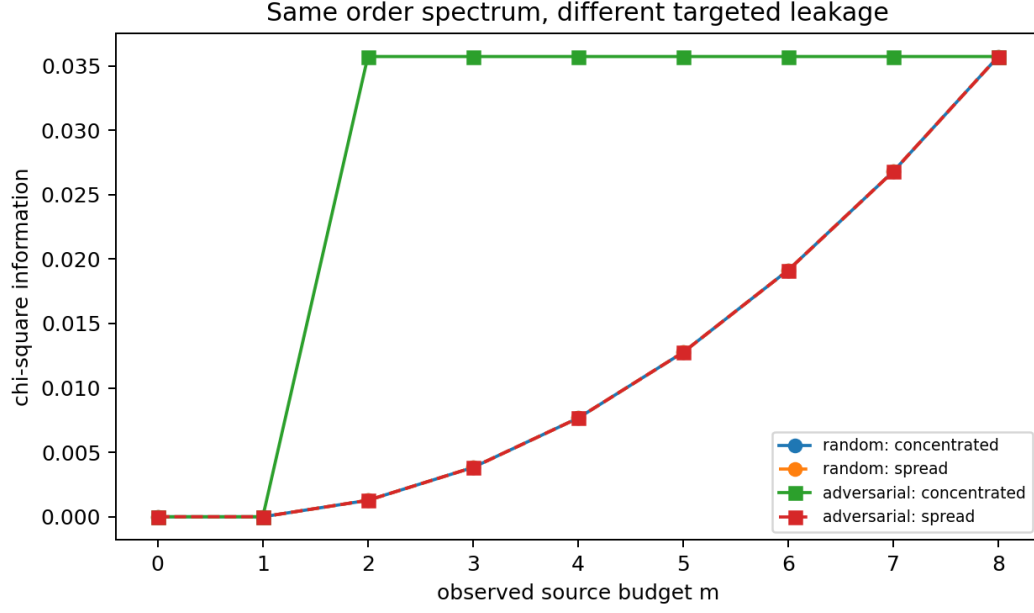


Figure 23: Same order spectrum, different adversarial leakage. Random budget curves coincide; targeted two-source observation fully exposes the concentrated pair and only a 1/28 share of the spread-pair system.

Anisotropic derivatives reveal the hidden geometry:

$$\frac{\partial \mathcal{J}}{\partial z_i} = \sum_{S \ni i} \omega_S \prod_{j \in S \setminus \{i\}} z_j.$$

At $z = 1$, this is a coordinate information-influence score. Higher mixed derivatives identify joint participation.

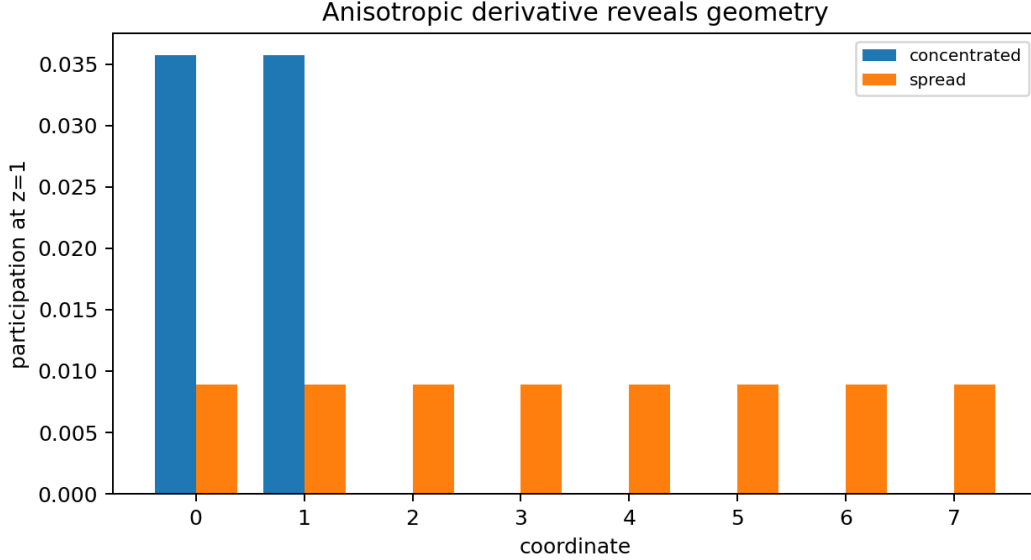


Figure 24: Coordinate participation distinguishes concentrated from spread pair information even when the isotropic spectrum cannot.

E.4 Sparse information-hypergraph recovery

Subset observations convert hypergraph recovery into a linear inverse problem:

$$b_A = \mathcal{J}(A) = \sum_S \mathbf{1}\{S \subseteq A\} \omega_S.$$

If candidate order is bounded and the true information hypergraph is sparse, random and designed ablation masks can recover individual hyperedges without measuring all 2^n subsets.

In a synthetic $n = 10$ experiment, all hyperedges of order at most three were used as candidates (175 columns). Seven true hyperedges were measured through 260 subset probes. Linear recovery found all seven hyperedges exactly in the noiseless case. With additive measurement noise of standard deviation 2.5×10^{-4} , it again found all seven with no false positives at threshold 0.003.

setting	candidates	probes	TP	FP	FN	ℓ_1 error
noiseless	175	260	7	0	0	3.28×10^{-15}
noisy	175	260	7	0	0	1.35×10^{-2}

Table 15: Sparse hypergraph recovery from subset probes.

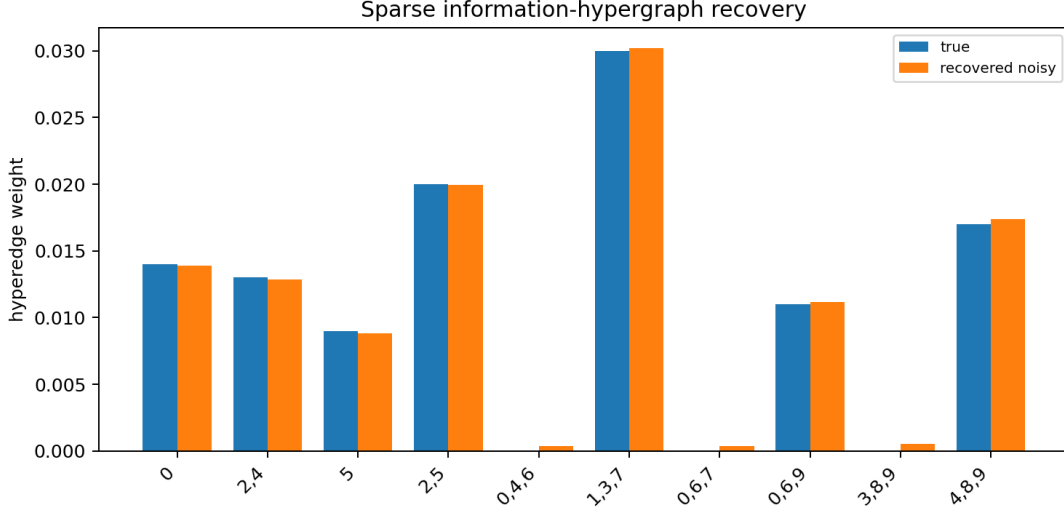


Figure 25: Recovered noisy weights match the true sparse information hypergraph.

This suggests a practical protocol: estimate $\mathcal{J}(A)$ for a designed collection of ablation masks, solve a constrained sparse inverse problem, and validate recovered hyperedges by targeted anisotropic noise interventions.

E.5 The phase problem: quadratic energy misses signed cycles

NRID atoms are squared amplitudes. Thus ω_S is invariant under sign changes of Fourier coefficients. This is appropriate for a nonnegative energy decomposition, but it creates a phase ambiguity: different target likelihoods can have the same ω_S for every S and hence the same χ^2 response under every independent product observation.

A minimal example uses three pair characters on a triangle:

$$\Pr(T = 1 \mid X = x) = p + a(s_{12}x_1x_2 + s_{23}x_2x_3 + s_{13}x_1x_3).$$

Changing the signed cycle product $s_{12}s_{23}s_{13}$ from $+1$ to -1 leaves all quadratic atoms unchanged but changes cubic score moments. This is analogous to signed graph frustration: edge signs are not all removable by coordinate flips because the product around a cycle is invariant.

quantity	balanced cycle	frustrated cycle
signed cycle product	$+1$	-1
Ω_2	0.0253125	0.0253125
max atom difference	1.74×10^{-18}	1.74×10^{-18}
max $\mathcal{J}_\rho$ difference	1.39×10^{-17}	1.39×10^{-17}
third moment at $\rho = 1$	1.898437×10^{-3}	-1.898437×10^{-3}

Table 16: Quadratic NRID cannot see the signed cycle; a cubic posterior score moment can.

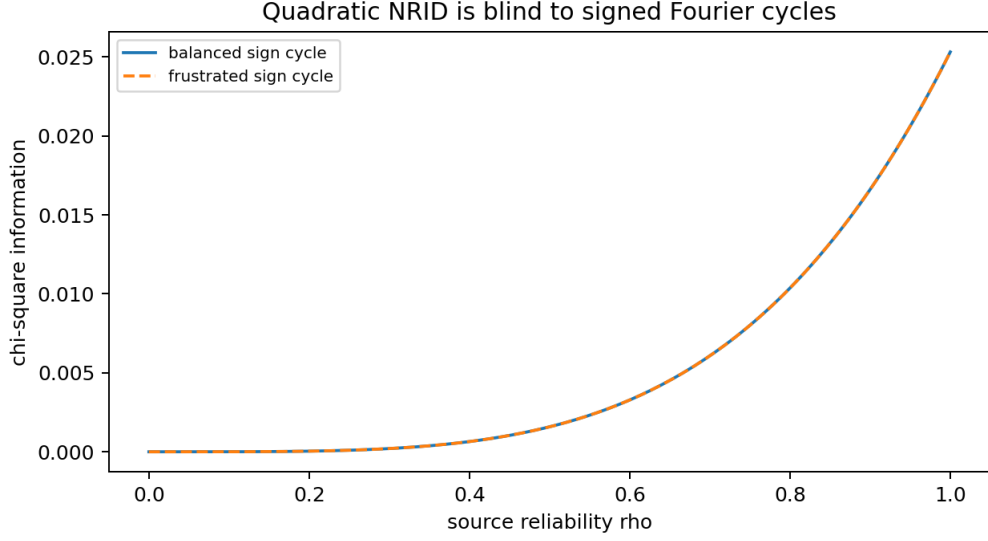


Figure 26: Balanced and frustrated sign cycles have indistinguishable $\mathcal{J}_\rho$ curves.

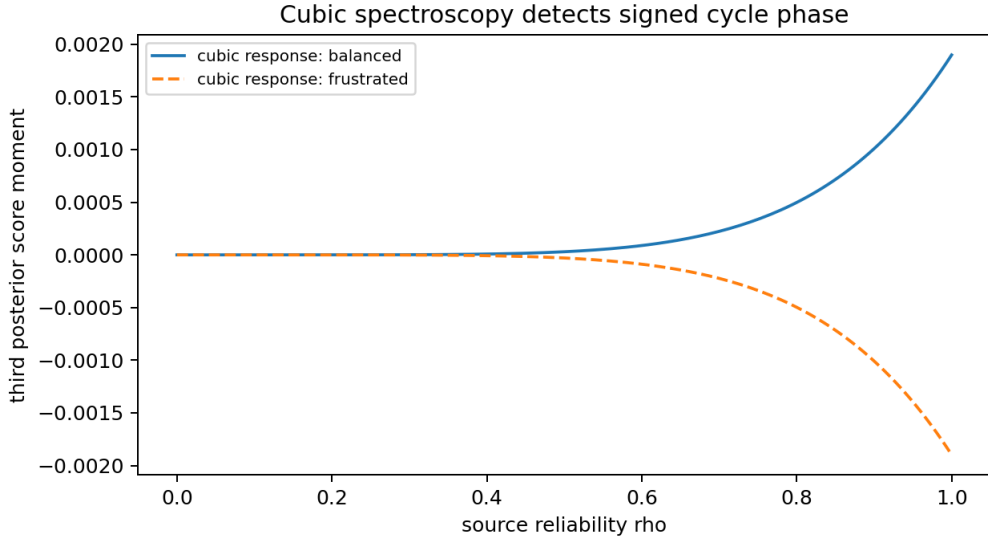


Figure 27: A cubic posterior-moment response separates the two targets and detects the signed cycle.

Let $a_{t,S} = \hat{m}_t(S)$. Define

$$K_3(\rho) = \mathbb{E}_Y \sum_t \frac{(P(T = t | Y) - p_t)^3}{p_t^2}.$$

Then symmetric Boolean reliability gives the exact identity

$$K_3(\rho) = \sum_t p_t^{-2} \sum_{A \oplus B \oplus C = \emptyset} a_{t,A} a_{t,B} a_{t,C} \rho^{|A|+|B|+|C|}.$$

The condition $A \oplus B \oplus C = \emptyset$ means that cubic spectroscopy detects closed cycles in the Fourier hypergraph. Exact enumeration matched this formula to 1.39×10^{-17} . This suggests a broader hierarchy: quadratic NRID is an unsigned energy spectrum, while higher posterior moments recover phase-sensitive signed hypergraph invariants.

E.6 Correlation immunity and applications

For balanced Boolean targets, NRID gives an operational form of correlation immunity. A target is m th-order correlation immune if and only if

$$\Omega_1 = \dots = \Omega_m = 0.$$

Equivalently, the low-reliability information response begins at z^{m+1} , or $\rho^{2(m+1)}$. In the budgeted view, $A_m = 0$ for all budgets at or below the immunity order. This connects classical Walsh-spectral criteria for cryptographic Boolean functions [28, 29] to an experimentally measurable noise-response exponent.

Applications suggested by this pass include:

- **Feature-interaction auditing.** Random masks estimate the order spectrum; targeted masks and anisotropic gradients reveal localized vulnerabilities.
- **Sensor networks.** The adversarial curve V_m distinguishes robustly distributed information from information concentrated in a small set of devices.
- **Privacy and secret sharing.** The hiddenness bound quantifies expected leakage from partial compromise.
- **Neuroscience and biology.** Sparse hypergraph recovery suggests perturbation designs for finding high-order population codes or epistatic interactions.
- **Cryptography.** Phase-sensitive cubic responses separate unsigned spectral strength from signed interaction structure.

New targets for further study are:

1. develop finite-sample estimators for A_m and V_m with nonnegative confidence bands;
2. design adaptive subset-probe schemes for sparse hypergraph recovery;
3. characterize the extremal gap between random and adversarial leakage for fixed order spectrum;
4. build a phase-sensitive NRID hierarchy from third and higher posterior central moments;
5. extend budget spectroscopy to nonbinary modal NRID, where a probe budget may select both variables and local singular modes;
6. compare NRID exponents against classical cryptographic criteria such as algebraic degree, nonlinearity, resiliency, and propagation criteria;
7. test whether training, compression, distillation, and adversarial regularization in neural networks move target information between high- and low-order spectra.

E.7 Research ledger update

The fifth pass changes the working theory in three ways. First, it makes NRID experimentally budget-aware: random subset access yields an exact binomial projection of the order spectrum. Second, it separates order from geometry: a radial response curve is insufficient for adversarial leakage analysis unless anisotropic derivatives or targeted ablations are also measured. Third, it identifies a phase hierarchy: χ^2 NRID is the second-moment layer, while cubic and higher posterior moments detect signed closure invariants that quadratic energy intentionally discards.

F Sixth pass: reliability fields, metadata, and active allocation

F.1 Blind spot

Earlier sections treat reliability as either isotropic, subset-restricted, or externally chosen. A remaining blind spot is the following: in real sensor systems, measurements do not merely become noisy; their *quality field* varies across sources and across trials. Sometimes the quality state is known, as in an erasure mask, sensor-health report, dropout mask, or timestamped calibration value. Sometimes it is hidden and must be averaged into the channel. This pass studies the distinction between source reliability, reliability metadata, and active reliability allocation.

The results below use the subset polynomial

$$G(\mathbf{z}) = \mathcal{J}_{\sqrt{\mathbf{z}}}(T; Y) = \sum_{\emptyset \neq S \subseteq [n]} \omega_S \prod_{i \in S} z_i, \quad z_i = \rho_i^2.$$

This polynomial is multi-affine and has nonnegative coefficients. It may be interpreted as an information-generating function over source hyperedges. This view is related to the use of Fourier spectra for influences and noise sensitivity in Boolean analysis [16], but here the coefficients are target-posterior chi-square energies rather than squared coefficients of a single Boolean observable.

F.2 Reliability-field thermodynamics

Definition 5 (Active information-edge distribution). *For any reliability field $\mathbf{z}$ with $G(\mathbf{z}) > 0$, define*

$$\mathbb{P}_{\mathbf{z}}(S) = \frac{\omega_S \prod_{i \in S} z_i}{G(\mathbf{z})}, \quad S \neq \emptyset.$$

This is a probability distribution over the information hyperedges that are active at reliability field $\mathbf{z}$.

Theorem 9 (Log-reliability calculus). *Let $u_i = \log z_i$. Then*

$$\frac{\partial}{\partial u_i} \log G(e^{\mathbf{u}}) = \mathbb{P}_{\mathbf{z}}(i \in S),$$

and

$$\frac{\partial^2}{\partial u_i \partial u_j} \log G(e^{\mathbf{u}}) = \text{Cov}_{\mathbf{z}}(\mathbf{1}\{i \in S\}, \mathbf{1}\{j \in S\}).$$

Consequently, $\log G(e^{\mathbf{u}})$ is convex in the log-reliability field, and its Hessian is a source complementarity/substitution matrix.

Proof. Write

$$G(e^{\mathbf{u}}) = \sum_{S \neq \emptyset} \omega_S \exp \left(\sum_{i \in S} u_i \right).$$

This is a finite log-partition function. Differentiating its logarithm gives the expectation of the sufficient statistic $\mathbf{1}\{i \in S\}$ under the Gibbs distribution $\mathbb{P}_{\mathbf{z}}$. Differentiating a second time gives the covariance matrix of these sufficient statistics. $\square$

The scalar symmetric response is the special case

$$D(z) = \frac{d \log G(z)}{d \log z} = \mathbb{E}_z[|S|], \quad \frac{dD}{d \log z} = \text{Var}_z(|S|).$$

Thus the effective order is an information-edge mean, and the curvature of the response is an order variance.

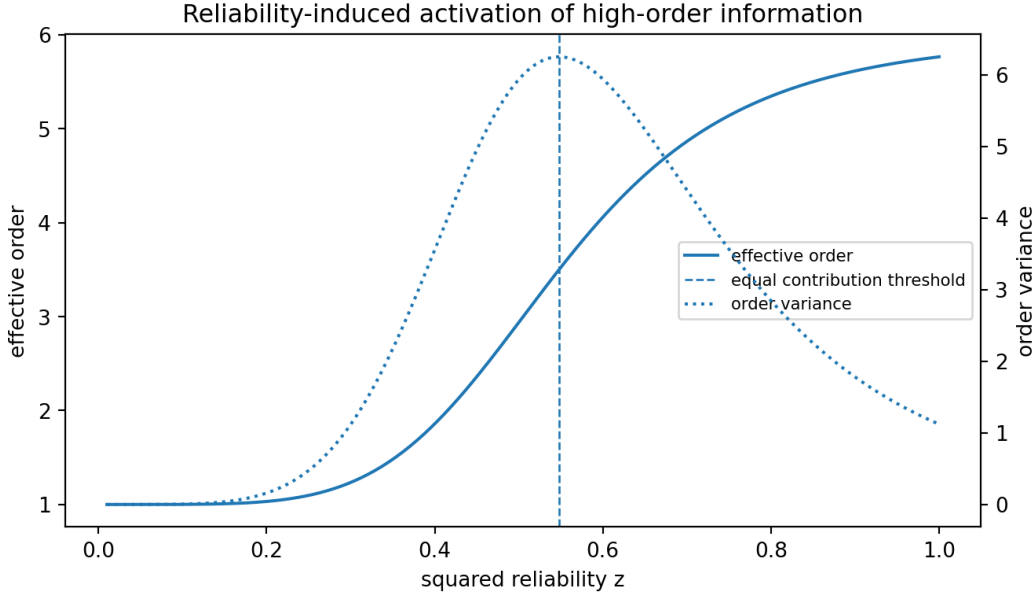


Figure 28: A realizable binary stochastic target with $\Omega_1 = 0.0064$ and $\Omega_6 = 0.1296$ has low-order leakage at weak reliability but activates mostly sixth-order information as reliability improves. The equal-contribution threshold is $z_c = (\Omega_1/\Omega_6)^{1/5} = 0.547917$.

The active-edge view also distinguishes complementary and substitutable sources. If two variables tend to occur in the same active hyperedges, their log-reliability covariance is positive. If they occur in alternative pathways, it is negative. In a planted hypergraph with a pair pathway, a four-way pathway, a bridge edge, and a singleton decoy, the analytic covariance matrix matched a finite-difference Hessian of $\log G$ with maximum absolute error 4.888×10^{-6} .

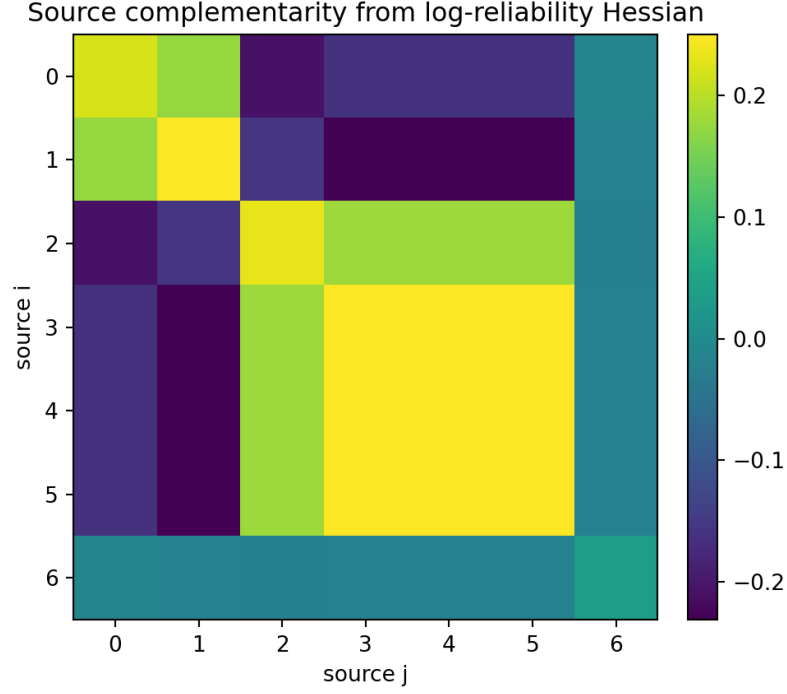


Figure 29: The Hessian of $\log G(e^{\mathbf{u}})$ is a covariance matrix over active information hyperedges. Positive entries indicate complementary sources; negative entries indicate substitutable pathways.

This suggests a new diagnostic: the first derivative of $\log G$ gives source participation, while the second derivative gives the geometry of information pathways. Order spectra say how many sources are needed. Hessian spectra say which sources are complements and which are substitutes.

F.3 Reliability-state observability

Consider a random reliability field $R = (R_1, \dots, R_n)$ with values in $[0, 1]^n$, independent of X and T . Conditional on R , source i is passed through a binary symmetric channel of reliability R_i .

There are two operational regimes. In the *state-known* regime, the observer sees both Y and the reliability state R . In the *state-hidden* regime, the observer sees only Y and the random reliability has been averaged into the channel.

Theorem 10 (Reliability-state observability law). *For independent fair Boolean sources under random nonnegative BSC reliabilities,*

$$\mathcal{J}_{\text{known}} = \sum_{S \neq \emptyset} \omega_S \mathbb{E} \left[\prod_{i \in S} R_i^2 \right]$$

and

$$\mathcal{J}_{\text{hidden}} = \sum_{S \neq \emptyset} \omega_S \left(\mathbb{E} \left[\prod_{i \in S} R_i \right] \right)^2.$$

Moreover, $\mathcal{J}_{\text{known}} \geq \mathcal{J}_{\text{hidden}}$ coefficientwise, with equality on a subset S exactly when $\prod_{i \in S} R_i$ has zero variance.

Proof. Conditioned on R , the Boolean character χ_S is attenuated by $\prod_{i \in S} R_i$. If R is observed, the squared posterior score is computed conditionally and then averaged, giving the first formula. If R is hidden, Y remains uniform and independent of R under the symmetric channel with uniform sources, so the averaged conditional-expectation operator attenuates χ_S by $\mathbb{E}[\prod_{i \in S} R_i]$ before squaring. Jensen's inequality gives $\mathbb{E}[U^2] \geq (\mathbb{E} U)^2$ for $U = \prod_{i \in S} R_i$. $\square$

This theorem exposes a new role for metadata. The reliability state is independent of the target, but revealing it can amplify target information by telling the observer which observations should be trusted. The effect grows rapidly with interaction order.

For a pure order- r target with $\Omega_r = 1$ and Bernoulli availability $R_i \in \{0, 1\}$ with marginal availability $\mu = 0.40$, exact enumeration produced the following survival values for order $r = 6$:

Reliability field	State known	State hidden
Independent availability	$\mu^6 = 0.004096$	$\mu^{12} = 1.67772 \times 10^{-5}$
Common-mode availability	$\mu = 0.400000$	$\mu^2 = 0.160000$

Table 17: Reliability metadata and reliability correlation can dominate interaction-order survival. For an order-six target at $\mu = .40$, known common-mode availability gives about 97.7 times more expected chi-square information than known independent availability, while revealing independent reliability metadata gives a factor 244.1 over the hidden independent channel.

Exact enumeration of parity targets of orders 2, 3, and 4 verified the known/hidden formulas. The maximum known-state formula error was numerically zero, and the maximum hidden-state formula error was 5.55×10^{-17} .

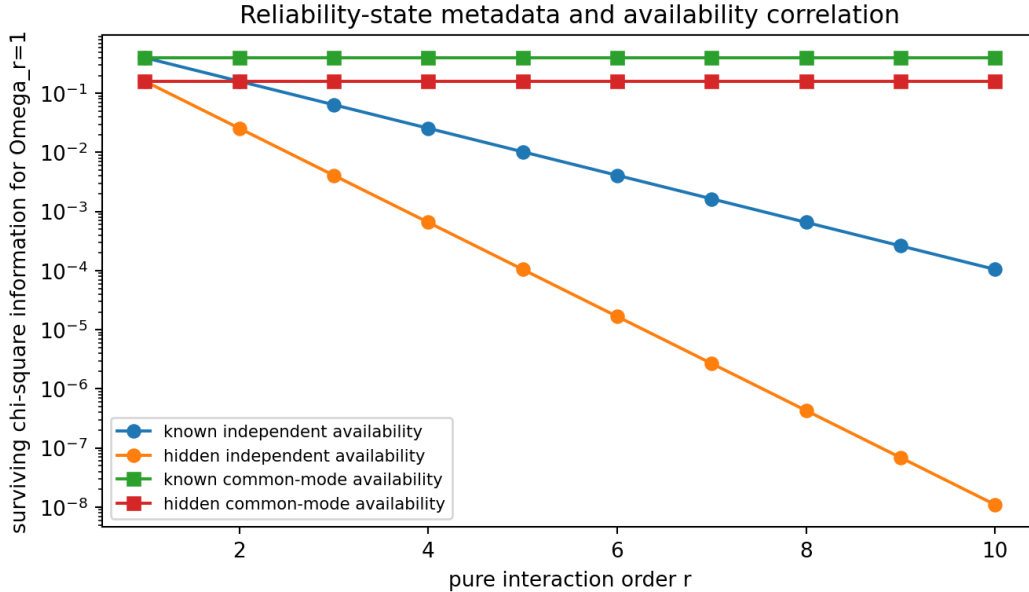


Figure 30: For pure order- r information with unit NRID mass and marginal availability $\mu = .40$, independent hidden availability decays as μ^{2r} , independent known availability decays as μ^r , hidden common-mode availability is μ^2 , and known common-mode availability is μ .

F.4 Availability-correlation law

For a pure information edge S of size r with Bernoulli availability indicators A_i , the state-known value is

$$\mathcal{J}_{\text{known}} = \omega_S \mathbb{P}(A_i = 1 \text{ for every } i \in S).$$

If every source has the same marginal availability μ , then elementary probability bounds give

$$\boxed{\max(0, r\mu - r + 1) \leq \frac{\mathcal{J}_{\text{known}}}{\omega_S} \leq \mu.}$$

The upper bound is achieved by common-mode availability $A_1 = \dots = A_r$. Independence gives μ^r , which may be far below the upper bound. A mutually exclusive schedule can drive the intersection probability to zero when $r\mu \leq r - 1$.

This is a sharp design law: marginal sensor uptime is not enough to characterize high-order information survival. For high-order targets, simultaneous uptime is the relevant resource. Positive availability correlation preserves synergy in expected chi-square information; independent availability suppresses it exponentially in order; negative dependence can erase it entirely.

F.5 Active sensing budget laws

A second use of the reliability-field polynomial is active sensing. Suppose z_i is a controllable squared-reliability investment and $\sum_i z_i \leq B$, with $0 \leq z_i \leq 1$.

Proposition 4 (Pure-edge budget law). *For a single information hyperedge S of size d and weight ω_S , the optimal value under budget B is*

$$\boxed{\max_{0 \leq z_i \leq 1, \sum_{i \in S} z_i \leq B} \omega_S \prod_{i \in S} z_i = \omega_S \left(\min \left\{ \frac{B}{d}, 1 \right\} \right)^d.}$$

For $B \leq d$, the optimum equalizes reliability within the edge: $z_i = B/d$ for $i \in S$.

Proof. The objective is monotone in each active coordinate. If $B \geq d$, setting all $z_i = 1$ attains ω_S . If $B < d$, the arithmetic-geometric mean inequality gives $\prod_{i \in S} z_i \leq (B/d)^d$, with equality at $z_i = B/d$. $\square$

Corollary 4 (Gradient blindness of pure synergy). *If every nonzero information hyperedge has size at least $d > 1$, then every coordinate derivative $\partial G / \partial z_i$ vanishes at $\mathbf{z} = 0$. The first nonzero local signal at the origin is a mixed derivative of order d . Thus any active-sensing rule that only asks for one-coordinate marginal improvement from zero is blind to pure d -way information.*

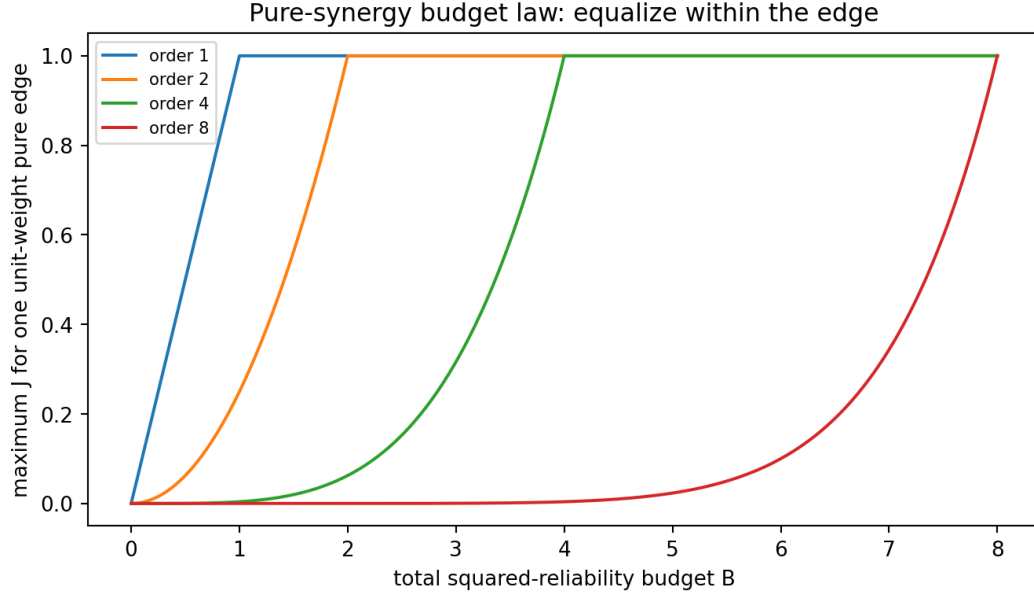


Figure 31: A unit-weight pure edge of order d obeys the budget law $J_{\max}(B) = (B/d)^d$ until saturation. High-order information requires spread across all participating sources before it becomes visible.

When pathways compete, optimal allocation can become winner-take-most. In an exact grid optimization with a pair pathway of weight 1 and a six-way pathway of weight 80, the optimum first saturated the pair pathway, then jumped to the six-way pathway, then returned leftover budget to the pair after the six-way edge saturated.

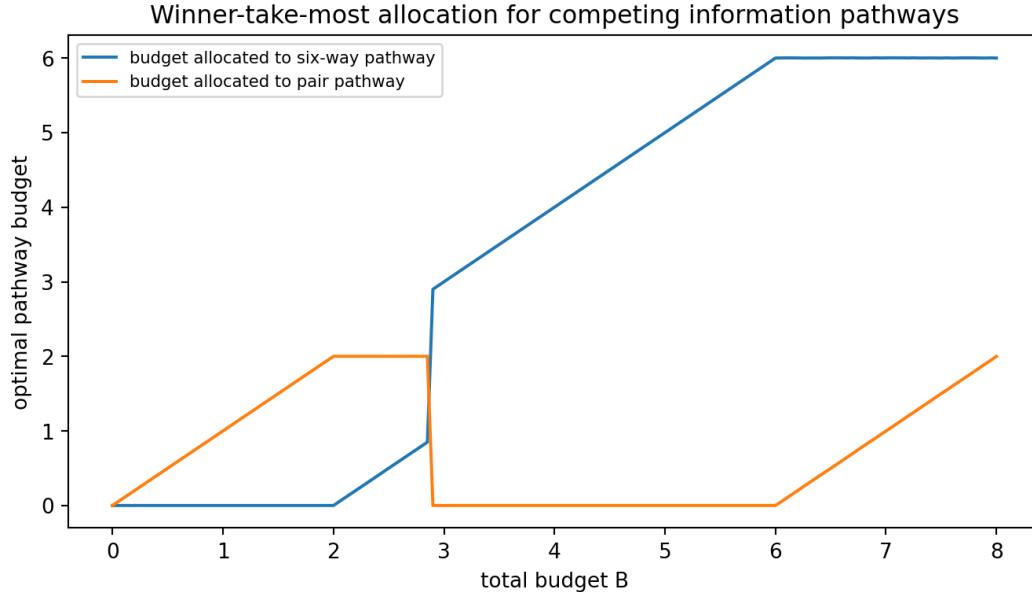


Figure 32: Budget allocation for a pair pathway and a stronger six-way pathway. High-order pathways can remain inactive until the budget is sufficient to fund the whole bundle, after which the optimum can switch abruptly.

This non-submodular behavior contrasts with classical sensor-placement settings where mutual-information objectives often admit useful submodular approximations or guarantees [32]. NRID budget objectives are positive multi-affine hypergraph polynomials; high-order monomials create complementarity rather than diminishing returns.

F.6 Applications suggested by this pass

Sensor networks and observatories. For high-order phenomena, marginal uptime is a weak design statistic. Coordinated uptime and explicit reliability metadata can improve expected information by orders of magnitude. A network designed to detect synergistic events should schedule simultaneous high-quality windows and transmit sensor-health states downstream.

Privacy and secure release. Visible dropout masks leak more high-order information than hidden independent dropout. For pure order- r information with availability μ , the known independent channel leaves μ^r chi-square mass, while the hidden independent channel leaves μ^{2r} . Reliability metadata can therefore be privacy-relevant even when it is statistically independent of the target.

Experimental design and perturbation biology. Gradient blindness shows why single-factor perturbations may fail to discover pure high-order interactions. Group probes, mixed derivatives, or seeded reliability fields are necessary when the first nonzero NRID order is greater than one.

Machine-learning interpretation. The Hessian of $\log G$ with respect to log-reliabilities is an interaction map over features. Positive entries indicate complementary feature bundles; negative entries indicate alternative feature pathways. This gives an information-theoretic analogue of a feature interaction graph.

F.7 Targets for further study

1. develop estimators for the known/hidden reliability gap when reliability states are partially observed or noisy;
2. characterize optimal reliability-state disclosure policies under privacy, robustness, and accuracy constraints;
3. solve active NRID allocation for sparse positive hypergraph polynomials with caps, costs, and group constraints;
4. extend the reliability-state observability law to finite-alphabet modal NRID and reversible Markov noise semigroups;
5. derive minimax availability-correlation bounds for nonuniform source availabilities and overlapping information hypergraphs;
6. test whether neural-network dropout masks, attention masks, and missingness indicators shift usable information from hidden to known reliability regimes;
7. combine cubic phase-sensitive spectroscopy with reliability-field Hessians to recover signed, geometry-aware information hypergraphs.

F.8 Research ledger update

The sixth pass changes the working theory in four ways. First, it turns the NRID polynomial into a log-partition function whose gradients and Hessians have direct source-participation and complementarity meanings. Second, it shows that reliability metadata is an information-bearing operational resource even when statistically independent of the target. Third, it separates marginal source quality from joint source quality: high-order information depends on intersection moments of the reliability field. Fourth, it shows that active sensing for high-order information is intrinsically bundle-based; local one-coordinate gradients are blind at the origin whenever the target is purely synergistic.

G Seventh pass: decision utility, posterior-amplitude shape, and replicated observation

G.1 Blind spot

The preceding sections establish an exact quadratic-information decomposition. The natural next question is whether the same spectrum also determines operational recovery. It does not. For binary targets, the posterior chi-square information is the second moment of a posterior score. Bayes accuracy, Shannon information, and other utilities depend on the entire amplitude distribution of that score. This pass isolates the resulting ambiguity and identifies a new object: the *posterior-amplitude shape* associated with a fixed NRID value.

Let $T \in \{-1, +1\}$ be balanced. For an arbitrary observation Y , define the posterior score

$$S(Y) = \mathbb{E}[T \mid Y].$$

Then

$$\mathcal{J}(T; Y) = \mathbb{E}[S(Y)^2],$$

while the Bayes advantage over random guessing is

$$A(T; Y) = \mathbb{P}(\hat{T}_{\text{Bayes}} = T) - \frac{1}{2} = \frac{1}{2} \mathbb{E} |S(Y)|.$$

The Shannon mutual information is

$$I(T; Y) = \mathbb{E} \psi(|S(Y)|),$$

where

$$\psi(a) = 1 - h_2\left(\frac{1+a}{2}\right) = \frac{(1+a) \ln(1+a) + (1-a) \ln(1-a)}{2 \ln 2}.$$

Thus quadratic NRID fixes only $\mathbb{E} S^2$, not $\mathbb{E} |S|$ or the full nonlinear average $\mathbb{E} \psi(|S|)$.

G.2 Binary posterior utility envelope

Theorem 11 (Utility envelope for balanced binary targets). *For any balanced binary target and any observation channel,*

$$\boxed{\frac{1}{2} \mathcal{J}(T; Y) \leq A(T; Y) \leq \frac{1}{2} \sqrt{\mathcal{J}(T; Y)}}.$$

Moreover,

$$\boxed{\frac{\mathcal{J}(T; Y)}{2 \ln 2} \leq I(T; Y) \leq \mathcal{J}(T; Y)}.$$

The lower Bayes envelope is attained by all-or-nothing posterior scores $|S| \in \{0, 1\}$, and the upper Bayes envelope is attained by constant-amplitude posterior scores $|S| = c$.

Proof. Since $|S| \leq 1$, one has $S^2 \leq |S|$, and therefore $A = \frac{1}{2} \mathbb{E} |S| \geq \frac{1}{2} \mathbb{E} S^2 = \mathcal{J}/2$. Cauchy's inequality gives $\mathbb{E} |S| \leq \sqrt{\mathbb{E} S^2}$, giving the upper bound. The equality cases are the equality cases of these two scalar inequalities. The Shannon inequalities follow from the scalar bounds

$$\frac{a^2}{2 \ln 2} \leq \psi(a) \leq a^2, \quad 0 \leq a \leq 1,$$

which are verified by elementary one-variable calculus. $\square$

This theorem gives a precise sense in which χ^2 NRID is utility-incomplete. At a fixed quadratic information level $\mathcal{J}$, Bayes decision advantage can vary between order $\mathcal{J}$ and order $\sqrt{\mathcal{J}}$. The gap is large in weak-information regimes and grows exponentially when the same local ambiguity is multiplied across a high-order parity edge.

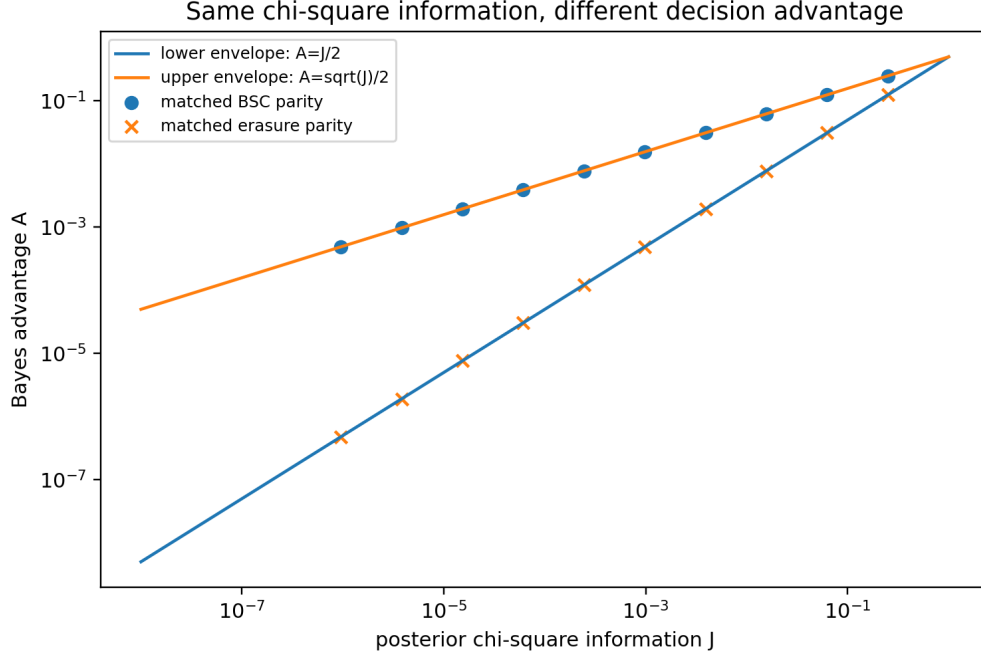


Figure 33: The Bayes utility envelope for balanced binary targets. Matched BSC and erasure parity examples have identical posterior chi-square information but lie on opposite decision envelopes.

G.3 Channel-shape split: same NRID, exponentially different Bayes utility

Consider a pure d -way parity target

$$T = \prod_{i=1}^d X_i.$$

Let each source be observed through an arbitrary independent local channel and write

$$\eta_i(Y_i) = \mathbb{E}[X_i | Y_i], \quad z_i = \mathbb{E}[\eta_i(Y_i)^2], \quad \ell_i = \mathbb{E}|\eta_i(Y_i)|.$$

Then the posterior score factorizes:

$$\mathbb{E}[T | Y_1, \dots, Y_d] = \prod_{i=1}^d \eta_i(Y_i).$$

Consequently,

$$\mathcal{J}(T; Y) = \prod_{i=1}^d z_i,$$

whereas

$$A(T; Y) = \frac{1}{2} \prod_{i=1}^d \ell_i.$$

Since $z_i \leq \ell_i \leq \sqrt{z_i}$ for $|\eta_i| \leq 1$,

$$\boxed{\frac{1}{2} \prod_{i=1}^d z_i \leq A(T; Y) \leq \frac{1}{2} \prod_{i=1}^d \sqrt{z_i}.}$$

The upper envelope is realized by a constant-magnitude channel such as a binary symmetric channel with $|\eta_i| = \sqrt{z_i}$. The lower envelope is realized by an erasure channel that reveals X_i perfectly with probability z_i and reveals nothing otherwise.

For identical local observability $z_i = q$, both channels have the same NRID value

$$\mathcal{J} = q^d.$$

But their Bayes advantages are

$$A_{\text{BSC}} = \frac{1}{2} q^{d/2}, \quad A_{\text{erasure}} = \frac{1}{2} q^d.$$

Thus

$$\boxed{\frac{A_{\text{BSC}}}{A_{\text{erasure}}} = q^{-d/2}.}$$

The ambiguity is exponential in interaction order.

order d	common $\mathcal{J}$	A_{BSC}	A_{erasure}	advantage ratio	$I_{\text{erasure}}/I_{\text{BSC}}$
2	6.2500×10^{-2}	1.2500×10^{-1}	3.1250×10^{-2}	4	1.37164
4	3.9063×10^{-3}	3.1250×10^{-2}	1.9531×10^{-3}	16	1.38539
6	2.4414×10^{-4}	7.8125×10^{-3}	1.2207×10^{-4}	64	1.38624
8	1.5259×10^{-5}	1.9531×10^{-3}	7.6294×10^{-6}	256	1.38629
10	9.5367×10^{-7}	4.8828×10^{-4}	4.7684×10^{-7}	1024	1.38629

Table 18: Matched local observability $q = 0.25$. The two channels have identical NRID values for every order, but Bayes advantage differs by $q^{-d/2} = 2^d$. The Shannon ratio tends to $2 \ln 2$.

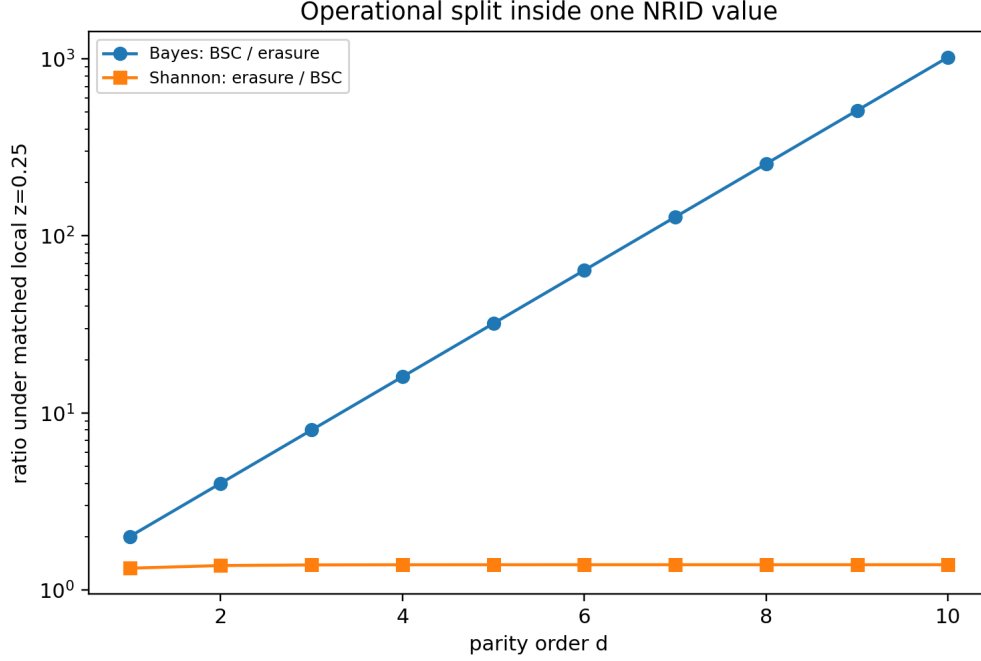


Figure 34: Operational split under matched local NRID. Bayes advantage separates exponentially with order, while Shannon information remains within the scalar envelope $\mathcal{J}/(2 \ln 2) \leq I \leq \mathcal{J}$.

Exact enumeration of parity targets of orders 2, 3, 4, 5 verified the formulas for both BSC and erasure channels with maximum absolute error zero at displayed precision.

G.4 Leading-layer decision exponent

Let $g(x) = \mathbb{E}[T \mid X = x]$ be the balanced binary target score and let

$$g(x) = \sum_{S \neq \emptyset} \hat{g}(S) \chi_S(x).$$

Suppose the first nonzero Fourier order is d , and define

$$H_d(y) = \sum_{|S|=d} \hat{g}(S) \chi_S(y).$$

Under symmetric BSC reliability ρ ,

$$S_\rho(Y) = T_\rho g(Y) = \rho^d H_d(Y) + O(\rho^{d+1}).$$

Therefore

$$\mathcal{J}_\rho = \rho^{2d} \mathbb{E}[H_d^2] + o(\rho^{2d}),$$

$$A_\rho = \frac{1}{2} \rho^d \mathbb{E}[H_d] + o(\rho^d),$$

while

$$I_\rho = \frac{\rho^{2d}}{2 \ln 2} \mathbb{E}[H_d^2] + o(\rho^{2d}).$$

Thus the Bayes-decision exponent is d , while the quadratic-information and Shannon weak-signal exponents are $2d$.

This motivates a leading-layer *decision flatness* coefficient

$$\Phi_d = \frac{(\mathbb{E} |H_d|)^2}{\mathbb{E}[H_d^2]} \in [0, 1].$$

It controls the asymptotic conversion of chi-square energy into Bayes edge:

$$\boxed{\frac{A_\rho}{\frac{1}{2}\sqrt{\mathcal{J}_\rho}} o\sqrt{\Phi_d}}.$$

A parity monomial has $\Phi_d = 1$. A spread-out leading layer has smaller Φ_d because posterior score magnitude varies across observations.

Target	first d	$\mathbb{E} H_d^2$	$\mathbb{E} H_d $	Φ_d	slope A	slope $\mathcal{J}$
copy	1	1.0000	1.0000	1.0000	1.0000	2.0000
XOR-2	2	1.0000	1.0000	1.0000	2.0000	4.0000
XOR-3	3	1.0000	1.0000	1.0000	3.0000	6.0000
majority-3	1	0.7500	0.7500	0.7500	1.0040	2.0001
2-bit-address multiplexer	1	0.2500	0.3750	0.5625	1.0450	2.0235

Table 19: Low-reliability exponents and leading-layer flatness from exact enumeration. Deviations from integer slopes are finite-range fitting artifacts.

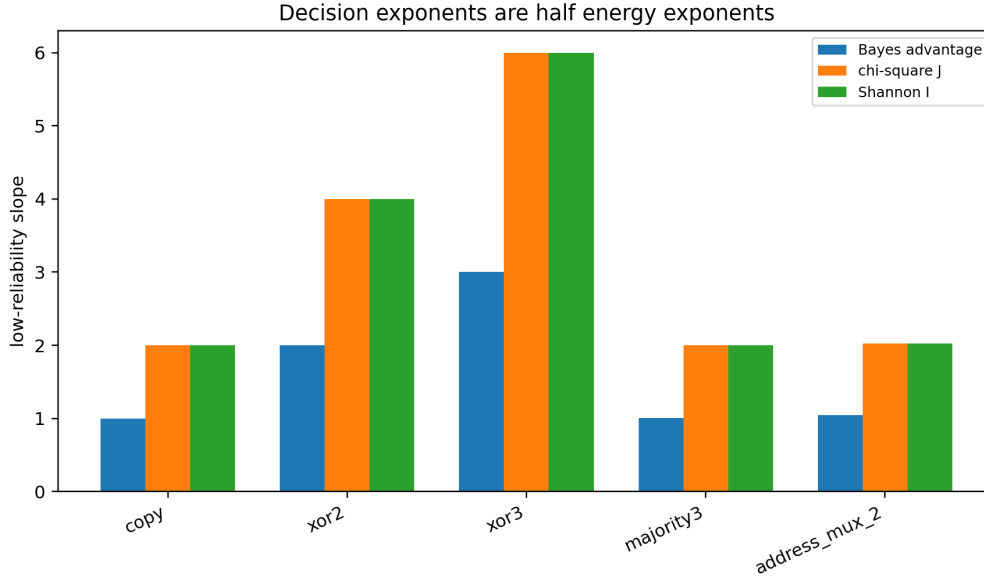


Figure 35: Fitted low-reliability slopes. Bayes advantage decays as ρ^d , whereas $\mathcal{J}$ and weak-signal Shannon information decay as ρ^{2d} .

A random search over normalized level-two Fourier layers on $n = 8$ variables produced flatness values centered near the Gaussian reference $2/\pi \approx 0.63662$. Over 2500 random trials, the mean

flatness was 0.64857, the median was 0.64852, and the lowest observed value was 0.54566. This suggests that highly distributed same-order information behaves like a random posterior amplitude field rather than a constant-magnitude parity score.

G.5 Replicated local observations

The local-observability theorem from earlier sections says that quadratic NRID only sees

$$z_i = \mathbb{E}[\mathbb{E}[X_i | Y_i]^2].$$

For decision utility, a second local statistic appears:

$$\ell_i = \mathbb{E} |\mathbb{E}[X_i | Y_i]|.$$

Repeated observations make the distinction explicit. Suppose one source X is observed m times through independent BSC channels with reliability ρ . Let

$$\lambda = \operatorname{arctanh} \rho, \quad K \sim \operatorname{Binomial} \left(m, \frac{1 + ho}{2} \right).$$

Conditional on $X = +1$, the posterior mean is

$$\eta_m = \tanh((2K - m)\lambda).$$

Hence

$$\begin{aligned} z_m(\rho) &= \sum_{k=0}^m \binom{m}{k} \left(\frac{1 + ho}{2} \right)^k \left(\frac{1 - ho}{2} \right)^{m-k} \tanh^2((2k - m) \operatorname{arctanh} \rho), \\ \ell_m(\rho) &= \sum_{k=0}^m \binom{m}{k} \left(\frac{1 + ho}{2} \right)^k \left(\frac{1 - ho}{2} \right)^{m-k} |\tanh((2k - m) \operatorname{arctanh} \rho)|. \end{aligned}$$

For weak observations,

$$z_m(\rho) = m\rho^2 + O(\rho^4),$$

whereas

$$\ell_m(\rho) = \rho \mathbb{E} |R_m| + O(\rho^3),$$

with R_m a sum of m independent Rademacher variables. For large m in the weak-observation regime,

$$\frac{\ell_m(\rho)^2}{z_m(\rho)} \approx \frac{2}{\pi}.$$

Thus repeated weak measurements naturally create a Gaussian posterior-amplitude penalty: the quadratic energy grows like the variance of accumulated evidence, while Bayes edge grows like its mean absolute value.

For a pure parity edge of order d with m repeated BSC readings per source,

$$\mathcal{J}_{m,d} = z_m(\rho)^d, \quad A_{m,d} = \frac{1}{2} \ell_m(\rho)^d.$$

The efficiency of converting $\sqrt{\mathcal{J}}$ into Bayes advantage is

$$\left(\frac{\ell_m(\rho)^2}{z_m(\rho)} \right)^{d/2}.$$

Small local amplitude-shape penalties are therefore magnified by interaction order.

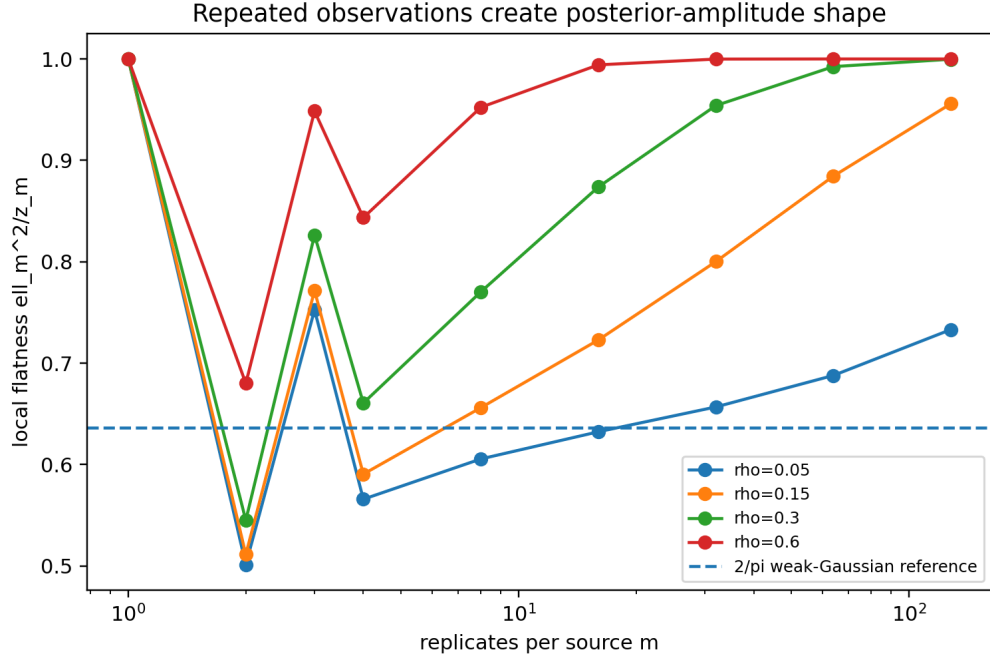


Figure 36: Local flatness ℓ_m^2/z_m for repeated BSC observations. The nonmonotone even/odd pattern comes from exact ties when an even number of weak readings split evenly. The weak-Gaussian reference $2/\pi$ appears as the accumulated evidence becomes approximately normal but still weak.

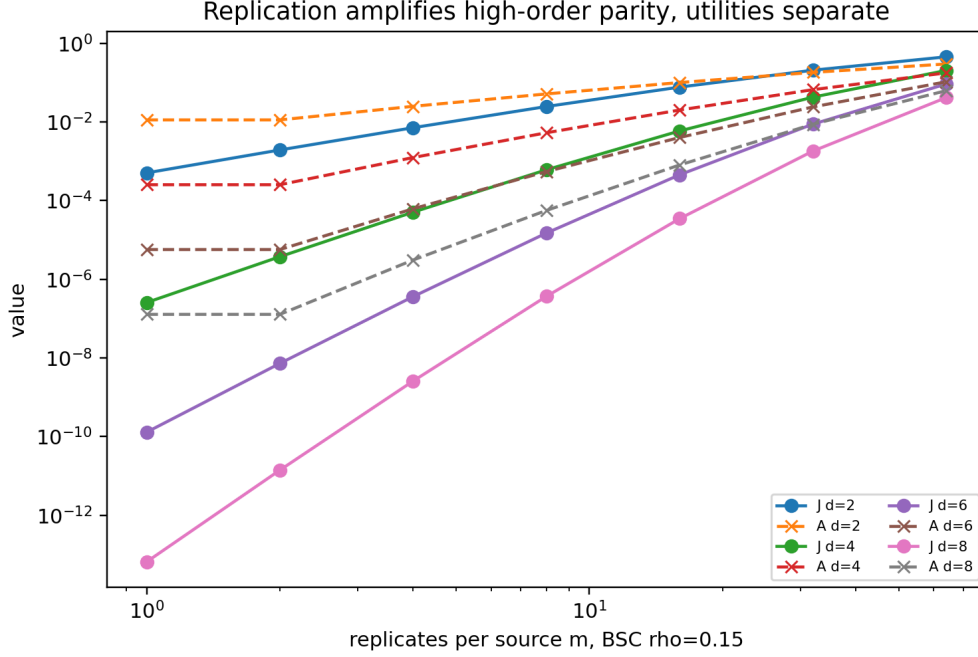


Figure 37: Replication amplifies high-order parity information, but $\mathcal{J}$ and Bayes advantage separate according to the posterior-amplitude shape.

G.6 Applications suggested by this pass

Sensor design. Two sensors can have the same quadratic observability z but very different decision value. A constant weak signal is better for immediate classification; an all-or-nothing erasure signal can carry more Shannon information per unit chi-square but may be decision-poor unless all participating sources are present. For high-order phenomena, this distinction is exponentially amplified.

Privacy and leakage auditing. Quadratic leakage can understate or overstate operational risk depending on the adversary’s utility. A small but persistent bias across all records may be more useful for decisions than rare perfect revelations with the same chi-square value. Conversely, rare revelations may carry more Shannon information than a diffuse weak bias at matched chi-square.

Machine learning and representation analysis. Feature attributions based only on second moments may miss the difference between diffuse evidence and intermittent certainty. Decision flatness Φ_d provides a candidate diagnostic for whether a learned representation stores an interaction as a stable distributed bias or as sparse high-confidence events.

Experimental design. Replicated measurements should be evaluated with both z_m and ℓ_m . Quadratic NRID predicts information energy, but Bayes utility requires amplitude-shape diagnostics. In high-order targets, a modest local flatness penalty becomes a large global decision penalty.

G.7 Targets for further study

1. develop an amplitude-resolved NRID that records the distribution of posterior score contributions, not only their squared Fourier energy;
2. characterize the full set of achievable Bayes and Shannon utilities for a fixed subset spectrum $\{\omega_S\}$;
3. extend the binary utility envelope to multiclass targets, general losses, and abstention policies;
4. determine extremal local channels for parity and non-parity hypergraphs under fixed z_i , fixed cost, and fixed Shannon leakage;
5. build estimators for decision flatness Φ_d from noisy intervention data;
6. analyze whether trained classifiers move from erasure-like to BSC-like posterior amplitude shapes during calibration, distillation, or adversarial training;
7. combine cubic phase-sensitive NRID with amplitude-resolved utility to distinguish signed structure from operational recoverability.

G.8 Research ledger update

The seventh pass changes the working theory in three ways. First, it proves that quadratic NRID is an exact energy theory but not a complete operational utility theory: the same $\mathcal{J}$ can imply Bayes advantages ranging from $\mathcal{J}/2$ to $\sqrt{\mathcal{J}}/2$. Second, it identifies posterior-amplitude shape as an independent axis of information structure. BSC-like diffuse evidence and erasure-like intermittent certainty can have identical NRID but exponentially different decision utility for high-order parity. Third, it introduces decision flatness and replicated-observation formulas, showing that the decision exponent is the interaction order d while the information-energy exponent is $2d$.

H Eighth pass: moment-closure spectroscopy and phase-gauge invariants

H.1 Blind spot

Quadratic NRID is exact for posterior chi-square information, but it is phase-blind: it sees squared Fourier amplitudes and discards signs. The seventh pass showed that this second-moment description is not operationally complete. The present pass studies the full posterior-score moment hierarchy. For balanced binary targets this hierarchy has a simple algebraic form: every moment is a sum over zero-XOR closures of Fourier supports. This makes higher-order posterior moments a form of phase-sensitive spectroscopy.

Let $T \in \{-1, +1\}$ be balanced and define the target score

$$g(x) = \mathbb{E}[T \mid X = x] = \sum_{A \neq \emptyset} a_A \chi_A(x).$$

Under symmetric BSC reliability ρ ,

$$S_\rho(y) = \mathbb{E}[T \mid Y_\rho = y] = \sum_A a_A \rho^{|A|} \chi_A(y).$$

Define posterior-score moments

$$M_k(\rho) = \mathbb{E}[S_\rho(Y)^k].$$

Theorem 12 (Posterior moment-closure law). *For every integer $k \geq 1$,*

$$M_k(\rho) = \sum_{A_1 \oplus \dots \oplus A_k = \emptyset} \left(\prod_{j=1}^k a_{A_j} \right) \rho^{\sum_j |A_j|}.$$

Equivalently,

$$M_k(\rho) = \sum_d C_{k,d} \rho^d,$$

where $C_{k,d}$ is the signed sum of all k -tuples of active Fourier supports with zero symmetric-difference sum and total degree d .

Proof. Expand $S_\rho(y)^k$. A product of Walsh characters satisfies

$$\chi_{A_1}(y) \cdots \chi_{A_k}(y) = \chi_{A_1 \oplus \dots \oplus A_k}(y).$$

Its expectation over the uniform cube is one if the symmetric difference is empty and zero otherwise. Collecting surviving terms gives the identity. $\square$

The quadratic case is precisely the binary balanced NRID layer:

$$M_2(\rho) = \sum_A a_A^2 \rho^{2|A|}.$$

Thus NRID is the second member of a larger moment-closure hierarchy. Cubic moments detect triangle closures $A \oplus B \oplus C = \emptyset$; quartic moments detect parallelogram-like closures; higher moments detect longer closed walks in the support matroid.

H.2 Minimal cubic phase bit

Let

$$g_\pm(x) = \frac{1}{3}(\chi_1 + \chi_2 \pm \chi_{12}).$$

Then both targets have the same quadratic response,

$$M_2(\rho) = \frac{1}{9}(2\rho^2 + \rho^4),$$

but opposite cubic response,

$$M_3(\rho) = \pm \frac{2}{9}\rho^4.$$

At $\rho = 0.8$, exact enumeration gives $M_2 = 0.187733333333$ for both targets, while $M_3 = \pm 0.091022222222$. The quadratic spectrum is identical; the cubic layer reveals the signed closure product.

H.3 Source-polarity gauge theorem

Let the active support be $H = \{A_1, \dots, A_m\}$ and let $B \in \mathbb{F}_2^{n \times m}$ be the incidence matrix with $B_{ij} = 1$ if $i \in A_j$. Write coefficient signs as $\sigma_j \in \{-1, +1\}$. A source-polarity recoding $X_i \mapsto \epsilon_i X_i$ changes signs by

$$\sigma_j \mapsto \sigma_j \prod_i \epsilon_i^{B_{ij}}.$$

Thus individual coefficient signs are gauge-dependent.

Theorem 13 (Phase-code gauge theorem). *For each relation $u \in \ker_{\mathbb{F}_2} B$, the closed sign product*

$$\Pi_u = \prod_{j: u_j=1} \sigma_j$$

is invariant under source-polarity gauge. Conversely, the collection of all such products determines the sign pattern up to source-polarity gauge. The number of independent gauge-invariant phase bits is

$$\boxed{m - \text{rank}_{\mathbb{F}_2}(B)}.$$

Proof. If $u \in \ker B$, then every source coordinate appears an even number of times in $\prod_{j: u_j=1} \prod_i \epsilon_i^{B_{ij}}$, so the gauge factor is one. Conversely, two sign patterns differ by a gauge transformation exactly when their sign-ratio vector lies in the row space of B . The orthogonal complement of the row space is $\ker B$, so equality of all closed products is equivalent to gauge equivalence. $\square$

This theorem identifies the phase object that higher-moment NRID can hope to recover. The squared spectrum gives unsigned energy; moment closures give gauge-invariant signs on zero-sum support cycles.

H.4 Odd-moment silence

There is a structural obstruction to odd moment spectroscopy. Suppose there exists a nonzero $\lambda \in \mathbb{F}_2^n$ such that

$$\lambda \cdot A = 1$$

for every active Fourier support A . Then the XOR of any odd number of active supports also has λ -dot value one, and therefore cannot be empty. Hence

$$\boxed{M_{2q+1}(\rho) = 0 \quad \text{for all } q \geq 0.}$$

This gives exact odd-moment silence. Exact enumeration with six-bit stochastic targets confirmed that an affine-support target had maximum absolute odd moment 1.73×10^{-18} across M_3, M_5, M_7 , while a non-affine support of the same size reached 1.49×10^{-2} .

H.5 Closure-rank experiments

For random supports in $\mathbb{F}_2^{10}$, the full phase dimension is $m - \text{rank}(B)$. The experiments estimated how much of this relation rank is generated by cubic closures alone and by cubic-plus-quartic closures. In the dense regime where the mean kernel dimension is at least eight, quartic closures first exceeded 99% of the full phase rank at support size $m = 40$, while cubic closures first exceeded 99% at $m = 100$. Thus quartic spectroscopy can become phase-complete much earlier than cubic spectroscopy.

Selected exact case studies:

case	m	$\text{rank}(B)$	kernel dim	triangle rank	≤ 4 rank
basis support	6	6	0	0	0
single triangle	3	2	1	1	1
dense first 32 masks	32	6	26	26	26
affine hyperplane support	20	6	14	0	14

H.6 Finite moment completion of utility

For fixed ρ , the BSC observation Y is uniform on 2^n states. Let the posterior-score multiset be $\{s_y : y \in \{-1, +1\}^n\}$. The first $N = 2^n$ power sums

$$p_k = \sum_y s_y^k, \quad k = 1, \dots, N,$$

determine the monic polynomial with roots s_y by Newton identities. Hence the first 2^n posterior-score moments determine the posterior-score distribution and all symmetric utilities of it, including Bayes advantage and Shannon information. This is an exact finite-support moment-completion theorem, but the reconstruction is numerically ill-conditioned for large n .

A three-bit test reconstructed eight posterior scores from the first eight moments with maximum score error 8.87×10^{-13} ; the reconstructed Bayes advantage and Shannon information matched the exact values to 2.36×10^{-16} and 6.94×10^{-18} respectively.

H.7 Applications and study targets

Potential applications include phase-sensitive feature-interaction discovery, cryptographic analysis of Walsh spectra beyond magnitudes, privacy leakage auditing for signed high-order dependencies, and experimental design for choosing whether third-order or fourth-order perturbations are sufficient to recover interaction phase.

Targets for further study are:

1. prove threshold laws for random Walsh supports: when do cubic or quartic closures span the support kernel;
2. derive sample-complexity bounds for estimating $C_{k,d}$ from noisy reliability sweeps;
3. extend the phase-gauge theorem to multiclass targets, where Fourier coefficients are vectors in target-probability tangent space;
4. combine phase-code spectroscopy with amplitude-resolved NRID to predict Bayes utility from finitely many moment layers;
5. study numerical conditioning of finite moment completion and derive stable utility bounds from partial moment data;
6. use matroid and coding-theoretic tools to classify supports with odd-moment silence or delayed phase observability.

H.8 Research ledger update

The eighth pass adds a third structural layer to NRID. The original theory gives unsigned information energy; the moment-closure hierarchy adds gauge-invariant phase; finite moment completion adds posterior-amplitude distribution. These three layers suggest a broader decomposition program: information should be resolved not only by source order, but also by phase closure and utility-relevant amplitude shape.

I Ninth pass: geometry-resolved NRID and information heat spectra

I.1 Blind spot

The first eight passes treat source degradation mainly as independent coordinate noise, finite-alphabet mode attenuation, subset masking, reliability fields, and higher posterior moments. A remaining blind spot is geometric: many information-bearing systems are not products of independent coordinates. They are graphs, manifolds, networks, metric data clouds, social systems, transportation systems, communication fabrics, and sensor fields. In such systems the admissible degradation is often not coordinate erasure but diffusion along a geometry.

This pass develops a geometry-resolved version of NRID. The order spectrum becomes a heat spectrum: information is decomposed by the eigenmodes of a reversible Markov semigroup. In this setting the response curve is a Laplace transform of a target-specific information spectral measure. The Boolean order polynomial appears as the special case where the geometry is the product cube and the generator eigenvalue of a Walsh character is its support size.

I.2 Reversible semigroup NRID theorem

Let $\mathcal{X}$ be a finite state space with stationary distribution π . Let $(K_t)_{t \geq 0}$ be a reversible Markov semigroup on $\mathcal{X}$, so K_t is self-adjoint on $L^2(\pi)$ and

$$K_t \phi_k = e^{-\lambda_k t} \phi_k, \quad 0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots,$$

for an orthonormal eigenbasis $\{\phi_k\}$ with $\phi_0 \equiv 1$. Let $X \sim \pi$, let Y_t be produced from X by K_t , and let T be any finite target with

$$q_a(x) = \Pr(T = a \mid X = x), \quad p_a = \Pr(T = a), \quad m_a(x) = q_a(x) - p_a.$$

Define coefficients

$$c_{a,k} = \langle m_a, \phi_k \rangle_\pi.$$

Theorem 14 (Geometry-resolved NRID). *For reversible Markov degradation,*

$$\mathcal{J}_t(T; Y_t) = \sum_a \frac{1}{p_a} \sum_{k \geq 1} c_{a,k}^2 e^{-2\lambda_k t}.$$

Equivalently, if

$$W_\ell(T) = \sum_a \frac{1}{p_a} \sum_{k: \lambda_k = \ell} c_{a,k}^2,$$

then

$$\mathcal{J}_t = \sum_{\ell > 0} W_\ell(T) e^{-2\ell t}.$$

Thus the posterior chi-square information curve is the heat-Laplace transform of an information spectral measure.

Proof. By reversibility,

$$\Pr(T = a \mid Y_t = y) - p_a = (K_t m_a)(y).$$

Therefore

$$\mathcal{J}_t = \sum_a \frac{1}{p_a} \|K_t m_a\|_{L^2(\pi)}^2.$$

Expanding m_a in the eigenbasis and using self-adjoint diagonalization gives

$$\|K_t m_a\|_\pi^2 = \sum_{k \geq 1} c_{a,k}^2 e^{-2\lambda_k t}.$$

Summing over target states completes the proof. $\square$

This theorem gives a direct bridge to the original Boolean result. For the continuous-time product bit-refresh process on $\{-1, +1\}^n$, the Walsh character χ_S has generator eigenvalue $\lambda_S = |S|$. Setting $\rho = e^{-t}$ gives

$$e^{-2\lambda_S t} = \rho^{2|S|},$$

so the original order spectrum is a geometry spectrum on the hypercube.

I.3 Information heat measure and spectral thermodynamics

Define the information heat measure

$$\nu_T = \sum_a \frac{1}{p_a} \sum_{k \geq 1} c_{a,k}^2 \delta_{\lambda_k}.$$

Then

$$\mathcal{J}_t = \int e^{-2\lambda t} d\nu_T(\lambda).$$

The heat curve has exact cumulant diagnostics:

$$\Lambda(t) = -\frac{1}{2} \frac{d}{dt} \log \mathcal{J}_t = \mathbb{E}_t[\lambda],$$

where the tilted distribution has mass proportional to $e^{-2\lambda t} d\nu_T(\lambda)$, and

$$\frac{1}{4} \frac{d^2}{dt^2} \log \mathcal{J}_t = \text{Var}_t(\lambda).$$

Thus the slope of the log-heat information curve is the currently active diffusion frequency, and the curvature is the spread of active diffusion frequencies. Diffusion “cools” the information spectrum: high-frequency information disappears first, leaving only the slowest active modes.

A large-time corollary follows immediately. Let

$$\lambda_*(T) = \min\{\lambda > 0 : W_\lambda(T) > 0\}.$$

Then

$$\mathcal{J}_t \sim W_{\lambda_*}(T) e^{-2\lambda_* t} \quad (t \rightarrow \infty).$$

The asymptotic information half-life is

$$\tau_{1/2} \sim \frac{\log 2}{2\lambda_*(T)}.$$

This is the geometric analogue of the order-fragility law.

I.4 Cycle experiment: topology separates low- and high-frequency information

The first experiment used the continuous-time random walk generated by $I - P$ on a 64-cycle, where P steps to either neighbor with probability $1/2$. Four targets were compared: identity $T = X$, a half-cycle label, a quarter-cycle label, and the alternating label. Direct enumeration of the joint law and the spectral formula agreed to numerical precision: the largest binary-target discrepancy across selected times was 8.88×10^{-16} , and the largest identity-target discrepancy was 2.13×10^{-14} .

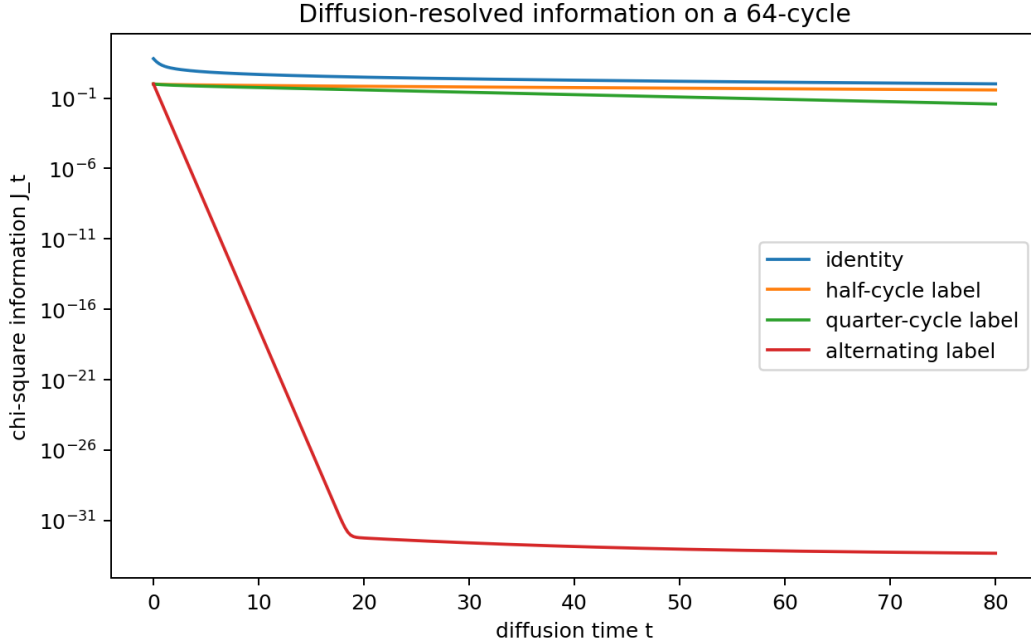


Figure 38: Diffusion-resolved information on a 64-cycle. The alternating label is a highest-frequency target and disappears almost immediately. The half-cycle label has substantial projection on the slowest nonconstant eigenmode and survives for a long time.

The slowest nonconstant eigenvalue was

$$\lambda_1 = 0.0048152733278,$$

while the alternating label lived at

$$\lambda_{\max} = 2.$$

At diffusion time $t = 10$, the half-cycle label retained $\mathcal{J}_t = 0.7784$, while the alternating label had fallen to 4.78×10^{-18} . Both labels begin with one bit-equivalent deterministic target energy, but the geometry makes one robust and the other thermally fragile.

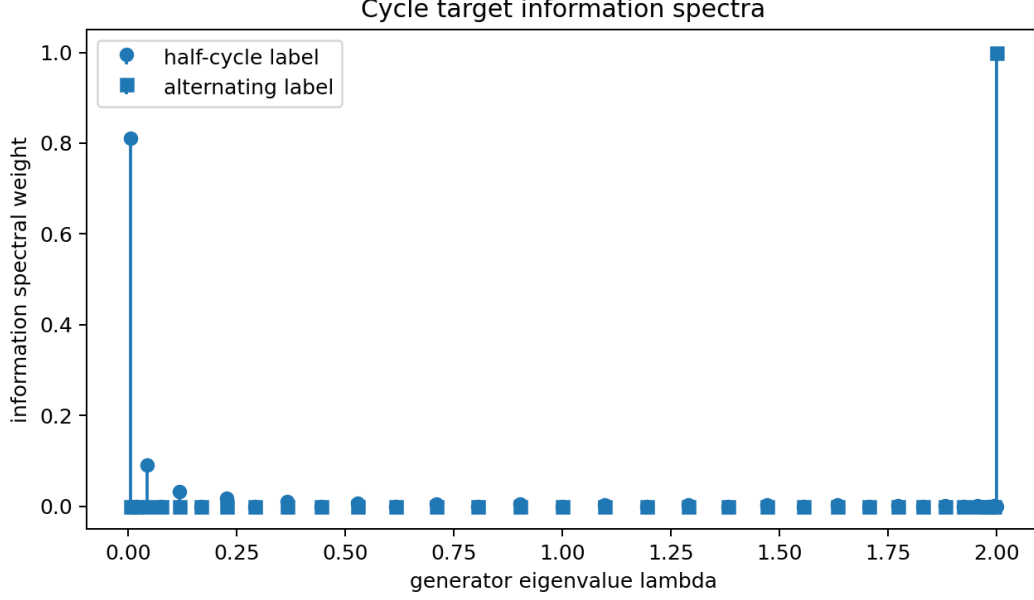


Figure 39: Information spectral weights for two 64-cycle targets. The alternating label is spectrally monochromatic at the largest generator eigenvalue. The half-cycle label is broadband but has nonzero weight at the slowest modes, which dominate late-time information.

The effective spectral rate $\Lambda(t)$ makes the cooling phenomenon visible. For broadband targets, early information contains many modes; late information is dominated by the smallest active eigenvalue.

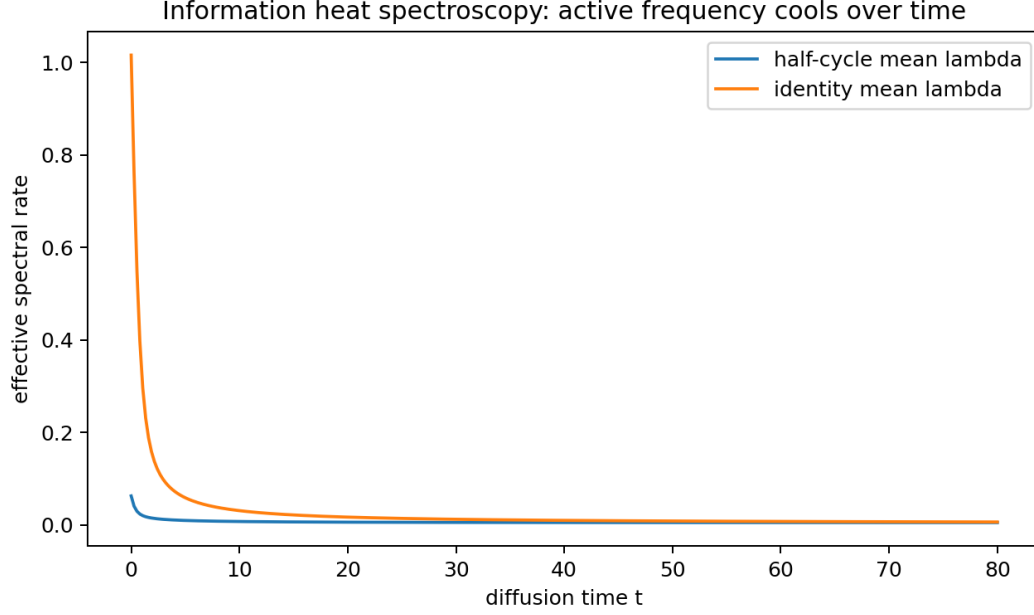


Figure 40: Information heat spectroscopy on the 64-cycle. The active spectral rate decreases over time as high-frequency information is erased by diffusion.

I.5 Bottleneck persistence law

A second experiment used a regular weighted two-block graph with 25 vertices per block. Within-block edge weights were 1 and cross-block weights were $\epsilon = 0.02$. The block/community label is nearly the slowest nonconstant eigenmode, with active eigenvalue

$$\lambda_{\text{block}} = 0.04081632653.$$

A within-block contrast with zero block mean had active mean eigenvalue

$$\lambda_{\text{within}} = 1.04081632653.$$

At $t = 5$,

$$\mathcal{J}_t(\text{block}) = 0.6648734228, \quad \mathcal{J}_t(\text{within}) = 2.9069713 \times 10^{-5},$$

so the community label was about

$$\boxed{2.287 \times 10^4}$$

times more recoverable under the same diffusion process.

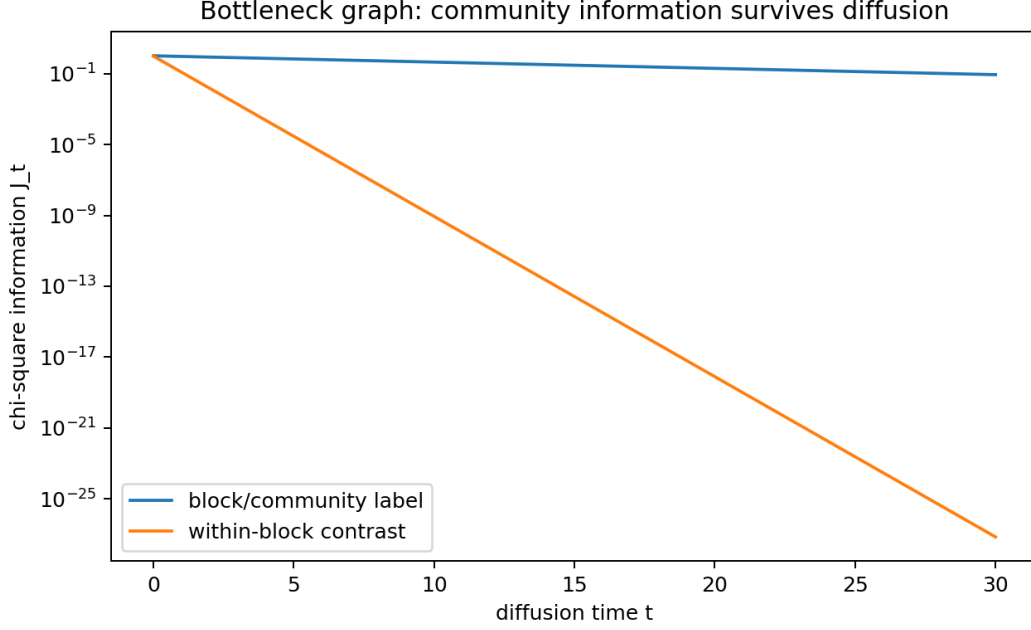


Figure 41: Bottleneck graph experiment. Diffusion destroys within-community detail while preserving the coarse block attribute for much longer.

This suggests a bottleneck persistence principle:

$$\frac{\mathcal{J}_t(A)}{\mathcal{J}_t(B)} \approx \frac{W_A}{W_B} \exp\{-2(\lambda_A - \lambda_B)t\}$$

when targets A and B are dominated by single spectral modes. Coarse attributes aligned with slow graph cuts can survive severe diffusion even when high-resolution information has vanished.

I.6 Eigenspace orientation blindness

The heat spectrum sees the amount of target information in each eigenspace, not its orientation inside a degenerate eigenspace. More precisely, let Π_λ be the projection onto the eigenvalue- λ eigenspace. Then

$$\mathcal{J}_t = \sum_{\lambda} e^{-2\lambda t} \sum_a \frac{1}{p_a} \|\Pi_\lambda m_a\|^2.$$

Any orthogonal rotation of the centered likelihoods inside a fixed eigenspace leaves $\mathcal{J}_t$ unchanged.

On the 64-cycle, the score functions

$$g_c(x) = 0.6 \cos(2\pi x/64), \quad g_s(x) = 0.6 \sin(2\pi x/64)$$

produce valid balanced stochastic binary targets. They had identical heat-information curves to numerical precision, with maximum observed difference 3.05×10^{-15} . But their local posterior scores at vertex 0 were different: at $t = 0$, $g_c(0) = 0.6$ while $g_s(0) = 0$. The global heat curve is therefore spectrally exact but orientation-blind inside degenerate eigenspaces. A vertex-frequency or windowed NRID would be needed to localize such information.

I.7 Heat-spectrum tomography

Because

$$\mathcal{J}_t = \sum_{\ell=1}^m W_{\ell} e^{-2\lambda_{\ell} t},$$

known eigenvalues and enough heat samples can in principle recover the spectral weights. For distinct λ_{ℓ} , the exponential sampling matrix

$$A_{j\ell} = e^{-2\lambda_{\ell} t_j}$$

forms a finite exponential system, so exact recovery is possible in noiseless arithmetic when A has full column rank. The derivative form is the moment sequence

$$\mu_k = (-1)^k 2^{-k} \mathcal{J}^{(k)}(0) = \int \lambda^k d\nu_T(\lambda).$$

Thus heat-response spectroscopy is an inverse Laplace problem.

The same experiment also shows the practical barrier. For the 64-cycle, the full 32-line nonzero spectral inversion matrix on a broad time grid had condition number

$$\kappa(A) = 2.03 \times 10^{17}.$$

With only 10^{-4} relative measurement noise, a spectrum containing a high-frequency line had relative recovery error 0.114. The high-frequency component is not hard because its eigenvalue is close to another line; it is hard because heat makes it visible only near $t = 0$.

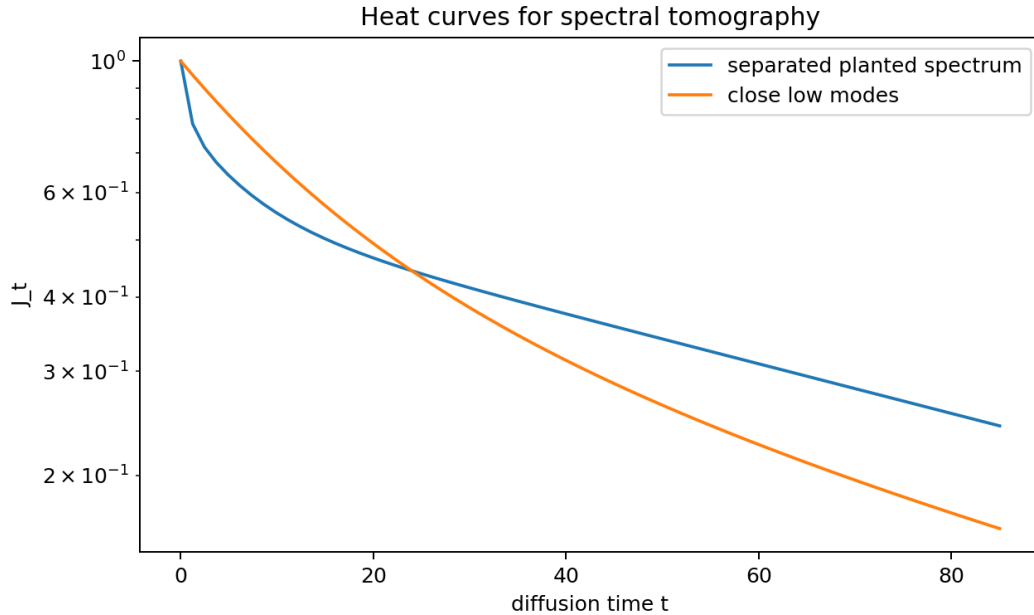


Figure 42: Two heat curves for planted spectral measures on the 64-cycle. Recovering the spectral measure is an inverse Laplace problem; high-frequency contributions are compressed into the earliest times.

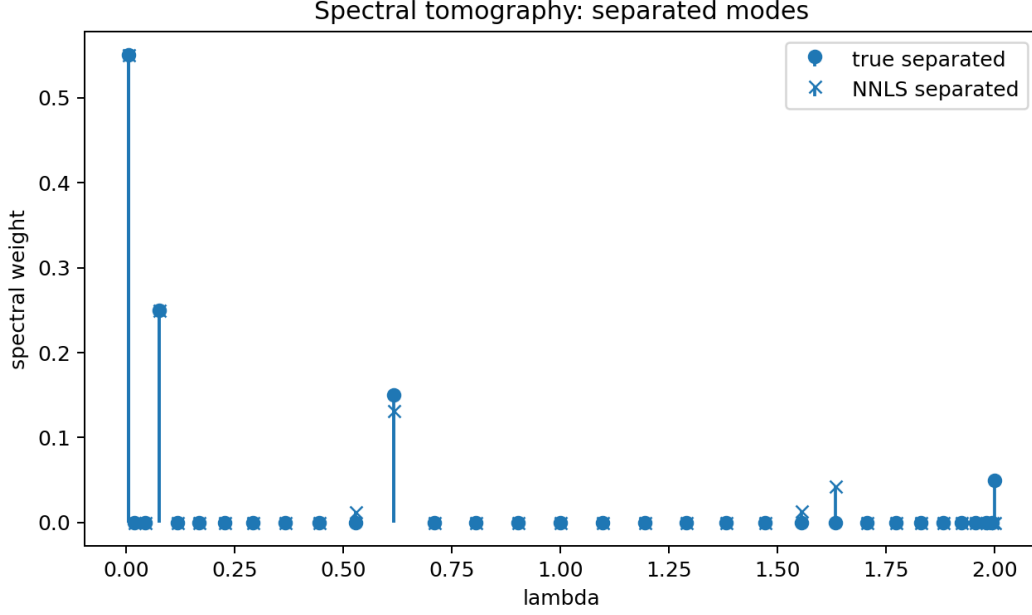


Figure 43: Noisy nonnegative least-squares recovery for a separated spectrum. Even with separated eigenvalues, high-frequency weights are difficult to recover from heat samples after diffusion suppresses them.

I.8 Interpretation and possible applications

Geometry-resolved NRID changes the meaning of robustness. In product systems, robustness is low interaction order. In geometric systems, robustness is alignment with slow diffusion modes. These are the same principle after replacing subset degree by generator eigenvalue.

Potential applications include:

1. *Privacy and anonymization.* Diffusion, smoothing, and random walks can remove local details while preserving attributes aligned with slow graph cuts. Heat NRID can identify which attributes remain recoverable after geometric obfuscation.
2. *Network sensing.* Sensor placement can prioritize sources that excite slow informative modes when long-range degraded observations are expected, or high-frequency modes when local detail matters.
3. *Graph machine learning.* Target labels can be audited by their information spectral measures, distinguishing labels learnable from smooth graph diffusion from labels requiring high-frequency graph filters.
4. *Scientific data compression.* Heat NRID can rank target variables by how much task-relevant information survives a chosen physical diffusion or coarse-graining process.
5. *Infrastructure resilience.* Attributes aligned with bottleneck modes may remain inferable under partial communication or transport failures; high-frequency operational details may vanish first.

I.9 Targets for further study

1. Develop vertex-frequency NRID by replacing eigenspace energy $\|\Pi_\lambda m\|^2$ with localized windowed energies, resolving the orientation blindness demonstrated above.
2. Extend heat NRID from reversible Markov kernels to nonreversible kernels using singular values of the conditional-expectation operator; determine when nonnormal transient structure creates additional observability phenomena.
3. Derive stable bounds on ν_T from finitely many noisy samples of $\mathcal{J}_t$, including confidence intervals for the slowest active eigenvalue $\lambda_*(T)$.
4. Connect bottleneck persistence to Cheeger-type graph cut quantities: characterize when a target attribute has long information half-life because it aligns with a low-conductance partition.
5. Combine geometry-resolved NRID with the phase-gauge hierarchy from the eighth pass, replacing Walsh support cycles by cycles among graph eigenspaces or representation-theoretic modes.
6. Study the continuum limit on manifolds, where $\mathcal{J}_t = \int e^{-2\lambda t} d\nu_T(\lambda)$ becomes an information heat trace over Laplace-Beltrami eigenvalues.

I.10 Research ledger update

The ninth pass upgrades NRID from an order theory on products to a spectral theory on geometries. The central object is no longer only the polynomial $G(z) = \sum_S \omega_S \prod_{i \in S} z_i$, but the information heat measure ν_T whose Laplace transform is $\mathcal{J}_t$. The original Boolean order spectrum is recovered when the geometry is the hypercube and generator eigenvalues equal Walsh degree. The new experimental lesson is that information can be high-order, high-frequency, or both; robustness is governed by the degradation semigroup's eigenstructure, not by scalar mutual information alone.

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Addendum v0.10: Structured Degradation, Walsh Transfer, and Noise-Shielded Synergy

Summary

The preceding passes treated independent coordinate noise, finite-alphabet modal attenuation, reliability fields, posterior moments, and geometry-resolved Markov diffusion. A further blind spot is that real degradation is often structured: a sensor array may have a common polarity error, a network may suffer a correlated outage, or an adversary may apply a constrained perturbation rather than independent bit flips. This addendum studies arbitrary multiplicative sign noise

$$Y = X \odot N, \quad X \in \{-1, +1\}^n,$$

where the noise vector N is independent of X but its coordinates may be strongly dependent.

The main result is an exact transfer theorem: every information hyperedge S is multiplied by the squared Walsh coefficient of the noise law at S . Thus the observed quadratic information is not governed by order alone; it is the overlap of a target-information spectrum and a noise-character spectrum. The independent BSC is only one special case, where all hyperedges of order r are attenuated by ρ^{2r} . Under common-mode noise, even parity information can survive perfectly while single-source information is destroyed. Under subgroup noise, the channel becomes an exact code-defined projector.

1 General structured-noise transfer theorem

Let X be uniform on $\{-1, +1\}^n$. Let N be an independent random vector on $\{-1, +1\}^n$ with law ν , and let

$$Y = X \odot N.$$

For a finite target T , write

$$q_a(x) = P(T = a \mid X = x), \quad p_a = P(T = a), \quad m_a(x) = q_a(x) - p_a.$$

Expand

$$m_a(x) = \sum_{S \subseteq [n]} \hat{m}_a(S) \chi_S(x), \quad \chi_S(x) = \prod_{i \in S} x_i.$$

Define the noise-transfer character

$$\lambda_\nu(S) = \mathbb{E}_\nu[\chi_S(N)]$$

and the target information hyperedge weight

$$\omega_S(T) = \sum_a \frac{\hat{m}_a(S)^2}{p_a}.$$

Theorem 1 (Walsh transfer law for correlated sign noise). *Under the channel $Y = X \odot N$,*

$$\mathcal{J}_\nu(T; Y) = \sum_{\emptyset \neq S \subseteq [n]} \omega_S(T) \lambda_\nu(S)^2.$$

Equivalently, for each target state a ,

$$P(T = a \mid Y = y) - p_a = \sum_{S \neq \emptyset} \hat{m}_a(S) \lambda_\nu(S) \chi_S(y).$$

Proof. Since X is uniform and $Y = X \odot N$, the posterior law of X given $Y = y$ is obtained by drawing $N \sim \nu$ and setting $X = y \odot N$. Hence

$$P(T = a \mid Y = y) - p_a = \mathbb{E}_\nu[m_a(y \odot N)].$$

Using $\chi_S(y \odot N) = \chi_S(y) \chi_S(N)$ gives

$$\mathbb{E}_\nu[m_a(y \odot N)] = \sum_S \hat{m}_a(S) \chi_S(y) \mathbb{E}_\nu[\chi_S(N)] = \sum_S \hat{m}_a(S) \lambda_\nu(S) \chi_S(y).$$

Orthogonality of Walsh characters over uniform Y then gives

$$\mathbb{E}_Y[(P(T = a \mid Y) - p_a)^2] = \sum_{S \neq \emptyset} \hat{m}_a(S)^2 \lambda_\nu(S)^2.$$

Dividing by p_a and summing over a proves the claim. □

This theorem reframes NRID as a spectral overlap:

$$\text{observable information} = \text{target hyperedge energy} \times \text{noise pass power}.$$

Thus “high order is fragile” is not an intrinsic statement about high-order information. It is a statement about independent coordinate degradation.

2 Canonical structured-noise filters

2.1 Independent BSC noise as a low-pass filter

If the coordinates of N are independent with $\mathbb{E}[N_i] = \rho_i$, then

$$\lambda_\nu(S) = \prod_{i \in S} \rho_i,$$

and therefore

$$\mathcal{J}_\nu(T; Y) = \sum_{S \neq \emptyset} \omega_S(T) \prod_{i \in S} \rho_i^2.$$

The earlier NRID law is the isotropic special case $\rho_i = \rho$.

2.2 Common-mode sign noise as a parity filter

Suppose $N_1 = \dots = N_n = Z$, with $\mathbb{E}Z = \rho$. Then

$$\lambda_\nu(S) = \mathbb{E}[Z^{|S|}] = \begin{cases} 1, & |S| \text{ even}, \\ \rho, & |S| \text{ odd}. \end{cases}$$

Consequently, even-order parity hyperedges are perfectly invariant to a global unknown sign flip, while odd-order hyperedges are attenuated only by ρ^2 , not by $\rho^{2|S|}$.

This is a sharp counterpoint to independent-noise fragility. At $\rho = 0.2$, an order-six parity target has

$$\mathcal{J}_{\text{independent}} = \rho^{12} = 4.096 \times 10^{-9}, \quad \mathcal{J}_{\text{common}} = 1.$$

Thus the common-mode channel reveals the same order-six target about 2.44140625×10^8 times more strongly than independent noise with the same single-coordinate reliability.

Table 1: Independent versus common-mode degradation for pure copy and parity targets.

ρ	independent copy	independent pair	independent order 6	common pair	common order 6
0.0	0.000000	0.000000	0.000000e+00	1.000000	1.000000
0.2	0.040000	0.001600	4.096000e-09	1.000000	1.000000
0.5	0.250000	0.062500	2.441406e-04	1.000000	1.000000
0.8	0.640000	0.409600	6.871948e-02	1.000000	1.000000
1.0	1.000000	1.000000	1.000000e+00	1.000000	1.000000

2.3 Exact- k flip noise as a Krawtchouk notch filter

Let N be uniform over sign vectors with exactly k negative coordinates. Then $\lambda_\nu(S)$ depends only on $r = |S|$:

$$\lambda_{n,k}(r) = \frac{1}{\binom{n}{k}} \sum_j (-1)^j \binom{r}{j} \binom{n-r}{k-j}.$$

The numerator is a binary Krawtchouk polynomial. Therefore exact- k noise is not merely weaker or stronger than independent noise; it is a structured order filter with zeros and sign changes.

For $n = 16$, the experiments found, among others,

k	orders r with $\lambda_{16,k}(r) = 0$
1	8
2	6, 10
6	2, 14
8	1, 3, 5, 7, 9, 11, 13, 15.

In particular, flipping exactly half the coordinates kills all odd orders while preserving selected even-order content.

2.4 Subspace noise as an exact code projector

View sign vectors as the additive group $\mathbb{F}_2^n$. Let $H \leq \mathbb{F}_2^n$ be a linear subspace and let N be uniform over H . Then

$$\lambda_\nu(S) = \begin{cases} 1, & S \in H^\perp, \\ 0, & S \notin H^\perp. \end{cases}$$

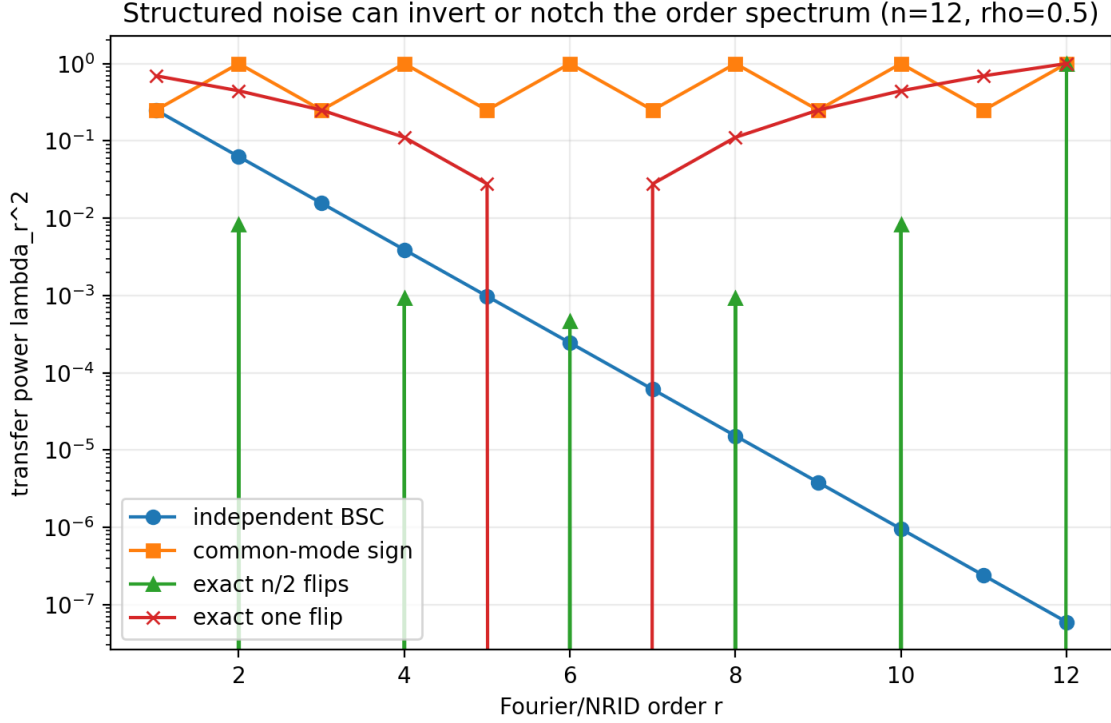


Figure 1: Transfer power under four noise geometries. Independent BSC noise is a monotone low-pass order filter. Common-mode sign noise preserves all even orders. Exact- k noise produces Krawtchouk notches.

Therefore

$$\mathcal{J}_H(T; Y) = \sum_{S \in H^\perp \setminus \{0\}} \omega_S(T).$$

Uniform subgroup noise is an exact information projector: it preserves precisely the hyperedges lying in the dual code and annihilates all others.

This gives a direct code-design interpretation. Choosing a noise code H is equivalent to choosing a pass code $H^\perp$. Task information aligned with $H^\perp$ survives; sensitive or nuisance information outside $H^\perp$ is removed.

3 Exact experiments

3.1 Random correlated-noise verification

The transfer law was tested by exact enumeration with $n = 5$, three target states, random stochastic target maps, and random correlated noise distributions on all 2^5 sign patterns. For six independent random seeds, direct posterior calculation matched the spectral formula with maximum absolute error

$$1.041e - 17$$

and maximum relative error

$$7.764e - 16.$$

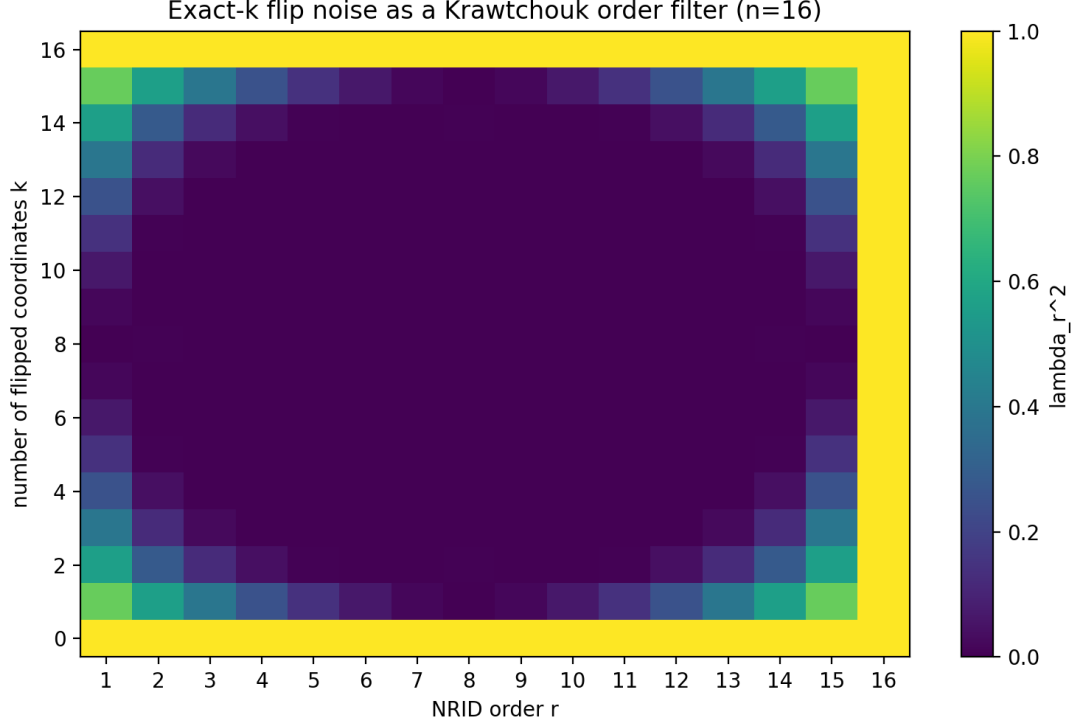


Figure 2: Exact- k flip noise on $n = 16$ sources. The pass power $\lambda_{n,k}(r)^2$ forms a Krawtchouk filter bank over NRID order.

The tested noise laws had both positive and negative Walsh coefficients, so the experiment was not restricted to monotone or product noise.

3.2 Hamming-code noise projector

A concrete subspace-noise test used uniform noise over the $[7, 4, 3]$ Hamming code. Its dual is the $[7, 3, 4]$ simplex code. The transfer theorem predicts that the channel passes exactly the seven nonzero simplex codewords, all of weight four, and blocks the other 120 nonzero Walsh supports.

The experiment constructed a valid stochastic binary target whose posterior score was the sum of one singleton mode, one pair mode, and one passed simplex mode, each with coefficient 0.28. Before degradation,

$$\mathcal{J}_0 = 0.235200.$$

After Hamming-code noise, both direct enumeration and the spectral projector law gave

$$\mathcal{J}_H = 0.078400,$$

with absolute error

$$0.0e + 00.$$

Only the order-four simplex component survived.

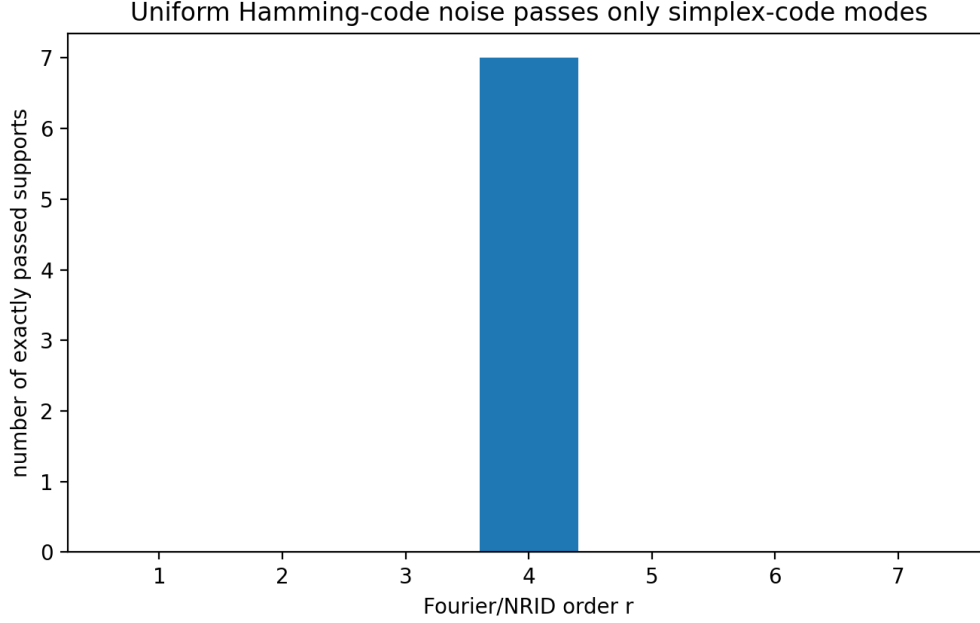


Figure 3: Uniform $[7, 4, 3]$ Hamming-code noise passes exactly the seven nonzero modes of the dual simplex code, all at order four.

4 New concepts suggested by the pass

Noise-transfer spectrum. The law

$$\mathcal{J}_\nu = \sum_S \omega_S \lambda_\nu(S)^2$$

suggests that a degradation channel has its own spectrum of observability. Independent BSC noise has a radially decreasing spectrum. Common-mode noise has a parity spectrum. Exact- k noise has a Krawtchouk spectrum. Subspace noise has a code-projector spectrum.

Noise-shielded synergy. High-order information can be fragile under independent noise but perfectly protected under a structured noise symmetry. Even parity is shielded against global sign flips. Code-aligned parity checks are shielded against noise from the primal code. Thus high-order information can be either fragile or robust depending on whether its support lies in the degradation passband.

Information resonance and notching. A target resonates with a noise law when its information hyperedges lie where $|\lambda_\nu(S)|$ is large. A target is notched when its hyperedges lie where $\lambda_\nu(S) = 0$. Exact- k noise supplies order notches; subgroup noise supplies algebraic support notches.

Duality between privacy and utility. For a useful target T and a sensitive target U , a structured-noise design problem can be written as

$$\max_{\nu \in \mathcal{V}} \sum_S (\omega_S(T) - \beta \omega_S(U)) \lambda_\nu(S)^2,$$

possibly with distortion or implementation constraints on ν . For subgroup noise, this becomes a code-selection problem:

$$\max_{L \subseteq \mathbb{F}_2^n} \sum_{S \in L} (\omega_S(T) - \beta \omega_S(U)), \quad L = H^\perp.$$

This is a possible bridge from NRID to privacy-preserving release, error correction, sensor masking, and robust feature design.

5 Applications

1. **Sensor arrays and calibration.** If all sensors may undergo a common polarity flip, then single-sensor readings are unreliable but even relational features are invariant. NRID can identify which learned features lie in the common-mode passband.
2. **Privacy-preserving data release.** Subspace noise can destroy an entire family of forbidden low-order or nuisance features while exactly preserving allowed parity/check features. This suggests code-based utility-preserving obfuscation.
3. **Fault-tolerant communication.** Targets encoded as dual-code invariants survive primal-code perturbations. The Hamming/simplex example is a minimal demonstration of an information projector: 120 nonzero modes are removed and seven selected order-four modes remain.
4. **Robust machine learning diagnostics.** Structured perturbation tests can reveal whether a classifier depends on low-pass coordinate evidence, parity-shielded relational evidence, or code-aligned invariants.
5. **Cryptography and secret sharing.** The transfer law gives a direct way to analyze which Walsh components leak through a permitted noise or masking distribution. It may be useful for constructing functions that are hidden under some perturbations but recoverable under authorized structured observations.

6 Targets for further study

1. Characterize the feasible set of transfer spectra $\{\lambda_\nu(S)\}_S$. Since λ_ν is the Walsh transform of a probability law, it must obey positive-definiteness and inverse-nonnegativity constraints.
2. Develop convex and combinatorial optimization methods for structured-noise privacy/utility design:

preserve task NRID while suppressing sensitive NRID.

3. Extend subgroup-noise projectors from Boolean groups to finite Abelian groups using characters, and then to non-Abelian symmetry groups using irreducible representations.
4. Combine the structured-noise transfer law with the posterior moment hierarchy. Quadratic response sees $|\lambda_\nu(S)|^2$; higher moments should reveal signed closure interactions among noise-filtered hyperedges.
5. Study empirical estimation. Given samples from a real degradation process, estimate $\lambda_\nu(S)$ for candidate hyperedges and predict the loss or preservation of task information before measuring full mutual information.

6. Investigate whether exact- k Krawtchouk notches can be used as experimental probes: deliberately impose fixed-weight perturbations to isolate or cancel specific NRID orders.
7. Generalize to correlated source distributions. The Walsh transfer law uses uniform independent X ; correlated sources require replacing characters with an appropriate harmonic or conditional-orthogonal basis.

7 Conclusion

The tenth pass changes the interpretation of NRID robustness. Interaction order alone does not determine survival. Survival is determined by the alignment between the target’s information hyperedges and the degradation channel’s Walsh transfer spectrum. Independent noise makes high-order information exponentially fragile. Common-mode noise can make even high-order parity perfectly robust. Exact- k noise creates Krawtchouk order notches. Subspace noise creates code-defined information projectors. The emerging principle is:

robust information is target spectrum matched to noise spectrum.

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11 Eleventh pass: context-assisted order collapse and supermodular acquisition

11.1 Blind spot

The earlier sections treat information order as a property of the target relative to a source-degradation channel. This pass isolates a further operational variable: *side information*. Exact context can turn high-order information into low-order information without changing the target itself. Consequently, interaction order is not only a property of a target; it is a property of a target together with an information state.

The central observation is simple but powerful. In the NRID reliability polynomial, exact knowledge of a source coordinate is equivalent to setting its squared reliability to one. Unknown noisy coordinates retain squared reliability $z = \rho^2$. Thus context does not merely add information; it changes the order of the remaining inference problem. This creates increasing returns: the more variables from a hidden hyperedge are already known, the more valuable the next variable becomes.

11.2 Context transform theorem

Let $X \in \{-1, +1\}^n$ be uniform and let T be a finite target generated from X . Let $C \subseteq [n]$ be a set of coordinates observed exactly. Coordinates outside C are observed through independent BSC reliability ρ . Denote the combined observation by

$$O_{C,\rho} = (X_C, Y_{[n]\setminus C}).$$

Let ω_S be the subset-level NRID coefficient from the original decomposition,

$$\omega_S = \sum_t \frac{\hat{m}_t(S)^2}{p_t}, \quad S \neq \emptyset.$$

Theorem 1 (Exact context transform). *For prior-relative posterior chi-square information,*

$$\boxed{\mathcal{J}(T; O_{C,\rho}) = \sum_{S \neq \emptyset} \omega_S \rho^{2|S \setminus C|}.$$

Equivalently, context substitutes $z_i = 1$ for known coordinates and $z_i = \rho^2$ for noisy coordinates in the multi-affine NRID reliability polynomial

$$G(z_1, \dots, z_n) = \sum_{S \neq \emptyset} \omega_S \prod_{i \in S} z_i.$$

Proof. For every target state t , the centered likelihood $m_t(X)$ has Walsh expansion

$$m_t(X) = \sum_S \hat{m}_t(S) \chi_S(X).$$

Taking conditional expectation given $O_{C,\rho}$ leaves characters on known coordinates untouched and attenuates characters on unknown noisy coordinates by ρ :

$$\mathbb{E}[\chi_S(X) \mid O_{C,\rho}] = \chi_{S \cap C}(X_C) \rho^{|S \setminus C|} \chi_{S \setminus C}(Y).$$

Orthogonality over the joint observation distribution then gives

$$\mathbb{E} \left[(\mathbb{E}[m_t(X) \mid O_{C,\rho}])^2 \right] = \sum_S \hat{m}_t(S)^2 \rho^{2|S \setminus C|}.$$

Dividing by p_t , summing over t , and dropping the empty centered mode proves the claim. $\square$

This theorem is a context-assisted version of the original NRID identity. It has a different interpretation: a hyperedge S does not disappear if some of its variables are known; its remaining order is reduced from $|S|$ to $|S \setminus C|$. If all of S is known, that hyperedge contributes at order zero because the context itself already exposes that component of the target.

11.3 Pure parity order collapse

For a pure d -way parity target, the only nonzero coefficient is $\omega_S = 1$ with $|S| = d$. If h participating variables are known exactly, then

$$\mathcal{J}_h = \rho^{2(d-h)} \quad (h < d),$$

while full revelation of the participating set gives $\mathcal{J}_d = 1$. Hence every revealed participating bit reduces the information exponent by two. The context amplification relative to no context is

$$\frac{\mathcal{J}_h}{\mathcal{J}_0} = \rho^{-2h}.$$

For $d = 10$ and $\rho = 0.30$, exact context produces the following values.

Known participating bits h	$\mathcal{J}_h$	amplification over $h = 0$
0	3.48678×10^{-11}	1
2	4.30467×10^{-9}	1.23457×10^2
5	5.90490×10^{-6}	1.69351×10^5
8	8.10000×10^{-3}	2.32306×10^8
9	9.00000×10^{-2}	2.58117×10^9
10	1.00000	2.86797×10^{10}

Table 1: A ten-way parity target at reliability $\rho = .30$ is nearly invisible with no context, but exact context collapses the remaining inference order.

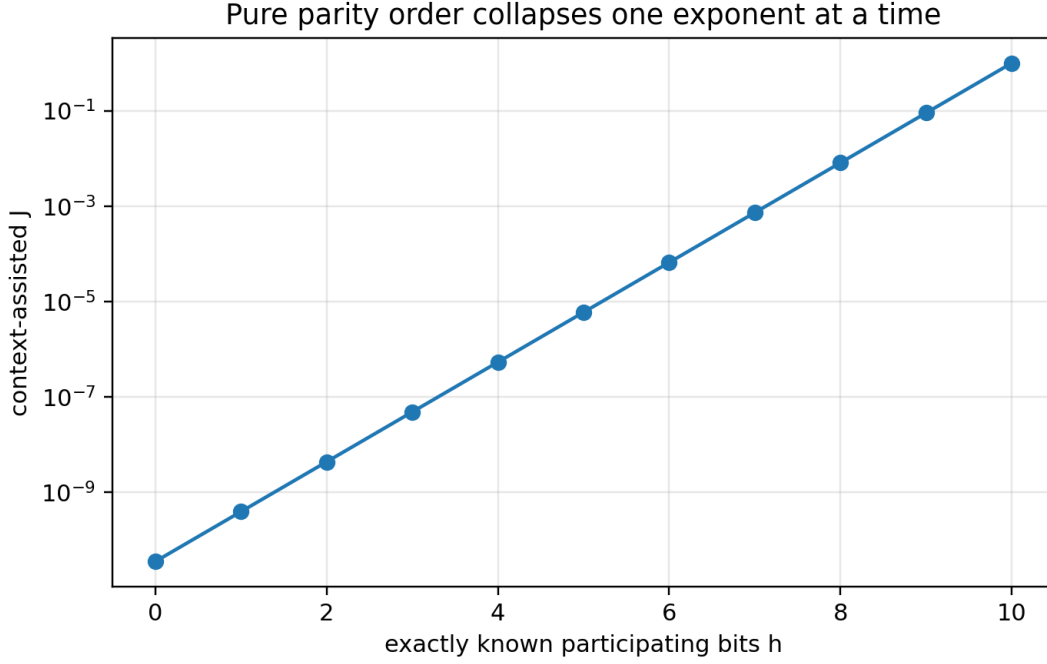


Figure 1: Each exact participating bit known in advance reduces the parity exponent by one NRID order. The vertical scale is logarithmic.

11.4 Multiplexer address context law

Consider a Boolean multiplexer with k address bits and 2^k data bits. The target is the addressed data bit. Without address context, the earlier multiplexer law gives

$$\mathcal{J}_0(\rho) = \rho^2 \left(\frac{1 + \rho^2}{2} \right)^k.$$

If h address bits are known exactly, the remaining unknown address dimension is $k - h$, and the same derivation gives

$$\boxed{\mathcal{J}_h(\rho) = \rho^2 \left(\frac{1 + \rho^2}{2} \right)^{k-h} .}$$

Thus address context does not reveal the data value by itself, but it hardens the data inference channel by shrinking the unresolved selector space.

For $k = 8$ and $\rho = 0.40$, full address context raises the posterior chi-square information from 0.002049 to 0.160000, a factor of 78.09.

Known address bits h	$\mathcal{J}_h$	amplification over $h = 0$
0	0.002049	1.000
2	0.006091	2.973
4	0.018106	8.837
6	0.053824	26.268
8	0.160000	78.087

Table 2: Address information is not target information by itself, but it converts a diffuse high-order selector problem into a more robust low-order data problem.

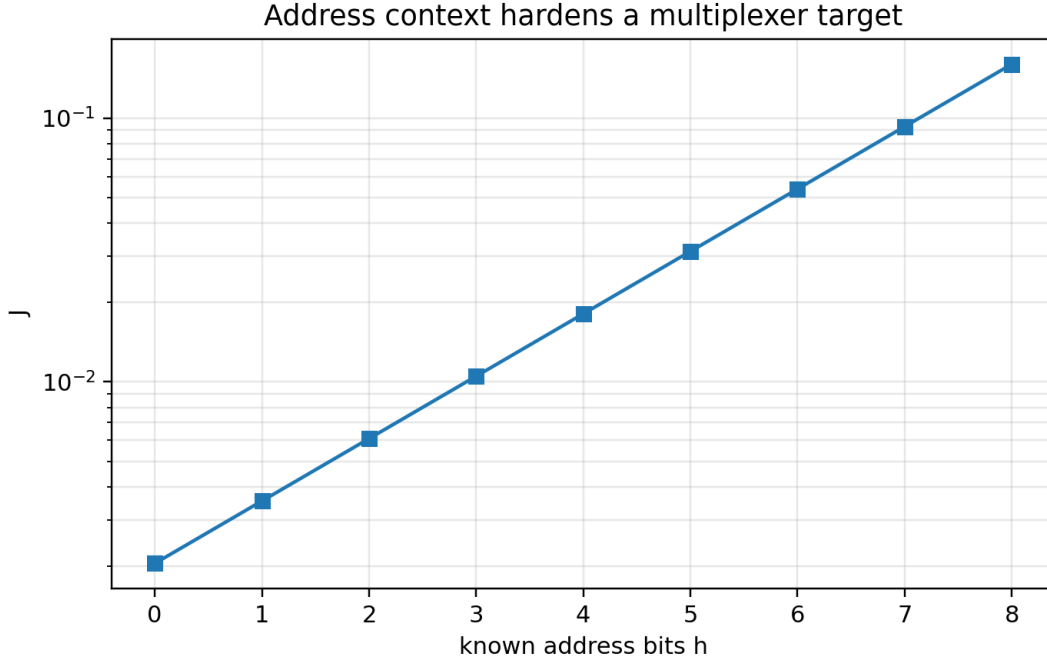


Figure 2: Context-assisted multiplexer information as address bits become known.

11.5 Marginal context gain and supermodularity

Let

$$F_z(C) = \mathcal{J}(T; O_{C,\rho}) = \sum_{S \neq \emptyset} \omega_S z^{|S \setminus C|}, \quad z = \rho^2.$$

For $i \notin C$, the marginal value of revealing coordinate i is

$$\Delta_i(C) = F_z(C \cup \{i\}) - F_z(C) = (1 - z) \sum_{S \ni i} \omega_S z^{|S \setminus C| - 1}.$$

Proposition 1 (Context acquisition is supermodular). *For $0 < z < 1$ and nonnegative NRID coefficients, $F_z(C)$ is monotone and supermodular:*

$$C \subseteq D, \quad i \notin D \quad \implies \quad \Delta_i(C) \leq \Delta_i(D).$$

Proof. It is enough to check each hyperedge monomial. If $i \notin S$, the marginal is zero. If $i \in S$, then

$$\Delta_{i,S}(C) = (1 - z)\omega_S z^{|S \setminus C| - 1}.$$

When $C \subseteq D$, $|S \setminus D| \leq |S \setminus C|$. Since $0 < z < 1$, the marginal under D is at least the marginal under C . Summing over S gives the result. $\square$

This is a sharp contrast with many classical information-gathering objectives where diminishing returns make greedy selection effective. In NRID context acquisition, high-order information creates increasing returns: the next member of a nearly completed hyperedge can be far more valuable than the first member.

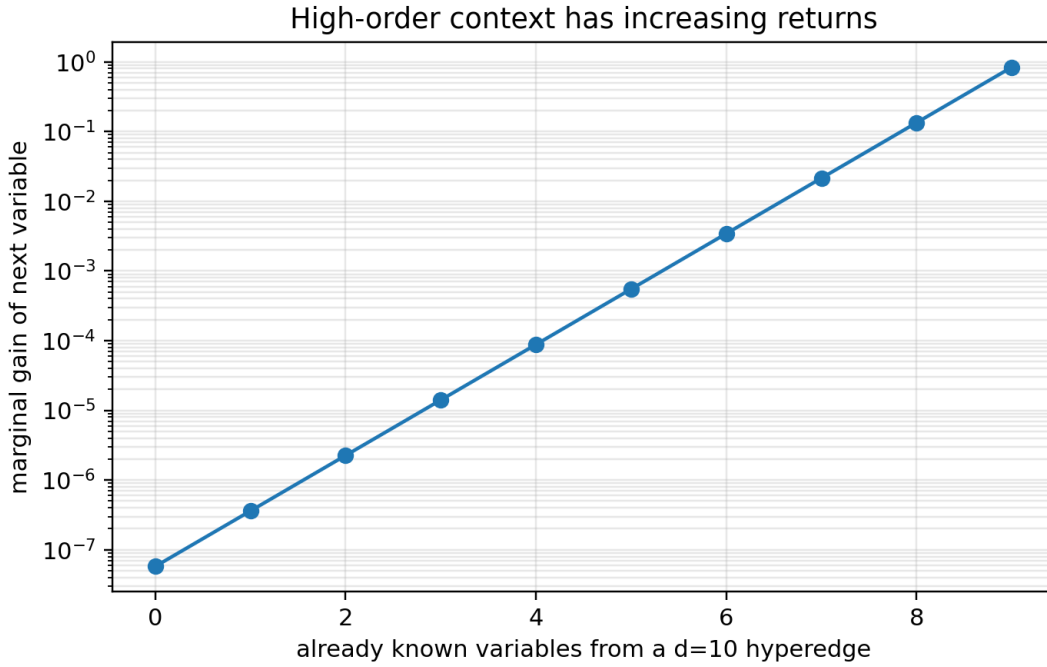


Figure 3: For a pure ten-way hyperedge with $z = .16$, the marginal gain of one more participating coordinate grows exponentially as the hyperedge becomes more complete.

11.6 Greedy blindness theorem

The supermodularity is not merely qualitative. It creates an explicit failure mode for one-step greedy context acquisition.

Theorem 2 (Exponential greedy blindness). *Fix $0 < z < 1$, $d \geq 2$, and $\epsilon > 0$. Construct one unit-weight d -way hyperedge H and d singleton distractor hyperedges, each of weight*

$$w = (1 + \epsilon)z^{d-1}.$$

With budget d , a one-step greedy policy selects all singleton distractors, while the optimal policy selects the d variables in H . The ratio of optimal gain to greedy gain is

$$\frac{1 - z^d}{d(1 - z)(1 + \epsilon)z^{d-1}}.$$

This ratio diverges exponentially in d for fixed $z < 1$.

Proof. At the empty set, each variable in H has marginal gain $(1 - z)z^{d-1}$, while each singleton distractor has marginal gain $(1 - z)w = (1 + \epsilon)(1 - z)z^{d-1}$. Hence greedy selects a distractor. Until greedy spends the full budget on distractors, the marginal value of the untouched d -way edge remains below the distractor marginal. Greedy gain is $d(1 - z)w$. The optimal context reveals all variables in H , gaining $1 - z^d$. Dividing gives the stated ratio. $\square$

For $z = .16$ and $\epsilon = .10$, the ratio already exceeds 5.0×10^4 at $d = 8$ and 5.1×10^7 at $d = 12$.

d	distractor weight	greedy gain	optimal/greedy
3	2.81600×10^{-2}	7.09632×10^{-2}	1.40341×10^1
5	7.20896×10^{-4}	3.02776×10^{-3}	3.30242×10^2
8	2.95279×10^{-6}	1.98427×10^{-5}	5.03962×10^4
10	7.55914×10^{-8}	6.34968×10^{-7}	1.57488×10^6
12	1.93514×10^{-9}	1.95062×10^{-8}	5.12657×10^7

Table 3: A constructed family where one-step greedy acquisition is exponentially blind to high-order information.

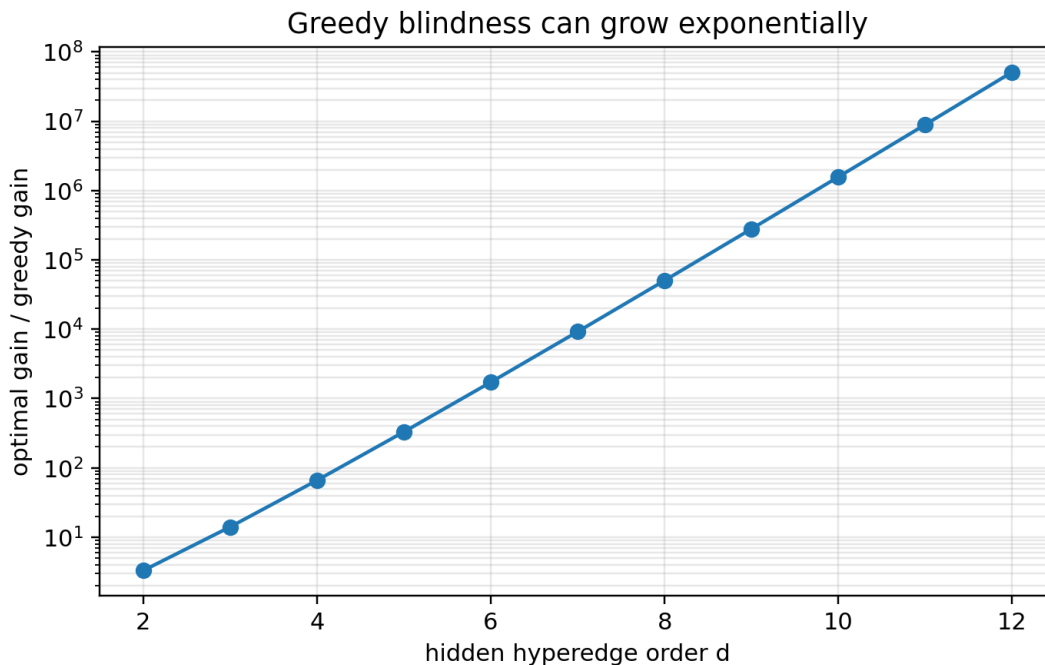


Figure 4: Optimal gain divided by greedy gain in the constructed blindness example.

11.7 Exact-enumeration verification

The context transform theorem was tested on random stochastic targets with $n = 5$ fair Boolean sources and three target states. For each state t , $P(T = t \mid X = x)$ was sampled from a Dirichlet

distribution. Direct enumeration of the joint distribution of $(T, O_{C,\rho})$ was compared with

$$\sum_{S \neq \emptyset} \omega_S \rho^{2|S \setminus C|}.$$

Across four context masks and five reliabilities, the maximum absolute discrepancy was

$$1.11 \times 10^{-16}.$$

This is numerical roundoff, confirming the exact transform for stochastic targets as well as deterministic ones.

11.8 Applications suggested by this pass

Active experimental design. Single-factor screening can miss high-order effects because the first member of a synergistic bundle may have tiny marginal value. Context-assisted NRID suggests designing group probes, beam search, or mixed-derivative tests aimed at hyperedge completion rather than one-step gain.

Privacy and side-information risk. A secret may be high-order and nearly invisible under independent degradation, yet become low-order once auxiliary context leaks. For a pure d -way secret, h leaked participating bits multiply chi-square visibility by ρ^{-2h} . Side-information auditing should therefore measure order collapse, not only direct leakage.

Sensor fusion and diagnostic protocols. Sensor networks designed for high-order events should coordinate acquisition over whole bundles. A source with small standalone value can become decisive after companion sources are known or trusted.

Machine-learning interpretation. Feature-importance methods based on isolated marginal gain can systematically understate features that participate in high-order rules. The supermodular context law gives a clean mathematical reason for this failure.

11.9 Targets for further study

1. Develop algorithms for maximizing $F_z(C)$ under budgets, costs, and matroid or routing constraints. The greedy blindness theorem suggests that sparse hypergraph recovery, branch-and-bound, or integer programming may be necessary in high-order regimes.
2. Generalize exact context to noisy context by setting known coordinates to intermediate reliabilities rather than one, linking this pass back to reliability-field thermodynamics.
3. Build adaptive group-testing protocols that identify which variables belong to the same information hyperedge before spending full observation budget.
4. Extend the supermodularity analysis from chi-square NRID to Shannon information, Bayes utility, and higher posterior-moment NRID.
5. Study side-information privacy: quantify how much auxiliary context collapses the order of protected information in databases, sensor logs, model embeddings, and communication protocols.

11.10 Research ledger update

This pass adds the principle that information order is context-relative. Exact side information substitutes reliability one into the NRID polynomial and reduces the remaining order of every hyperedge it touches. This produces supermodular acquisition, increasing returns, and an explicit exponential failure mode for one-step greedy selection. The resulting design rule is: high-order information should be acquired and protected as bundles, not as isolated coordinates.

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Addendum v0.12: Source-Dependence Heterodyning and Spectral Aliasing in NRID

Draft of May 7, 2026

Abstract

Earlier NRID identities assume independent uniform source coordinates. This addendum studies what breaks, and what becomes newly visible, when the source distribution itself contains high-order dependence. The main result is an exact posterior-mean convolution law: a source-density Walsh mode A mixes with a target Walsh mode B into the observed mode $A \oplus B$. Thus source correlations can act as a heterodyne oscillator, shifting high-order target information into lower observed order. A one-tilt source model gives a closed-form chi-square response curve and an order-collapse law. Exact enumeration verifies the formula to numerical precision. A random-spectrum calculation gives a spectral-crowding law: if many dependence modes are present, low-order aliases of a fixed target mode become likely at a Hamming-ball threshold. The addendum motivates dependence-corrected, interventional, and privacy-oriented extensions of NRID.

1 Blind spot: source dependence as an information lens

The base NRID theorem uses independent uniform Boolean sources. Under that assumption the independent BSC noise operator is diagonal in the Walsh basis, and a target coefficient supported on $B \subseteq [n]$ is attenuated by $\rho^{|B|}$. This makes the observable chi-square information scale as $\rho^{2|B|}$ for a pure parity target.

However, real source coordinates are often dependent. In a dependent source, a high-order target mode may be statistically coupled to a different high-order source-density mode. Under noisy observation these modes can mix, producing lower-order posterior evidence. This creates a distinction between two notions of order:

- *interventional order*: the order measured after source randomization or whitening;
- *observational order*: the order measured under the natural source distribution.

The new phenomenon is that observational order can be much smaller than interventional order.

2 Source-density Walsh expansion

Let $X \in \{-1, +1\}^n$. Write Walsh characters as

$$\chi_A(x) = \prod_{i \in A} x_i, \quad A \subseteq [n].$$

Assume the source law is absolutely continuous with respect to the uniform law, and write its density ratio as

$$w(x) = 2^n P_X(x) = \sum_{A \subseteq [n]} \theta_A \chi_A(x), \quad \theta_\emptyset = 1, \quad w(x) \geq 0.$$

The set

$$\Theta = \{A : \theta_A \neq 0\}$$

is the Walsh support of the source-density field. When sources are independent uniform, $\Theta = \{\emptyset\}$.

Let Y be generated from X by an independent BSC reliability ρ in every coordinate:

$$\Pr(Y_i = X_i) = \frac{1+\rho}{2}, \quad \Pr(Y_i = -X_i) = \frac{1-\rho}{2}.$$

3 The heterodyne convolution theorem

Theorem 1 (Source-density heterodyne law for parity targets). *Let $T_B = \chi_B(X)$ be a parity target. For every $y \in \{-1, +1\}^n$,*

$$\mathbb{E}[T_B \mid Y = y] = \frac{\sum_A \theta_A \rho^{|A \oplus B|} \chi_{A \oplus B}(y)}{\sum_A \theta_A \rho^{|A|} \chi_A(y)}.$$

Here $A \oplus B$ denotes symmetric difference, equivalently addition in $\mathbb{F}_2^n$.

Proof. Bayes' rule gives

$$\mathbb{E}[T_B \mid Y = y] = \frac{\mathbb{E}_U[\chi_B(X) w(X) P(Y = y \mid X)]}{\mathbb{E}_U[w(X) P(Y = y \mid X)]},$$

where U is the uniform law. Up to a y -independent constant,

$$P(Y = y \mid X = x) \propto \prod_{i=1}^n (1 + \rho x_i y_i).$$

The Walsh identity

$$\mathbb{E}_U \left[\chi_C(X) \prod_i (1 + \rho X_i y_i) \right] = \rho^{|C|} \chi_C(y)$$

gives the denominator by setting $C = A$ and the numerator by setting $C = A \oplus B$. $\square$

For a general real-valued target score

$$g(x) = \sum_B \hat{g}_B \chi_B(x),$$

the same calculation yields

$$\mathbb{E}[g(X) \mid Y = y] = \frac{\sum_{A,B} \theta_A \hat{g}_B \rho^{|A \oplus B|} \chi_{A \oplus B}(y)}{\sum_A \theta_A \rho^{|A|} \chi_A(y)}.$$

Thus source-density modes and target modes combine by XOR. This is the basis for calling the effect *information heterodyning*: a high-order target frequency can be mixed with a nearby high-order source-density frequency into a low-order observable frequency.

4 Observed order as spectral Hamming distance

At low reliability, the denominator is $1 + O(\rho)$ unless the source law is singular. Ignoring exact cancellations, the first posterior-amplitude order for a target score with Walsh support $\Gamma = \{B : \hat{g}_B \neq 0\}$ is

$$d_{\text{het}} = \min_{A \in \Theta, B \in \Gamma} |A \oplus B|.$$

The corresponding chi-square information begins at order $\rho^{2d_{\text{het}}}$. Under independent uniform sources, $\Theta = \{\emptyset\}$, so this reduces to the usual Fourier degree. Under dependent sources, d_{het} can be much smaller.

This suggests a revised fragility principle:

Observed fragility measures the Hamming distance from the target spectrum to the source-density spectrum, not merely the target's intrinsic Fourier order.

5 Exact one-tilt response law

A particularly transparent source family is the single-mode tilted law

$$P_X(x) = 2^{-n} (1 + \theta \chi_A(x)), \quad |\theta| < 1.$$

Assume $A \neq B$ so that $T_B = \chi_B(X)$ is balanced under P_X . Let

$$a = |A|, \quad b = |B|, \quad c = |A \oplus B|.$$

For a balanced binary target the posterior chi-square information is

$$\mathcal{J}_\rho = \mathbb{E}[\mathbb{E}[T_B | Y]^2].$$

Proposition 1 (Closed-form one-tilt law). *For $P_X(x) = 2^{-n}(1 + \theta \chi_A(x))$ with $A \neq B$,*

$$\mathcal{J}_\rho(A, B, \theta) = \frac{\rho^{2b} + \theta^2 \rho^{2c} - 2\theta^2 \rho^{a+b+c}}{1 - \theta^2 \rho^{2a}}.$$

Proof. The heterodyne law gives

$$\mathbb{E}[T_B | Y = y] = \frac{u(y) + \theta v(y)}{1 + \theta w(y)},$$

where

$$u = \rho^b \chi_B(y), \quad v = \rho^c \chi_{A \oplus B}(y), \quad w = \rho^a \chi_A(y).$$

The observation law is $P_Y(y) = 2^{-n}(1 + \theta w(y))$. Hence

$$\mathcal{J}_\rho = \mathbb{E}_U \left[\frac{(u + \theta v)^2}{1 + \theta w} \right].$$

Because $uv = \rho^{b+c-a}w$ and $w = \rho^a \chi_A$,

$$\mathbb{E}_U \frac{1}{1 + \theta w} = \frac{1}{1 - \theta^2 \rho^{2a}}, \quad \mathbb{E}_U \frac{w}{1 + \theta w} = \frac{-\theta \rho^{2a}}{1 - \theta^2 \rho^{2a}}.$$

Substitution gives the stated formula. □

Corollary 1 (Dependence-induced order collapse). *If $\theta \neq 0$ and $A \neq B$, then, as $\rho \rightarrow 0$,*

$$\mathcal{J}_\rho(A, B, \theta) \asymp \begin{cases} \rho^{2b}, & b < c, \\ (1 + \theta^2)\rho^{2b}, & b = c, \\ \theta^2\rho^{2c}, & c < b. \end{cases}$$

Thus a b -way parity target behaves observationally as a c -way target whenever the source density contains a mode A at Hamming distance $c < b$ from B .

6 Exact experiments

All experiments used exact enumeration or closed-form sums over the Boolean cube; no Monte Carlo was used for the one-tilt identity verification. Random exact tests with $n = 5, 6, 7, 8$ compared the closed-form one-tilt law to direct enumeration of the full joint law of (X, Y, T) . The maximum absolute error was

$$1.11 \times 10^{-16}.$$

Table 1: Representative exact-enumeration checks for the one-tilt law.

n	$ A $	$ B $	$ A \oplus B $	θ	ρ	absolute error
5	5	2	3	0.7195	0.2998	1.73×10^{-18}
6	3	5	2	-0.7478	0.6543	1.39×10^{-17}
7	5	4	1	0.7175	0.2089	1.04×10^{-17}
8	4	6	4	0.6275	0.9002	0

6.1 A 10-way target becomes observationally first-order

For $n = 12$, take B to be a 10-coordinate parity target. Under independent sources,

$$\mathcal{J}_\rho = \rho^{20}.$$

Now choose a source-density mode A obtained by removing one coordinate from B , so $|A| = 9$ and $|A \oplus B| = 1$, with $\theta = 0.35$. The closed form predicts leading scaling

$$\mathcal{J}_\rho \sim 0.35^2 \rho^2.$$

The fitted log-log slopes were 20.0000 under independence and 2.0000 under the tilted source. At $\rho = 0.1$,

$$\mathcal{J}_{\text{independent}} = 10^{-20}, \quad \mathcal{J}_{\text{tilted}} = 1.225 \times 10^{-3},$$

a visibility ratio of approximately

$$1.225 \times 10^{17}.$$

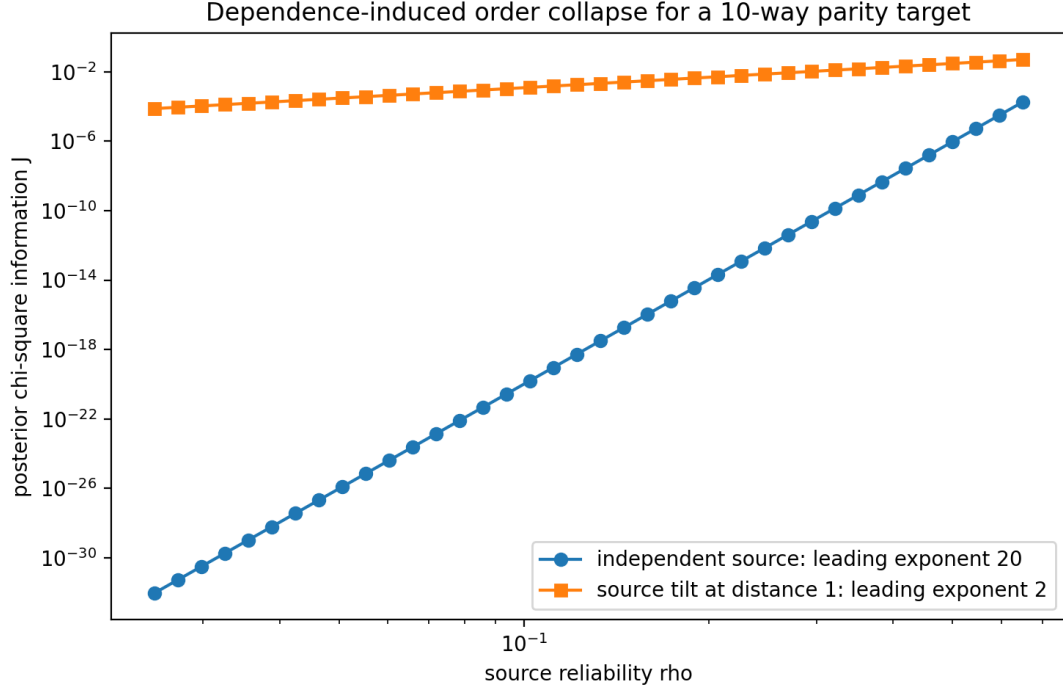


Figure 1: A high-order target can appear low-order under a correlated source distribution. The target is a 10-way parity. A ninth-order source-density mode at Hamming distance one from the target mode shifts the leading chi-square exponent from 20 to 2.

6.2 Nearest source-spectrum mode controls the observed order

A source with several weak density modes at Hamming distances 6, 3, and 1 from a 10-way target was tested. The low-reliability slope was 2.0000189, matching twice the nearest distance. This supports the spectral-distance law beyond a single tilt.

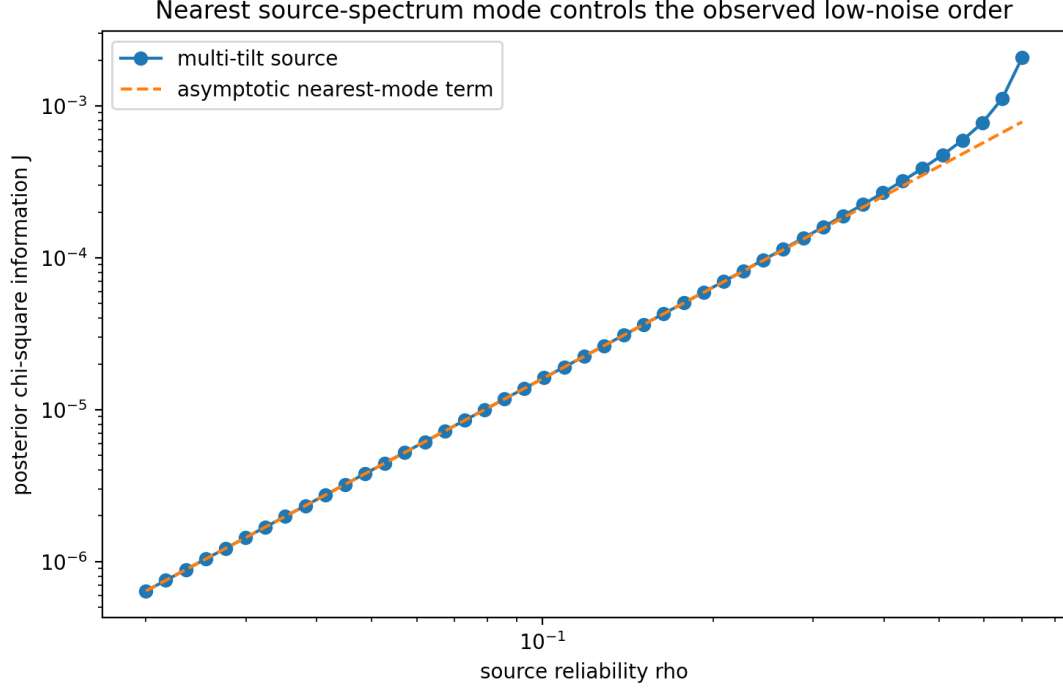


Figure 2: For multiple weak source-density modes, the nearest mode in symmetric-difference distance controls the low-reliability response.

6.3 Hamming heterodyne map

For a target mode of order 10 in $n = 16$, the alias order created by a source-density mode A is

$$|A \oplus B| = |A| + |B| - 2|A \cap B|.$$

Thus the source mode need not be low-order; it must be geometrically close to the target support.

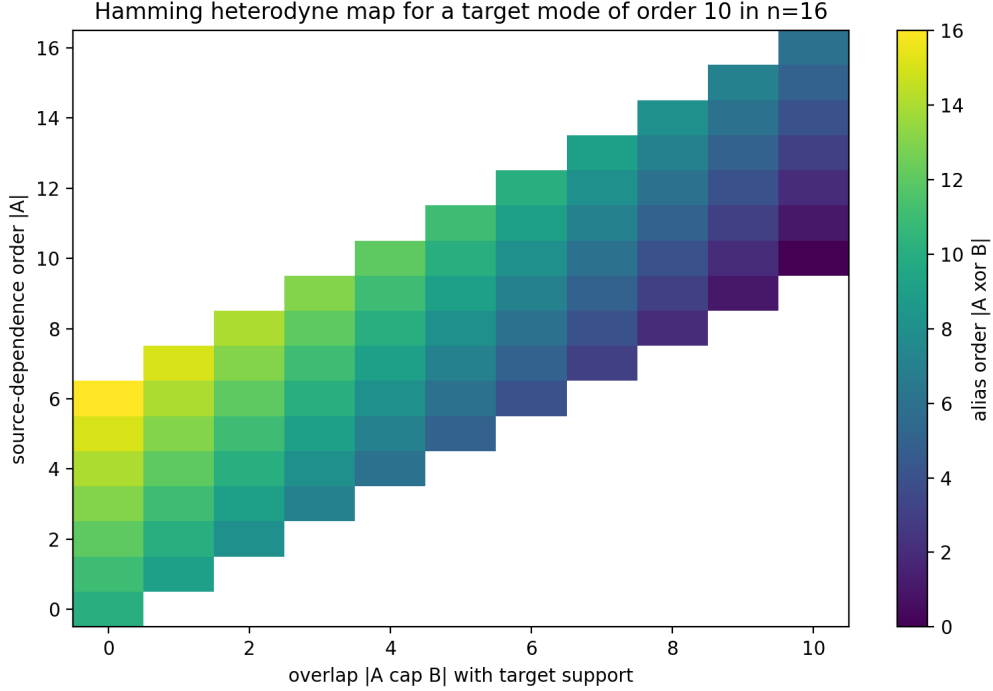


Figure 3: Alias order is symmetric-difference distance, not source order alone. A high-order source correlation almost aligned with the target can create a low-order observable.

7 Spectral crowding law

The heterodyne formula suggests a probabilistic question: if the source-density spectrum is rich, how likely is it that a fixed target mode has a low-order alias?

Proposition 2 (Random source-spectrum crowding). *Fix a target support $B \in \mathbb{F}_2^n$. Let $A_1, \dots, A_m$ be independent uniformly random source-density supports in $\mathbb{F}_2^n$. Define*

$$D_m = \min_{1 \leq j \leq m} |A_j \oplus B|.$$

Then

$$\mathbb{P}(D_m \leq r) = 1 - \left(1 - \frac{V(n, r)}{2^n}\right)^m, \quad V(n, r) = \sum_{j=0}^r \binom{n}{j}.$$

Proof. For each random A_j , the event $|A_j \oplus B| \leq r$ is membership in a Hamming ball of radius r centered at B , which has volume $V(n, r)$. Independence gives the complement probability. $\square$

For $n = 20$, the approximate number of random source modes required for a 50% chance of an alias within radius r is:

r	1	2	3	4
$m_{1/2}$	34610.0	3444.3	537.6	117.0

This gives a covering-threshold view of dependence-induced leakage: as the source-density spectrum becomes dense, a high-order target is increasingly likely to have a low-order observational shadow.

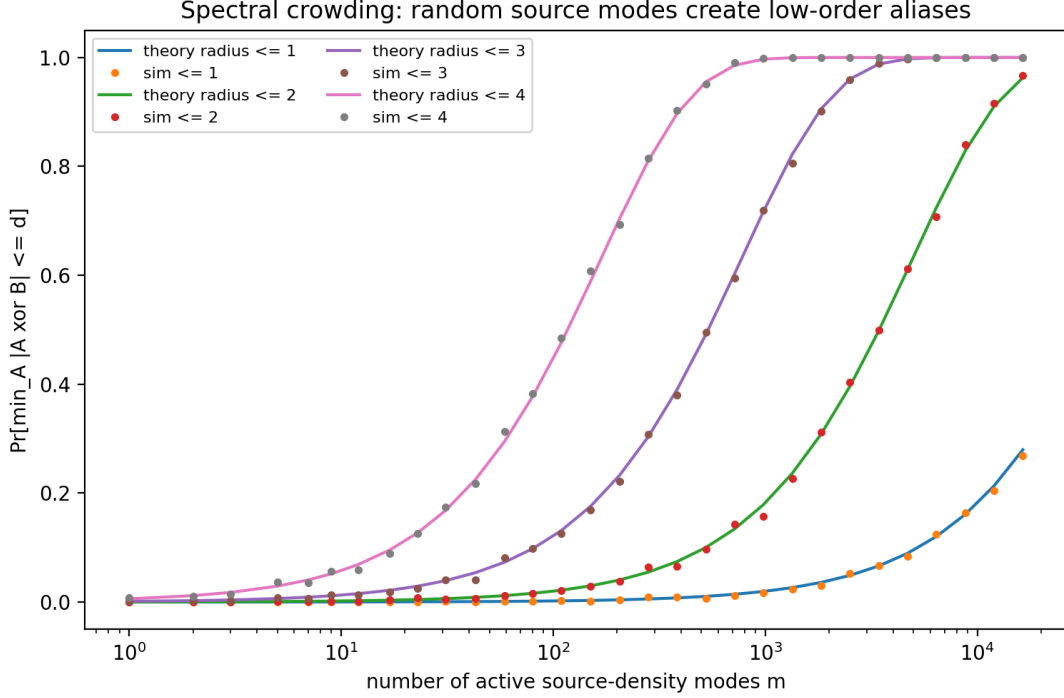


Figure 4: Simulation agrees with the Hamming-ball crowding law. Dense source-dependence spectra make low-order aliases likely even for a fixed target mode.

8 Interpretation and proposed principle

The new phenomenon can be summarized as follows:

Robust or fragile information is not determined by the target spectrum alone. It is determined by the relative geometry of the target spectrum, the source-density spectrum, and the observation channel.

Under independent sources, the source-density spectrum contains only the zero mode, and NRID order equals target Fourier order. Under dependent sources, source modes shift target modes by XOR. High-order source dependence can therefore function as a local oscillator that reveals high-order target structure through low-order posterior evidence.

This creates a caution for observational information analysis. A target may appear to be supported by simple low-order features in natural data, yet require high-order coordination after source randomization or under distribution shift. Conversely, a sensitive high-order target may leak through low-order measurements when the population distribution contains nearby dependence modes.

9 Possible applications

Distribution-shift diagnostics. The gap between observational order and interventional order can diagnose models that exploit accidental correlations. A feature that appears first-order under the natural source law may disappear after source randomization if it was produced by heterodyne aliasing.

Privacy leakage auditing. A sensitive parity or aggregate may be designed to be high-order under independent inputs, but population correlations can reduce its observational order. Privacy audits should therefore inspect the Hamming neighborhood of sensitive target modes in the empirical source-density spectrum.

Experimental design and causal information. Interventional NRID can be defined by replacing the natural source law with randomized, blocked, or knockoff distributions. The difference between observational and interventional spectra identifies which information is carried by source dependence rather than direct target mechanism.

Sensor calibration. Correlated sensor errors, calibration fields, or environmental common modes can mix with target information. The same target may be easy or hard to recover depending on whether the source-dependence spectrum is aligned with the target spectrum.

10 Targets for further study

1. **Dependence-corrected NRID.** Develop an estimator that separates target mechanism coefficients from source-density heterodyne aliases.
2. **Interventional order.** Define d_{do} by measuring the target under randomized or whitened source distributions, and compare it with observational d_{obs} .
3. **Sparse density-spectrum recovery.** Combine subset probes, Walsh regression, and positivity constraints to estimate the source-density support Θ .
4. **Cancellations and phase.** Characterize when multiple pairs (A, B) with the same XOR output cancel, and connect this to the phase-gauge hierarchy from v0.8.
5. **Beyond Boolean sources.** Replace XOR support geometry with tensor-product singular-mode convolution for finite alphabets and with Clebsch-Gordan-like product rules for non-Abelian source symmetries.
6. **General channels.** Study source-density heterodyning under correlated sign noise, erasures, finite-alphabet channels, and graph diffusion.

11 Addendum status

The formulas above are exact within the stated Boolean model. The broader interpretation - that source dependence acts as an information heterodyne lens - is proposed as a research direction. It should be compared against existing work on Walsh analysis, noisy Boolean functions, higher-order statistical interactions, source redundancy, and causal/interventional information measures.

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Noise-Resolved Information Decomposition

Addendum v0.13: Interventional NRID and Source–Mechanism Unaliasing

NRID Research Program

May 2026

Abstract

Earlier addenda showed that source dependence can make high-order target information appear at low observational order. This addendum develops an interventional correction. The central observation is that an observed posterior mode is not necessarily a mechanism mode: it can be a heterodyne product of a source-density mode and a mechanism mode. For a parity mechanism B and a source-density mode A , the observed alias support is $A \oplus B$. Randomizing or otherwise scrubbing source coordinates kills source-density modes that touch the intervention set, while preserving the mechanism under the controlled experiment. This yields an exact source–mechanism separation principle, a derivative spectroscopy formula, a one-tilt scrubbing law, and a hitting-set formulation of experimental design. Exact Boolean-cube experiments show a 12-way parity mechanism whose observational chi-square response scales as ρ^2 , while a one-coordinate intervention restores the true ρ^{24} response; at $\rho = 0.1$ the apparent-to-interventional visibility ratio is about 1.45×10^{20} .

1 Blind spot: apparent order is not necessarily mechanism order

The original NRID law assumes independent uniform sources. In that regime, the order of an information hyperedge has a direct operational meaning: if the first nonzero coefficient has order d , then small-reliability posterior chi-square information scales as ρ^{2d} . Addendum v0.12 showed that this statement becomes observational rather than mechanistic when the source distribution is itself structured. A source-density mode can mix with a target mechanism mode and produce a low-order posterior shadow.

This addendum asks how that aliasing can be removed. The answer is to distinguish two spectra:

mechanism spectrum := NRID spectrum of $Q(T \mid X)$ under controlled source randomization,
observational spectrum := NRID-like response induced by the ambient source law P_X .

The gap between the two is a diagnostic object. It records dependence-induced information shadows.

2 Mechanism NRID under source randomization

Let $X \in \{-1, +1\}^n$ and let $Q(t \mid x)$ be a fixed mechanism. Under a source randomization intervention, X is sampled from the uniform product law and then passed through the same mechanism Q . Define

$$q_t(x) = Q(t \mid x), \quad p_t^U = 2^{-n} \sum_x q_t(x), \quad m_t(x) = q_t(x) - p_t^U.$$

Let $\hat{m}_t(S)$ be the Walsh coefficient under the uniform measure and define the mechanism hyperedge energy

$$\Omega_S^{\text{mech}} = \sum_t \frac{\hat{m}_t(S)^2}{p_t^U}, \quad S \neq \emptyset.$$

Theorem 1 (Interventional whitening theorem). *Under the controlled experiment $X \sim U(\{-1, +1\}^n)$, followed by the fixed mechanism $Q(T \mid X)$ and independent BSC observation of each coordinate with reliability ρ ,*

$$\boxed{\mathcal{J}_\rho^{\text{do-U}}(T; Y) = \sum_{S \neq \emptyset} \Omega_S^{\text{mech}} \rho^{2|S|}.}$$

The formula is independent of the observational source distribution.

Proof. This is the original NRID theorem applied not to the ambient joint law, but to the controlled mechanism experiment. The Boolean noise operator multiplies a Walsh mode on support S by $\rho^{|S|}$; orthogonality gives the stated chi-square energy polynomial. $\square$

Interpretation. Mechanism NRID is the spectrum measured after source correlations have been experimentally neutralized. Observational NRID is what passive data see. A discrepancy between them is not a numerical nuisance; it is a source-dependence signature.

3 Heterodyne derivative spectroscopy

Let the ambient source density ratio be expressed in Walsh form

$$w(x) = 2^n P_X(x) = 1 + \sum_{A \neq \emptyset} \theta_A \chi_A(x),$$

and introduce a source-temperature path

$$w_\tau(x) = 1 + \tau \sum_{A \neq \emptyset} \theta_A \chi_A(x),$$

for values of τ where the density remains nonnegative. The parameter τ continuously turns source dependence on and off while leaving the target mechanism fixed.

Consider the deterministic parity mechanism

$$T_B = \chi_B(X).$$

For a BSC observation with reliability ρ , the posterior mean is

$$M_\tau(y) := \mathbb{E}_\tau[T_B \mid Y = y].$$

Theorem 2 (Source-temperature heterodyne derivative). *For the parity mechanism $T_B = \chi_B(X)$,*

$$\boxed{M_\tau(y) = \frac{\rho^{|B|} \chi_B(y) + \tau \sum_A \theta_A \rho^{|A \oplus B|} \chi_{A \oplus B}(y)}{1 + \tau \sum_A \theta_A \rho^{|A|} \chi_A(y)}}.$$

Consequently,

$$\boxed{\left. \frac{\partial M_\tau(y)}{\partial \tau} \right|_{\tau=0} = \sum_A \theta_A \left(\rho^{|A \oplus B|} - \rho^{|A|+|B|} \right) \chi_{A \oplus B}(y).}$$

Proof. The source-density numerator for the posterior mean is $K_\rho(w_\tau \chi_B)$ and the denominator is $K_\rho(w_\tau)$, where K_ρ is the Boolean BSC noise operator. Since $K_\rho \chi_S = \rho^{|S|} \chi_S$ and $\chi_A \chi_B = \chi_{A \oplus B}$, the ratio formula follows. Differentiating the ratio at $\tau = 0$ gives the derivative formula. $\square$

The term $A \oplus B$ is the alias support. The formula is analogous to heterodyne mixing: a source-density frequency A and a mechanism frequency B produce an observed frequency $A \oplus B$.

Law 1 (Source–mechanism alias kernel). *For a single source-density mode A and a parity mechanism B , the first-order source-temperature alias amplitude is*

$$K_{A,B}(\rho) = \rho^{|A \oplus B|} - \rho^{|A|+|B|} = ho^c(1 - \rho^{2h}),$$

where $c = |A \oplus B|$ and $h = |A \cap B|$. For $c > 0$ and $h > 0$, its maximum occurs at

$$\rho_* = \left(\frac{c}{c + 2h} \right)^{1/(2h)}.$$

The kernel shows why severe aliasing is a mid-to-high reliability phenomenon when h is large. If A nearly matches B , then c is small and the apparent order collapses, but the alias amplitude is suppressed again as $\rho \rightarrow 1$ because the full source is then nearly visible and the numerator-denominator cancellation becomes exact.

4 One-tilt scrubbing law

Let

$$w_\tau(x) = 1 + \theta\tau^{|A|}\chi_A(x),$$

where τ may be interpreted as independent source scrubbing that damps an $|A|$ -way source-density mode by $\tau^{|A|}$. Let

$$a = |A|, b = |B|, \quad c = |A \oplus B|, \quad \eta = \theta\tau^a.$$

Theorem 3 (Exact one-tilt scrubbing law). *For $A \neq B$ and parity target $T_B = \chi_B(X)$,*

$$\mathcal{J}_\rho(\tau) = \frac{\rho^{2b} + \eta^2\rho^{2c} - 2\eta^2\rho^{a+b+c}}{1 - \eta^2\rho^{2a}}.$$

Proof. Using the ratio in the previous theorem with a single active A , write

$$M_\tau(y) = \frac{\rho^b\chi_B(y) + \eta\rho^c\chi_{A \oplus B}(y)}{1 + \eta\rho^a\chi_A(y)}.$$

The observation law is $P_Y(y) = 2^{-n}(1 + \eta\rho^a\chi_A(y))$. Hence

$$\mathcal{J}_\rho(\tau) = 2^{-n} \sum_y \frac{(\rho^b\chi_B(y) + \eta\rho^c\chi_{A \oplus B}(y))^2}{1 + \eta\rho^a\chi_A(y)}.$$

Expanding over the two parity classes of χ_A and using Walsh orthogonality yields the stated rational form. $\square$

Two limits are important. At $\tau = 0$,

$$\mathcal{J}_\rho(0) = \rho^{2b},$$

which is the mechanism-only response. For any nonzero η and $c < b$, the small- ρ leading term becomes

$$\mathcal{J}_\rho(\tau) \sim \eta^2\rho^{2c}.$$

Thus even a weak source-density mode near the mechanism support can collapse the apparent order.

5 Intervention hitting theorem

Let $R \subseteq [n]$ be a set of coordinates randomized before the mechanism is applied. In Walsh form this intervention transforms the source density ratio by averaging over the randomized coordinates:

$$w^R(x) = \mathbb{E}_{U_R} w(U_R, x_{\bar{R}}).$$

Theorem 4 (Intervention hitting theorem). *If*

$$w(x) = 1 + \sum_A \theta_A \chi_A(x),$$

then

$$w^R(x) = 1 + \sum_{A: A \cap R = \emptyset} \theta_A \chi_A(x).$$

For a weak/generic parity mechanism on support B , the first apparent posterior-amplitude order after the intervention is

$$d_R = \min \left\{ |B|, \min_{A: A \cap R = \emptyset} |A \oplus B| \right\}.$$

Proof. Averaging over a randomized coordinate kills any Walsh character that contains that coordinate. Therefore only source-density supports disjoint from R remain. The heterodyne derivative theorem then gives the surviving alias supports $A \oplus B$; the mechanism mode B remains available under the controlled experiment. $\square$

The theorem turns interventional unaliasing into a hitting-set problem: to remove a set of low-order aliases, choose R that intersects the corresponding source-density modes A .

If R is chosen uniformly among all h -coordinate subsets, a single source mode A survives with probability

$$\Pr(A \cap R = \emptyset) = \frac{\binom{n-|A|}{h}}{\binom{n}{h}}.$$

Dense source-dependence modes are therefore fragile under randomized controls.

6 Virtual experiments

6.1 Experiment 1: a 12-way mechanism with a one-way observational shadow

A 12-bit parity mechanism $B = \{0, \dots, 11\}$ was tested under three source-density modes:

$$A_1 = B \setminus \{0\}, \quad \theta_1 = 0.12,$$

$$A_2 = B \setminus \{1, 2\}, \quad \theta_2 = 0.10,$$

$$A_3 = B \setminus \{3, 4, 5\}, \quad \theta_3 = 0.08.$$

Their aliases have orders 1, 2, and 3. The exact posterior chi-square response was computed from the Walsh ratio formula on the full Boolean cube.

The intervention $R = \{0\}$ misses the mode A_1 , so the one-coordinate alias survives. The intervention $R = \{6\}$ hits all three source-density modes, so the response returns to the true 12-way mechanism exponent, $2b = 24$.

Table 1: Interventional unaliasing of a 12-way parity mechanism. Slopes are fitted chi-square exponents in small reliability.

case	fitted exponent	$\mathcal{J}_{0.1}$	ratio to $R = \{6\}$
observational	2.0086	1.450064×10^{-4}	1.450064×10^{20}
$R = \{0\}$	2.0000	1.440000×10^{-4}	1.440000×10^{20}
$R = \{6\}$	24.0000	1.000000×10^{-24}	1
$R = \{0, 1\}$	24.0000	1.000000×10^{-24}	1

6.2 Experiment 2: derivative spectroscopy recovers alias supports

The source-temperature derivative was checked at $\rho = 0.62$ by finite differencing $\tau = \pm 10^{-6}$. The maximum absolute difference between the finite-difference Walsh coefficients and the theorem was

$$1.78 \times 10^{-13}.$$

The three nonzero derivative supports were exactly the expected aliases:

$$\{0\}, \quad \{1, 2\}, \quad \{3, 4, 5\}.$$

Table 2: Heterodyne derivative coefficients at $\rho = 0.62$.

alias support	theorem	finite difference
$\{0\}$	0.074397985394	0.074397985394
$\{1, 2\}$	0.038437292196	0.038437292196
$\{3, 4, 5\}$	0.019062746060	0.019062746060

6.3 Experiment 3: random interventions kill dense alias modes

For the same $n = 12$ construction, one randomized coordinate kills all three source-alias modes with probability 0.5. Two randomized coordinates kill all three with probability 0.9393939. The individual survival probabilities for a one-coordinate random intervention were

$$A_1 : 0.08333, \quad A_2 : 0.16667, \quad A_3 : 0.25.$$

This is consistent with

$$\Pr(A \cap R = \emptyset) = \binom{n - |A|}{h} / \binom{n}{h}.$$

6.4 Experiment 4: partial scrubbing exposes an apparent-order transition

For one source-density tilt $A = \{0, \dots, 8\}$ and mechanism $B = \{0, \dots, 9\}$, the alias order is $c = 1$ and the mechanism order is $b = 10$. At $\rho = 0.1$ and $\theta = 0.35$,

$$\mathcal{J}_{\text{observational}} = 1.225 \times 10^{-3}, \quad \mathcal{J}_{\text{interventional}} = 1.0 \times 10^{-20},$$

so the passive visibility is about

$$1.225 \times 10^{17}$$

times larger than the source-randomized mechanism visibility.

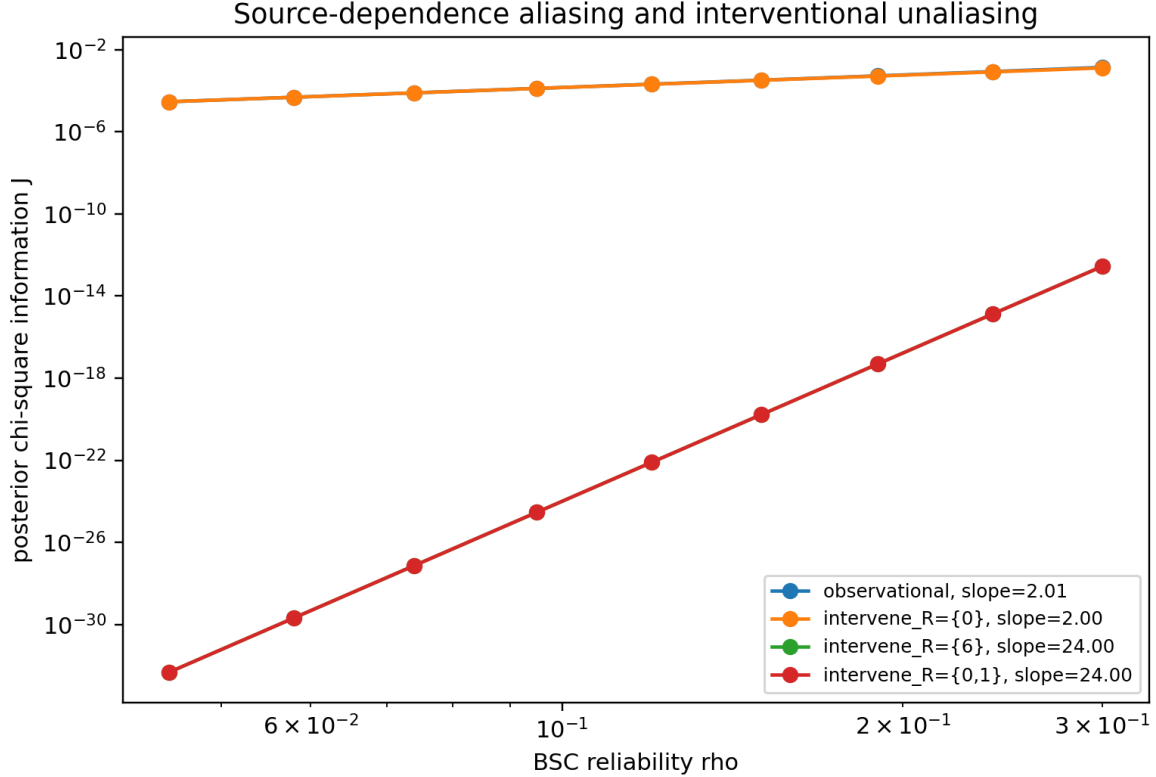


Figure 1: A 12-way parity mechanism appears as an order-one observational shadow until a source-density hitting intervention removes the aliasing modes.

7 Exceptional observations

Order can be causal rather than observational. The same target mechanism can have a true interventional order of 12 and an observational order of 1. The discrepancy is not an error in NRID; it records dependence-induced heterodyne aliasing.

Tiny dependence can dominate high-order mechanisms. A weak source-density mode close to the mechanism support can dominate the small-reliability response by many orders of magnitude. The one-tilt example produced a 1.225×10^{17} visibility ratio at $\rho = 0.1$.

Interventions can be more efficient than observation. Passive spectroscopy sees only the alias. A single well-chosen randomized coordinate can remove the alias and reveal the true mechanism order. This suggests that NRID should be treated as an experimental design theory, not only a passive decomposition.

8 Possible applications

Causal information auditing. Interventional NRID can test whether a low-order feature signal is part of the mechanism or merely a shadow of correlated inputs. This is relevant to causal feature analysis, model debugging, and scientific measurement systems.

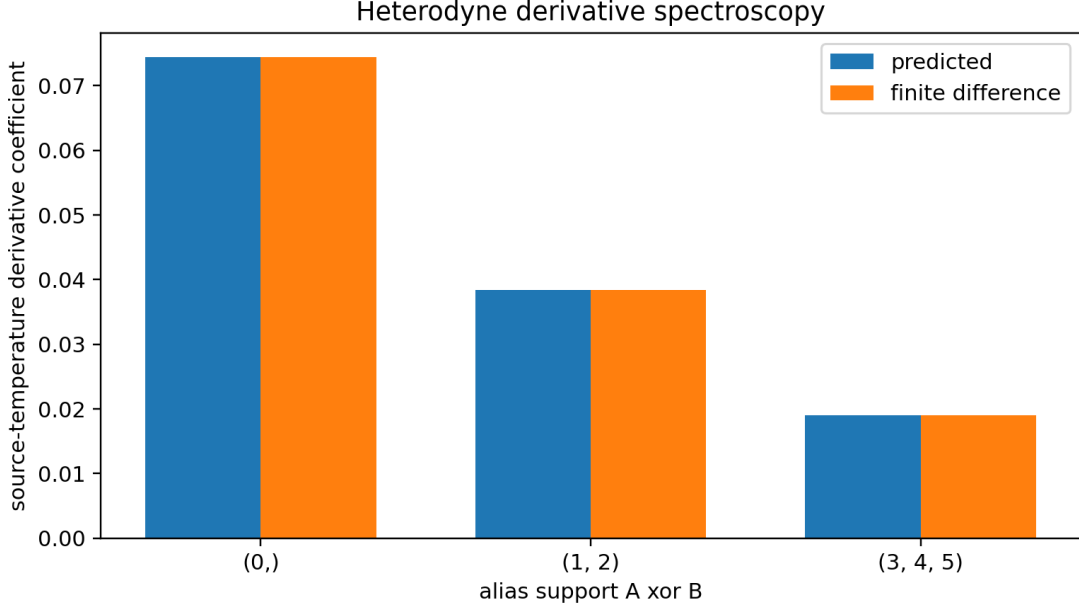


Figure 2: Predicted and finite-difference source-temperature derivative coefficients are indistinguishable at plotting precision.

Table 3: Partial source scrubbing: τ damps the source-density tilt by $\tau^{|A|}$.

τ	fitted chi-square exponent	$\mathcal{J}_{0.1}$
1.00	2.0000	1.225000×10^{-3}
0.70	2.0000	1.994807×10^{-6}
0.40	2.0000	8.418136×10^{-11}
0.20	2.1833	3.211364×10^{-16}
0.05	18.5665	1.000000×10^{-20}
0.00	20.0000	1.000000×10^{-20}

Privacy and leakage diagnosis. A private high-order mechanism may leak through low-order source dependencies. The alias formula identifies which dependence modes create low-order leakage and which interventions, perturbations, or distribution shifts remove it.

Sensor calibration. In sensor networks, correlated environmental structure can make a high-order event look locally detectable. Controlled randomization or calibration masks can separate environmental correlation from genuine multi-sensor mechanism.

Machine-learning interpretability. Feature-interaction methods often operate on observational data, where correlated features can induce spurious low-order explanations. Mechanism NRID suggests an intervention-based correction for interaction attribution.

9 Targets for further study

1. Extend source-mechanism unaliasing from Boolean parity mechanisms to finite-alphabet modal NRID.

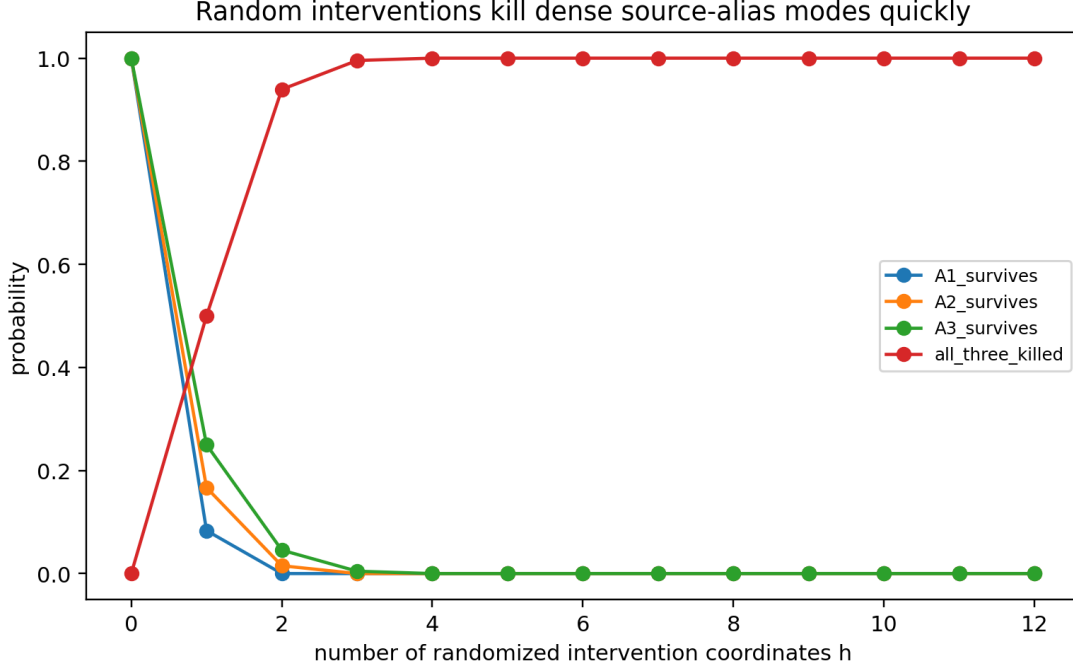


Figure 3: Dense source modes have low survival probability under random intervention masks.

2. Develop positivity-preserving source-temperature paths for arbitrary empirical distributions.
3. Estimate the sample complexity of heterodyne derivative spectroscopy under finite data.
4. Connect intervention hitting sets to adaptive experimental design and compressed sensing.
5. Develop a causal NRID ledger with three spectra: observational, source-randomized, and target-garbled.
6. Study whether source-mechanism alias kernels have analogs in graph heat NRID and Gaussian-chaos NRID.

10 Research ledger

- New proposed object: mechanism NRID, the source-randomized spectrum of $Q(T | X)$.
- New proposed law: source-mechanism heterodyne aliasing, $A, B \mapsto A \oplus B$.
- New exact equation: the one-tilt scrubbing law for $\mathcal{J}_\rho(\tau)$.
- New experimental design principle: unaliasing low-order shadows is a hitting-set problem over source-density modes.
- New diagnostic: disagreement between observational and interventional NRID is evidence of distribution-induced information shadowing.

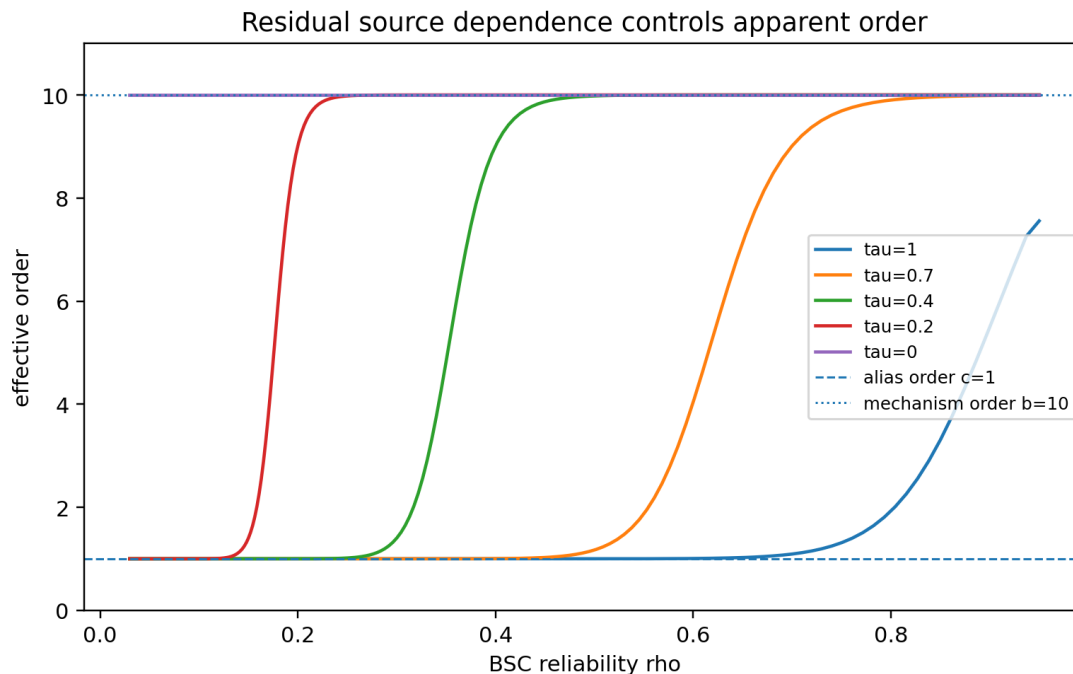


Figure 4: As source dependence is scrubbed away, the apparent information order moves from the alias order 1 back toward the mechanism order 10.

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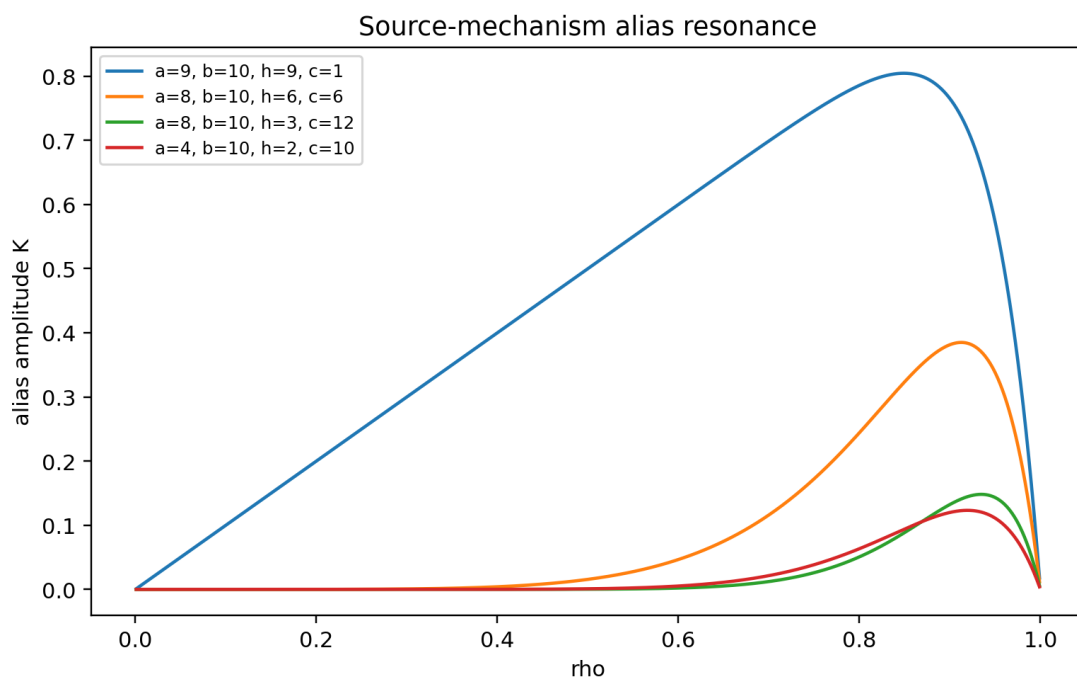


Figure 5: The source–mechanism alias kernel has a reliability–band maximum. When A and B strongly overlap, aliases can be most visible at intermediate-to-high reliability.

Addendum v0.14: Matroidal Access Spectra and q-ary Linear NRID

Noise-Resolved Information Decomposition Working Paper

May 7, 2026

Abstract

This addendum studies the special but structurally revealing case in which the target is a linear syndrome of finite-field sources. In this regime, the NRID response is not merely analogous to coding theory: it is exactly the nonzero weight enumerator of the row code evaluated at the squared channel-character reliability. Exact context and erasure access are controlled by a finer matroid rank-defect function. This produces a new separation: two linear targets can have identical isotropic NRID polynomials and identical mean fixed-budget access curves, while having different adversarial access-tail distributions. The experiments below verify the q-ary theorem by direct enumeration and identify a concrete pair of weight-equivalent binary targets with different access geometry.

1 Blind spot: spectra are not access distributions

Earlier addenda showed that a linear binary target $T = HX$ has an NRID spectrum equal to the nonzero weight distribution of the row code generated by H . This makes linear targets unusually transparent. It also exposes a limitation: a one-variable polynomial in reliability records how many information characters have each support size, but it does not record how these supports are arranged in the coordinate set. Random access to m coordinates depends mostly on support sizes. Adversarial or spatially localized access depends on support geometry.

The natural object is therefore not only the weight enumerator but a matroidal access spectrum: the distribution of rank defects of the unobserved columns. This connects NRID to the code-matroid relationship studied through Greene's theorem and rank/Tutte polynomials, but gives the quantities a direct operational meaning as posterior chi-square information under partial observation.

2 q-ary linear NRID theorem

Let $X = (X_1, \dots, X_n)$ be uniform on $\mathbb{F}_q^n$, where q is a prime power, and let

$$T = HX \in \mathbb{F}_q^k,$$

where $H \in \mathbb{F}_q^{k \times n}$ has rank k . Fix a nontrivial additive character χ of $\mathbb{F}_q$. Suppose each coordinate is observed through an independent additive-noise channel

$$Y_i = X_i + N_i,$$

with the noise independent of X . For $b \in \mathbb{F}_q$, define the local character transfer coefficient

$$\lambda_i(b) = \mathbb{E}[\chi(bN_i)].$$

Let $C = \{aH : a \in \mathbb{F}_q^k\}$ be the row code of H .

Theorem 1 (q-ary linear NRID transfer law). *For the linear target $T = HX$,*

$$\mathcal{J}(T; Y) = \sum_{a \in \mathbb{F}_q^k \setminus \{0\}} \prod_{i=1}^n |\lambda_i((aH)_i)|^2.$$

In particular, for a q -ary symmetric channel whose nontrivial character eigenvalue is λ ,

$$\mathcal{J}_\lambda(T; Y) = \sum_{c \in C \setminus \{0\}} |\lambda|^{2 \text{wt}(c)} = W_C(1, |\lambda|^2) - 1,$$

where $W_C(x, y) = \sum_{c \in C} x^{n - \text{wt}(c)} y^{\text{wt}(c)}$ is the homogeneous weight enumerator.

Proof. Because H has rank k , T is uniform on $\mathbb{F}_q^k$. The nontrivial characters of the target are indexed by $a \in \mathbb{F}_q^k \setminus \{0\}$ and are given by $\chi(a \cdot T) = \chi((aH) \cdot X)$. Conditional expectation through the additive channel multiplies each source character by its local transfer coefficient. Hence

$$\mathbb{E}[\chi(a \cdot T) \mid Y] = \prod_i \lambda_i((aH)_i) \chi((aH)_i Y_i).$$

Target characters are orthogonal, and posterior chi-square information is the sum of squared magnitudes of all nontrivial posterior character amplitudes. Summing over $a \neq 0$ gives the first formula. If all nonzero local character eigenvalues are equal to λ , each nonzero coordinate of aH contributes $|\lambda|^2$, giving the weight-enumerator specialization. $\square$

Interpretation. The result says that linear NRID is a character-level channel transfer law. The target mechanism supplies a code C ; the observation channel supplies a transfer spectrum; the recoverable information is their overlap. The binary theorem from earlier sections is the $q = 2$ case.

3 Exact subset access and matroid rank defects

Let $E = \{1, \dots, n\}$. Suppose coordinates in $U \subseteq E$ are observed exactly and coordinates in $E \setminus U$ are unobserved. This is the limiting transfer profile $\lambda_i = 1$ on U and $\lambda_i = 0$ off U .

Theorem 2 (matroidal subset-access law). *For a q -ary linear target $T = HX$,*

$$\mathcal{J}(U) = |\{c \in C \setminus \{0\} : \text{supp}(c) \subseteq U\}| = q^{k - \text{rank}(H_{E \setminus U})} - 1.$$

Here $H_{E \setminus U}$ is the submatrix containing the unobserved columns.

Proof. A code character $c = aH$ survives exact observation of U and blindness outside U exactly when $c_i = 0$ for every $i \notin U$, equivalently $aH_{E \setminus U} = 0$. The set of such a is a vector space of dimension $k - \text{rank}(H_{E \setminus U})$. It contains $q^{k - \text{rank}(H_{E \setminus U})}$ vectors, one of which gives the zero character. Subtracting one gives the chi-square information. $\square$

Corollary 1 (random fixed-budget access). *Let $A_r(C) = |\{c \in C : \text{wt}(c) = r\}|$. If U is chosen uniformly among all m -subsets of E , then*

$$\mathbb{E}_{|U|=m} \mathcal{J}(U) = \sum_{r=1}^m A_r(C) \frac{\binom{n-r}{m-r}}{\binom{n}{m}}.$$

Each nonzero codeword of weight r survives precisely when its support is contained in U .

Corollary 2 (rank-defect access distribution). *The entire access distribution at budget m is*

$$D_m(s) = |\{U \subseteq E : |U| = m, k - \text{rank}(H_{E \setminus U}) = s\}|,$$

and $\mathcal{J}(U) = q^s - 1$ on those subsets. Equivalently, $D_m(s)$ is a slice of the rank distribution of the column matroid represented by H .

Increasing returns. The exponent

$$L(U) = \log_q(1 + \mathcal{J}(U)) = k - \text{rank}(H_{E \setminus U})$$

is a monotone supermodular set function because it is minus the rank of the complement. Thus exact access to a coordinate is more valuable when the unobserved complement is already close to losing rank. Linear targets therefore give another exact model of the context-supermodularity observed in v0.11.

4 Consequences

Distance is the first access threshold. Let $d(C)$ be the minimum nonzero codeword weight. Then $\mathcal{J}(U) = 0$ for every $|U| < d(C)$, and some U with $|U| = d(C)$ has $\mathcal{J}(U) > 0$. In NRID language, the minimum distance is the first exact-access order.

Weight enumerator gives mean access, not access risk. The ordinary NRID polynomial $W_C(1, z) - 1$ determines the mean random subset access curve through binomial thinning. It does not determine the full access-tail distribution $D_m(s)$. The latter requires rank-profile information, or equivalently a finer matroid invariant.

Tutte-polynomial bridge. Greene’s theorem identifies the weight enumerator of a linear code as a specialization of the Tutte polynomial of its vector matroid. The present result gives an information-theoretic interpretation: that specialization is the isotropic linear NRID response, while other rank-polynomial slices govern exact-access distributions.

5 Exact experiments

All experiments were exact enumerations over finite fields and finite observation spaces, except for the random selection of full-rank test matrices.

Table 1: Verification of the q-ary transfer theorem for a random ternary target with $q = 3, k = 2, n = 5$ and q-ary symmetric character reliability $\lambda = 0.47$.

Quantity	Value
Direct posterior enumeration	0.03584520265505644
Weight-enumerator formula	0.035845202655056585
Absolute error	1.46×10^{-16}
Weight enumerator	$1 + 2y^3 + 6y^4$

Random subset theorem. For the same ternary target, 50 random subsets were checked. The maximum discrepancy between the rank-defect expression $q^{k - \text{rank}(H_{E \setminus U})} - 1$ and the direct count of surviving nonzero row-code characters was exactly zero.

5.1 Hamming syndrome access

The $[7, 3, 4]$ simplex row code generated by the Hamming parity-check matrix has

$$W_C(1, y) - 1 = 7y^4.$$

Consequently, exact subset access is delayed until four coordinates are observed. The mean access curve for $m = 0, \dots, 7$ is

$$0, 0, 0, 0, 0.2, 1, 3, 7.$$

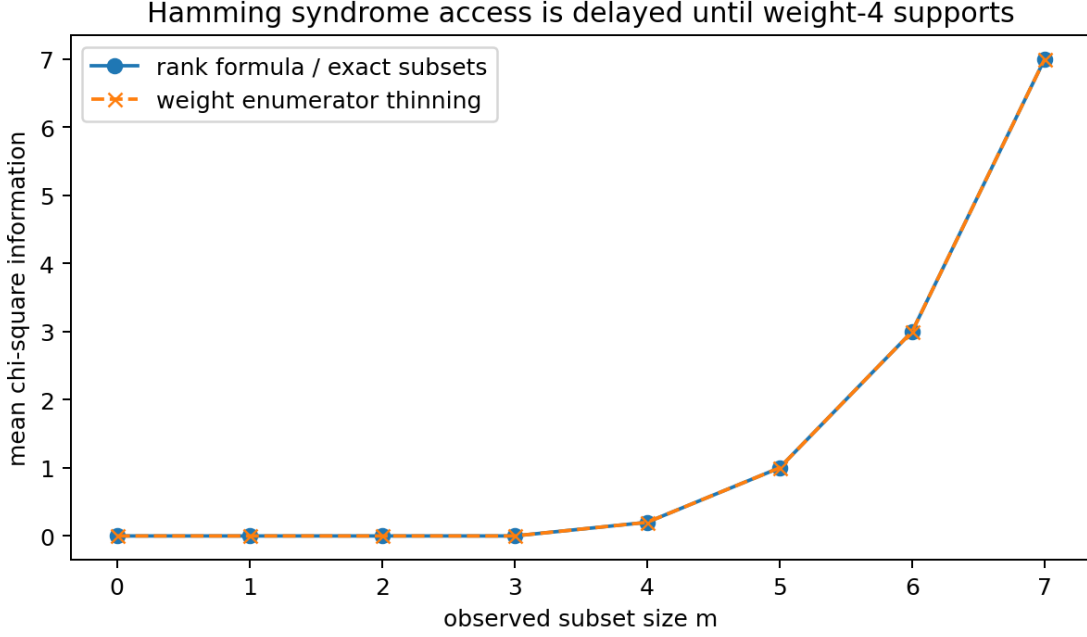


Figure 1: The Hamming syndrome target has no exact subset access until a full weight-4 simplex support is observed. The rank-defect formula and the weight-enumerator thinning formula coincide for the mean curve.

5.2 q-ary access curves

Figure ?? shows exact mean access curves for sampled q-ary linear targets. The final value is $q^k - 1$, the total posterior chi-square information under full observation.

6 A counterexample: identical NRID, different access tails

The following two binary 4×10 matrices define rank-four linear targets. Their row codes have identical weight enumerator

$$W_C(1, y) = 1 + 2y^3 + 2y^4 + 4y^5 + 4y^6 + 2y^7 + y^8.$$

Therefore they have the same isotropic NRID polynomial and the same random fixed-budget mean access curve.

Target A	Target B
1011110010	1101101110
1110010111	0111111110
1100111111	0000011101
1110111001	0001100100

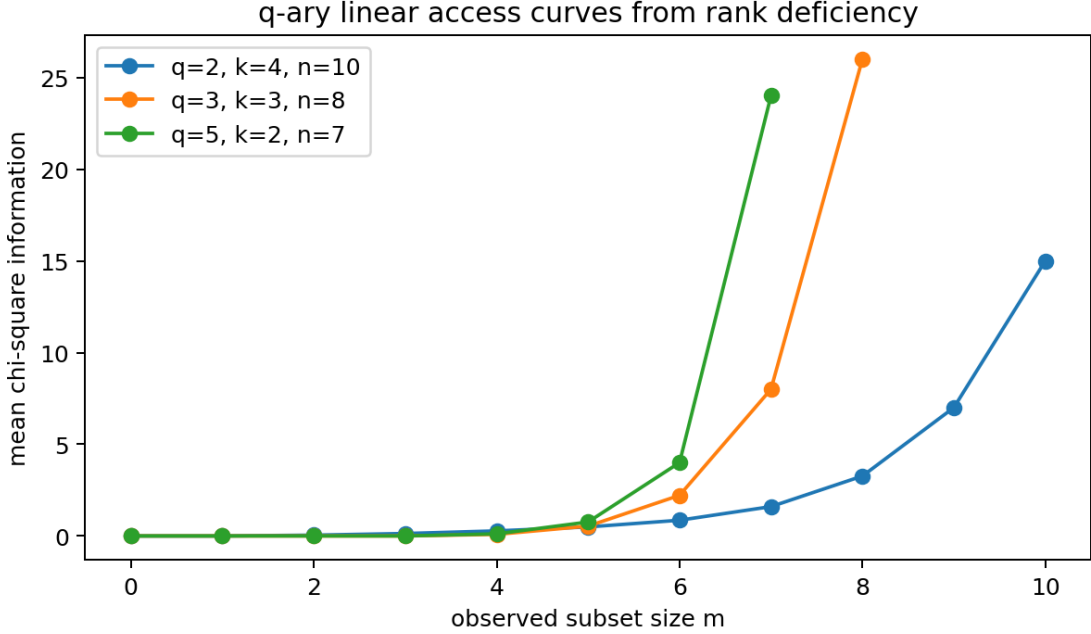


Figure 2: Mean exact-access curves for sampled q -ary linear targets. The curves are determined by the row-code weight enumerator, while the full access distribution is determined by the column-matroid rank profile.

At budget $m = 5$, both have mean access $58/252 \approx 0.230159$. But their access-tail distributions differ:

	$\mathcal{J}(U) = 0$	$\mathcal{J}(U) = 1$	$\mathcal{J}(U) = 3$
Target A	196	55	1
Target B	194	58	0

Thus Target A has an adversarial five-coordinate subset revealing $\mathcal{J} = 3$, while Target B has no such five-coordinate subset.

Observation 1 (access-incompleteness of isotropic NRID). *The one-dimensional NRID polynomial $W_C(1, z) - 1$ is not sufficient to characterize partial-access risk. It captures support-size counts, hence average random access, but not how supports overlap. Access tails require rank-defect distributions of the represented matroid.*

7 Exceptional observations

Linear NRID is a code enumerator. For q -ary linear targets, posterior chi-square information under symmetric memoryless noise is exactly the row-code weight enumerator evaluated at squared character reliability. This gives a direct operational information meaning to a classical code polynomial.

Exact access is a rank-defect event. A subset reveals information precisely when the unobserved columns fail to span the target syndrome space. The quantity of revealed information is exponential in the rank defect.

Same robustness, different compromise risk. Two targets can be indistinguishable under isotropic noise spectroscopy but differ under targeted access. This is a sharper linear-code version of the recurring NRID theme: order spectra are not geometry.

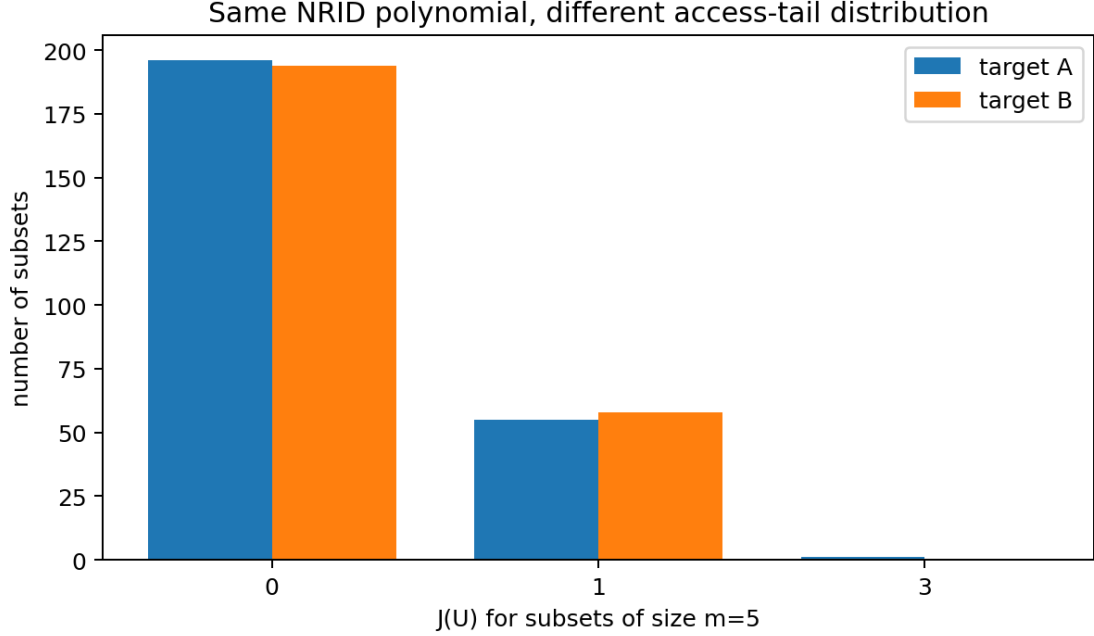


Figure 3: Two targets with the same NRID polynomial and the same mean subset-access curve have different access-tail distributions. The ordinary weight spectrum is therefore not access-complete.

8 Possible applications

Syndrome diagnostics and distributed sensing. The theorem gives exact formulas for how much information about a linear diagnostic syndrome survives q -ary sensor noise, erasures, or exact partial inspection.

Privacy and leakage auditing. For linear releases, the access distribution $D_m(s)$ measures the amount of leakage available to an observer controlling m coordinates. Mean leakage can be benign while tail leakage is not.

Code design for information shaping. Minimum distance controls the first access threshold. Weight distribution controls isotropic robustness. Rank-defect distribution controls targeted compromise risk. These three design objectives need not coincide.

Secret-sharing and erasure models. The access law resembles linear secret-sharing rank criteria: information appears when observed shares force a rank defect in the unobserved complement. NRID supplies a graded chi-square version rather than only an authorized/unauthorized dichotomy.

9 Targets for further study

1. Construct infinite families of weight-equivalent linear targets with exponentially different access-tail risk.
2. Express the full NRID access distribution in standard Tutte or rank-polynomial coordinates and identify which specializations control mean, variance, and maximum leakage.

3. Extend the q-ary theorem to nonadditive channels using singular modes rather than additive characters.
4. Develop algorithms that optimize linear targets jointly for distance, isotropic robustness, and access-tail safety.
5. Connect matroidal NRID to generalized Hamming weights, wiretap-II leakage, and ramp secret-sharing thresholds.
6. Study whether nonlinear targets admit matroid-like access spectra after lifting to their active Fourier or modal support hypergraphs.

10 Research ledger

- New exact theorem: q-ary linear NRID transfer law for additive finite-field channels.
- New exact theorem: subset access equals $q^{k-\text{rank}(H_{E \setminus U})} - 1$.
- New operational bridge: isotropic linear NRID is a weight-enumerator/Tutte specialization.
- New blind spot: weight spectra determine mean random access but not access-tail risk.
- New counterexample: two rank-four binary targets with identical NRID polynomial and different five-coordinate access distributions.

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Noise-Resolved Information Decomposition: Working Paper Addendum v0.15

Linear-Sketch NRID, Grassmann Leakage, and Measurement-Basis Preconditioning

Abstract. This addendum studies information that is observed through finite-field linear sketches rather than through labeled coordinates. For a q -ary linear target $T=HX$ and an exact linear sketch $Z=AX$, posterior chi-square information is exactly $q^{\dim(C \cap U)} - 1$, where C is the row code of H and U is the sketch rowspace. Uniform random sketches obey a Gaussian-hypergeometric intersection law, yielding a closed-form mean leakage curve independent of code geometry. With noisy sketch coordinates, however, the chosen measurement basis becomes information-relevant: two sketches with the same noiseless rowspace can have identical exact information and different noisy NRID polynomials. The resulting principle is that compressed information is rowspace-controlled without noise, but basis-weight-controlled after independent measurement noise.

1. Blind spot: compressed access is neither coordinate access nor ordinary noise

Earlier NRID addenda treated coordinate-wise degradation, exact coordinate subsets, symmetry hiding, source dependence, and finite-field linear targets. A missing case is a common one in sensing, telemetry, privacy, and network coding: the observer does not see coordinates, but linear mixtures of coordinates. Linear sketches preserve algebraic structure while erasing coordinate identity. The central question is: which target information hyperedges survive a sketch?

The answer is particularly sharp for linear finite-field targets. Coordinate access is controlled by codeword supports and minimum distance. Linear-sketch access is controlled by intersections of subspaces. The two can behave very differently, even for the same target.

2. Exact linear-sketch intersection theorem

Let X be uniform on F_q^n . Let $T=HX$ in F_q^k , where H has rank k , and let $C=\text{row}(H) \leq F_q^n$ be the target row code. Let $Z=AX$ be an exact linear sketch with $U=\text{row}(A)$ and $\dim U=m$. The posterior chi-square information between T and Z is denoted $J(T;Z)$.

Theorem 15.1 (linear-sketch intersection law).

$$J(T; Z) = q^{\dim(C \cap U)} - 1.$$

Proof sketch. Nontrivial characters of the target are indexed by nonzero codewords c in C . A character $\chi(c \cdot X)$ is measurable from $Z=AX$ exactly when c lies in $\text{row}(A)=U$. All nontrivial target characters have unit posterior energy when measurable and zero energy otherwise. Therefore J counts the nonzero vectors in $C \cap U$, giving $q^{\dim(C \cap U)} - 1$.

2.1 Adversarial envelope

For a fixed sketch dimension m , the possible leakage is bounded exactly by the possible intersection dimensions of a k -subspace and an m -subspace in F_q^n :

$$q^{\max(0, k+m-n)} - 1 \leq J(T;AX) \leq q^{\min(k,m)} - 1.$$

The lower bound is a codimension cliff: if $m > n-k$, every m -dimensional sketch leaks nonzero information, no matter how the sketch is chosen. If $m \leq n-k$, zero-leak sketches exist. The upper bound is reached by choosing sketch rows inside the target row code.

2.2 Uniform random sketches

If U is a uniformly random m -dimensional subspace of F_q^n , then $S = \dim(C \cap U)$ has the Gaussian-hypergeometric distribution

$$P(S=s) = \frac{q^{k-s} \binom{m-s}{k-s}_q}{\binom{n-k}{k-s}_q \binom{n}{m-s}_q} / \binom{n}{m}_q.$$

Consequently,

$$E[J] = (q^k - 1)(q^m - 1) / (q^n - 1).$$

The mean follows by linearity: each nonzero codeword c in C lies in a uniform random m -subspace with probability $(q^m - 1) / (q^n - 1)$. A striking consequence is that the *mean* random linear-sketch leakage depends only on n , k , m , and q , not on the code weight distribution. Code geometry returns in coordinate access, sparse sketches, constrained sketches, and noisy bases.

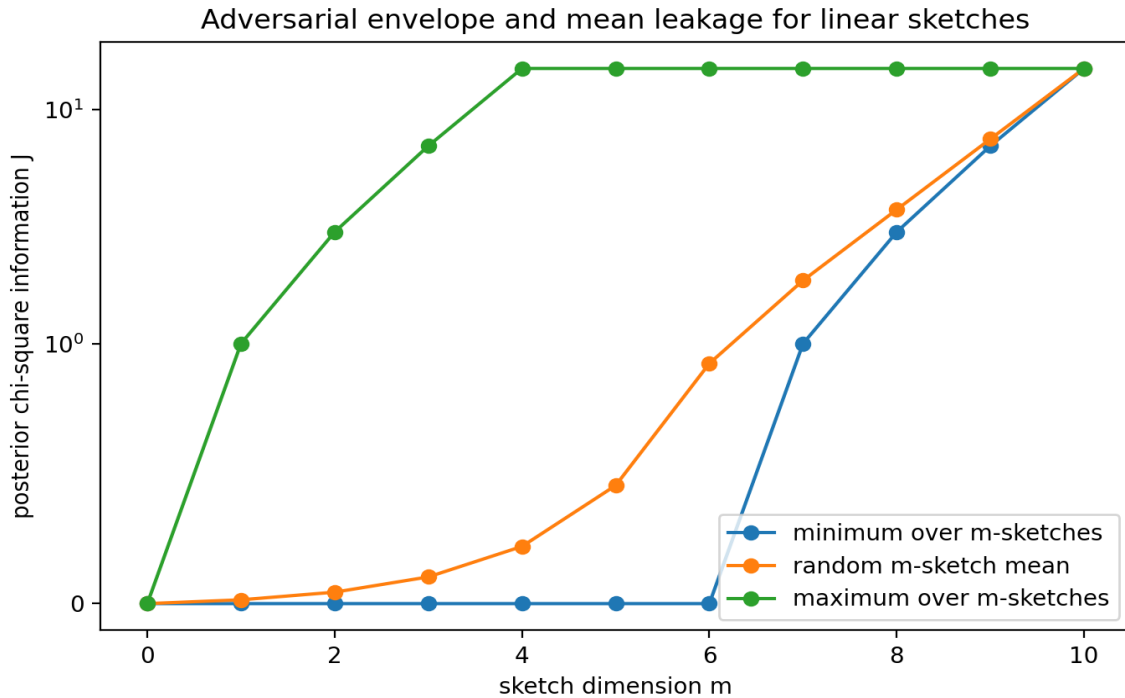


Figure 15.1. Exact minimum, random mean, and maximum posterior chi-square information for binary linear sketches with $n=10$ and target rank $k=4$. The minimum curve shows the codimension cliff at $m > n-k$.

3. Coordinate access versus linear sketches

For coordinate subsets, a codeword survives when its support is contained in the observed subset. For linear sketches, a codeword survives when it lies in the sketch row-space. These are different geometries. In the binary simplex $[15,4,8]$ example, every nonzero codeword has coordinate weight 8; coordinate access is delayed until eight coordinates. A random 8-coordinate subset has mean chi-square information 0.002331. A random 8-dimensional linear sketch has mean 0.116733, about 50.08 times larger.

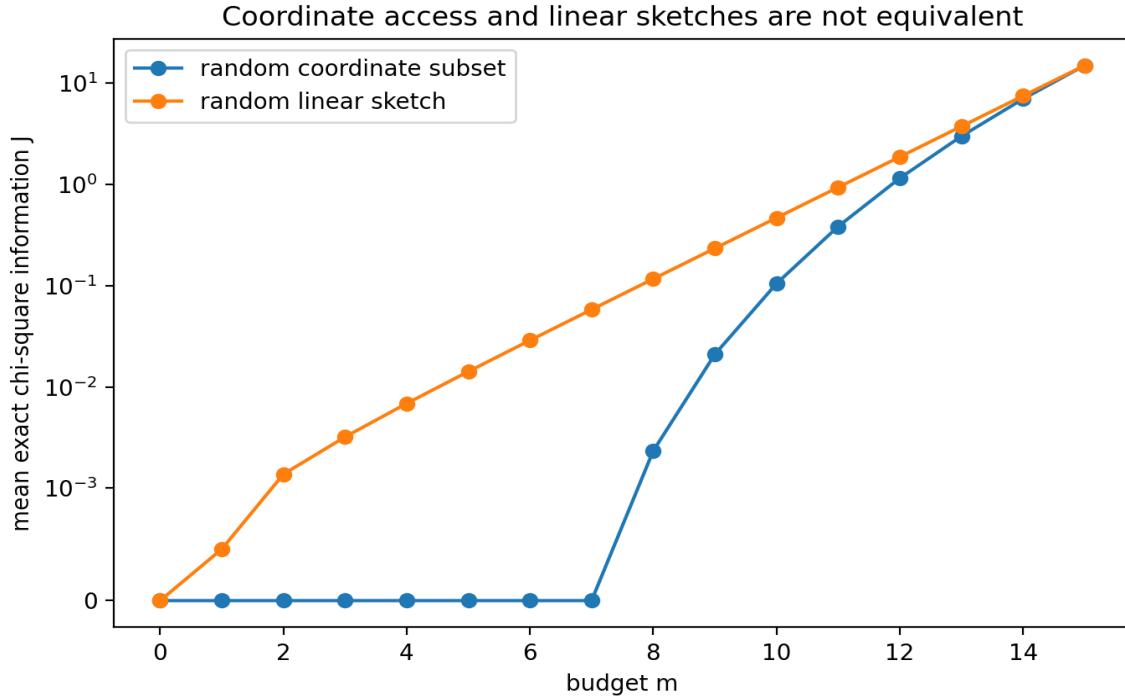


Figure 15.2. Same target, two forms of budgeted access. Coordinate access is governed by supports; linear sketches are governed by subspace intersections.

4. Noisy linear sketches

Now let $Z=AX+N$, where A has rank m and the measurement noises N_j are independent additive q -ary noises. For each measurement coordinate j , let $\lambda_j(b)=E[\chi(b, N_j)]$ be the character transfer coefficient.

Theorem 15.2 (noisy sketch transfer law).

$$J(T;Z) = \sum_{\{v \in F_q^m : v_A \in C, v_A \neq 0\}} \prod_{j=1}^m |\lambda_j(v_j)|^2.$$

For a q -ary symmetric measurement channel with common nontrivial character reliability λ , define $V=\{v \in F_q^m : v_A \in C\}$. Then

$$J(T;Z) = \sum_{\{v \in V, v \neq 0\}} |\lambda|^{2 \cdot \text{wt}(v)} = W_V(1, |\lambda|^2) - 1.$$

Thus exact sketch information depends only on $\dim V = \dim(C \cap U)$, while noisy sketch information depends on the measurement-coordinate weight enumerator of V . The rowspace determines what can be known in principle; the measurement basis determines how fast it appears through independent measurement noise.

5. Measurement-basis preconditioning

The noisy theorem exposes a new design degree of freedom. Take the same target $C=\text{span}(e_1, e_2)$ and the same noiseless sketch rowspace $U=\text{span}(e_1, e_2, e_3)$. Two bases of U give the same exact information $J=3$, but different noisy NRID polynomials.

Sparse basis: $A=(e_1, e_2, e_3) \rightarrow V$ has weights $1, 1, 2 \rightarrow J(\rho)=2\rho^2 + \rho^4$.
Dense basis: $A=(e_1+e_3, e_2+e_3, e_3) \rightarrow V$ has weights $2, 2, 2 \rightarrow J(\rho)=3\rho^4$.

At $\rho=0.55$, exact enumeration gives $J=0.69650625$ for the sparse basis and $J=0.27451875$ for the dense basis. At $\rho=0.35$, the sparse basis reveals about 5.78 times as much information. Both sketches become equivalent only at $\rho=1$, when measurement noise vanishes.

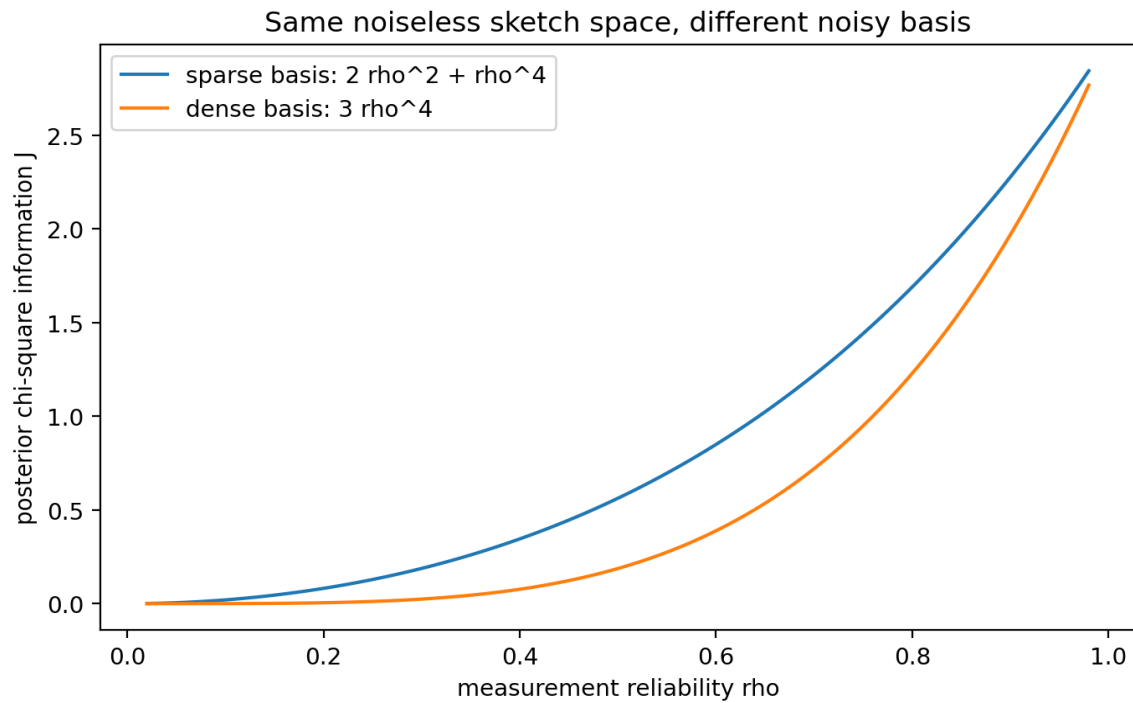


Figure 15.3. Independent noise after measurement makes the measurement basis information-relevant. A dense preconditioning of the same rowspace can suppress low-reliability leakage by moving surviving target characters to higher measurement weight.

6. Exact experiments

Experiment	Result
Exact sketch theorem	25 random binary targets/sketches, max abs error 0.00e+00
Noisy sketch theorem	Structured binary noisy sketch, max abs error 6.66e-16
Basis preconditioner at $\rho=0.55$	sparse 0.696506250000; dense 0.274518750000; max error 3.33e-16
Grassmann distribution	enumerated 11811 subspaces; formula total 11811; max count error 0
Simplex $m=8$ comparison	coordinate mean 0.002331002331; linear mean 0.116733298746; ratio 50.079x

s	enumerated count	formula count
0	7680	7680
1	3920	3920
2	210	210
3	1	1

Table 15.2. Exact enumeration of all 11811 three-dimensional subspaces of F_2^7 against the Gaussian-hypergeometric intersection law for $k=3$.

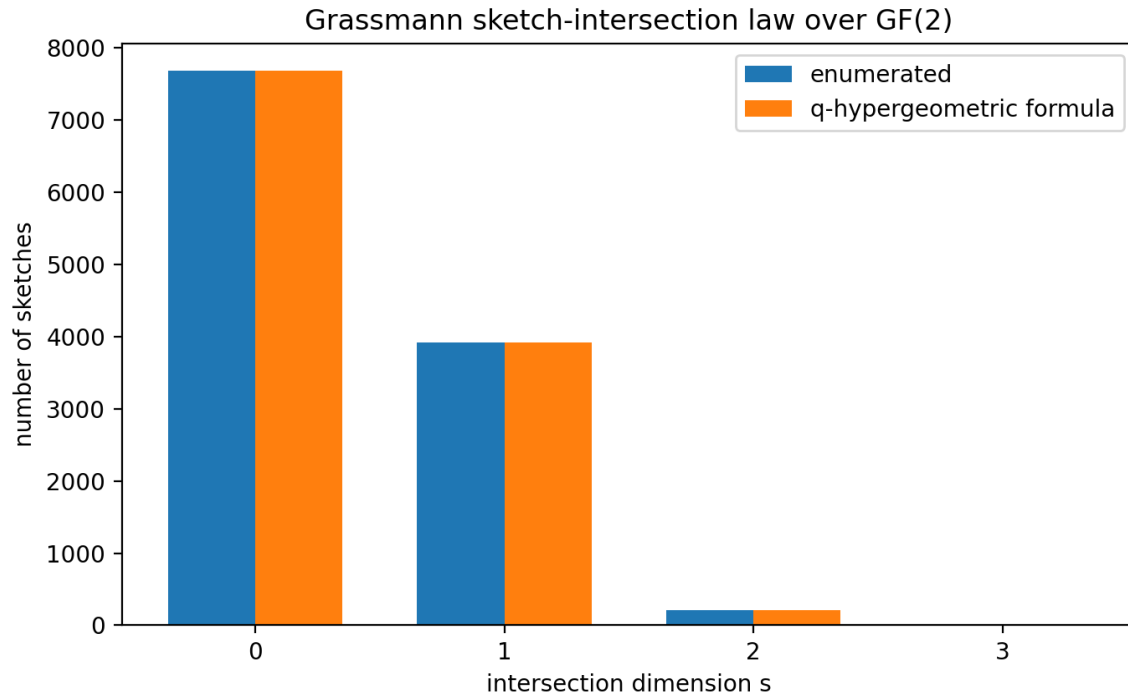


Figure 15.4. The enumerated intersection-dimension distribution exactly matches the Gaussian-hypergeometric formula.

7. Exceptional observations

Compressed information is subspace intersection. For exact linear sketches, chi-square information is not a function of coordinate support. It is the number of nonzero target characters lying in the sketch rowspace.

Random linear sketches erase code geometry in the mean. The mean random-sketch leakage depends only on dimensions. This contrasts with coordinate subsets, where the weight enumerator and support overlap structure matter.

Independent noise makes basis choice physical. If noise is added after measurement, two invertible re-encodings of the same noiseless sketch can have different NRID response curves. Basis design can therefore amplify robustness or privacy without changing the noiseless information content.

Codimension replaces distance. Coordinate access thresholds are controlled by minimum distance. Linear-sketch worst-case thresholds are controlled by codimension $n-k$. This creates a new dual fragility law.

8. Possible applications

Secure compressed telemetry. The formulas quantify how much information about a linear diagnostic syndrome leaks through compressed finite-field measurements. Dense noisy preconditioning can intentionally suppress low-reliability leakage.

Network coding and distributed sensing. In systems where packets or measurements span subspaces, recoverable target information is controlled by subspace intersections. NRID supplies a chi-square information interpretation of these intersections.

Privacy auditing for linear releases. Exact release risk is $q^{\dim(C \cap U)} - 1$. Random-release mean risk is dimension-only, but adversarial risk depends on whether the release rowspace intersects sensitive target characters.

Measurement-basis design. For fixed noiseless measurement span, choose a basis whose intersection code V has large minimum weight to reduce noisy leakage, or small minimum weight to improve robustness.

9. Targets for further study

- Develop optimization algorithms that choose a noisy sketch basis to minimize or maximize $W_V(1, \lambda^2)$ under sensing constraints.
- Extend the sketch law from linear finite-field targets to nonlinear targets by lifting active Fourier supports into subspace arrangements.
- Study sparse random sketches, where geometry no longer averages out and weight distribution should re-enter the leakage law.
- Relate sketch-NRID to generalized Hamming weights, wiretap-II leakage, rank-metric codes, and subspace codes.
- Develop Shannon and Bayes-utility analogues of the chi-square sketch laws, especially under rare-certain versus diffuse posterior amplitudes.
- Test whether measurement-basis preconditioning gives practical privacy gains in finite-field compressed sensing and telemetry systems.

10. Research ledger

- New exact theorem: $J(T; AX) = q^{\dim(\text{row}(H) \cap \text{row}(A))} - 1$ for exact finite-field linear sketches.
- New random-sketch law: Gaussian-hypergeometric intersection distribution and dimension-only mean leakage.
- New duality: coordinate access is support-controlled, while linear-sketch access is subspace-controlled.
- New noisy transfer theorem: noisy sketch information is the measurement-coordinate weight enumerator of $V = \{v: vA \text{ in } C\}$.
- New design principle: noiseless rowspace controls possibility; noisy basis controls reliability response.

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Addendum v0.16: Sequential NRID, Adaptive Querying, and Multiplexer Amplification

Abstract. This addendum studies NRID under sequential and adaptive observation. Posterior chi-square information is shown to be a submartingale with an exact Pythagorean increment formula. For exact coordinate-reveal processes, each Fourier information hyperedge activates at the first time all of its participating sources have been observed; in a random reveal order, an order- r hyperedge has mean activation time $r(n+1)/(r+1)$. The main constructive discovery is an adaptive multiplexer law: a k -address-bit multiplexer can have an exponential adaptive/nonadaptive information gap at the same query budget. With exact address access and budget $k+1$, an adaptive policy recovers the target perfectly while the best nonadaptive policy obtains chi-square information 2^{-k} . A noisy extension gives closed-form adaptive and nonadaptive response curves and an address-acquisition phase transition at $\alpha=\sqrt{2}-1$ for the full-address policy.

1. Posterior-energy martingale

Let F_t be the transcript generated by a sequential observation policy. Define the random posterior chi-square score

$$j_t = \sum_a (P(T=a | F_t) - p_a)^2 / p_a.$$

For any refinement $F_t \subset F_{t+1}$, the conditional expected increment is exactly a squared posterior jump:

$$E[j_{t+1} | F_t] - j_t = E[\sum_a (P(T=a | F_{t+1}) - P(T=a | F_t))^2 / p_a | F_t] \geq 0.$$

Thus expected posterior chi-square information is a submartingale under sequential observation. This is not merely monotonicity; it gives an exact energy budget for each observation. The identity follows by viewing posterior probability vectors as Hilbert-space projections of the target indicator vector and applying the Pythagorean theorem for nested σ -fields.

2. Hyperedge activation clocks

For independent uniform Boolean sources, exact revelation of a coordinate set C preserves precisely those Fourier/NRID hyperedges S that are contained in C :

$$J(C) = \sum_{\{S \subseteq C, S \neq \emptyset\}} \omega_S.$$

If coordinates are revealed in order π , hyperedge S activates at the stopping time

$$\tau_S = \max \{ \text{position}_{\pi}(i) : i \in S \}.$$

For a uniformly random reveal order and $|S|=r$,

$$\Pr\{\tau_S=t\} = C(t-1, r-1) / C(n, r), \quad t=r, \dots, n,$$

$$E[\tau_S] = r(n+1)/(r+1).$$

Consequently high-order information is not only fragile under noise; it is also late under random sequential access. In an $n=20$ source system, the mean activation times are 10.5 for order 1, 15.75 for order 3, 18.0 for order 6, and 19.09 for order 10.

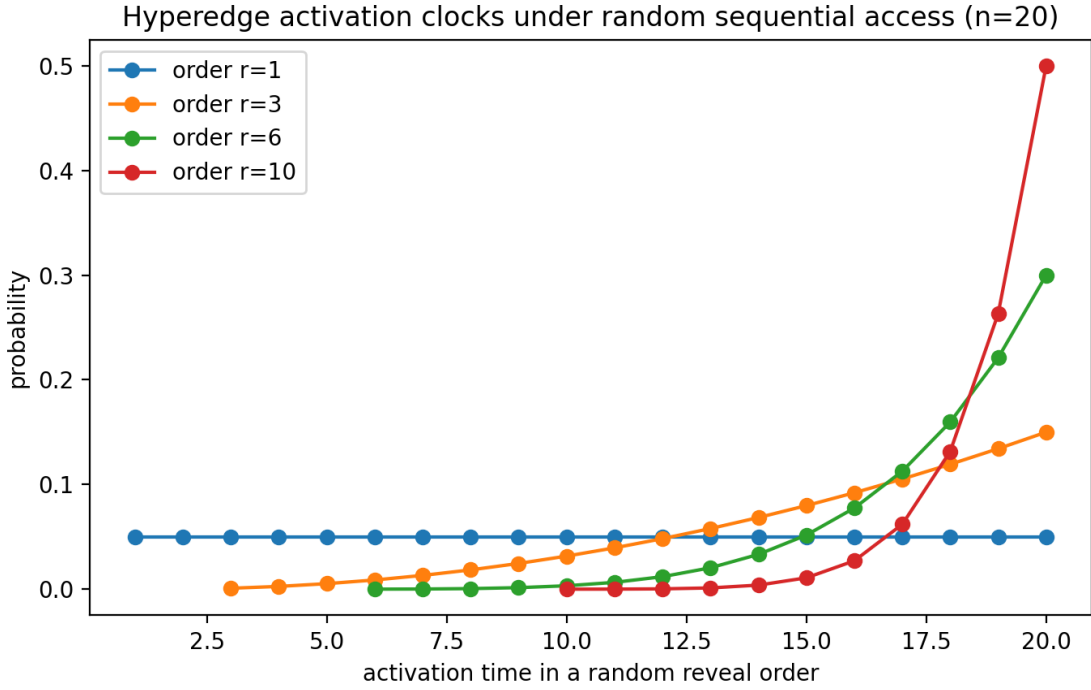


Figure v0.16.1. Activation-time distributions for order-r hyperedges under random sequential coordinate access.

3. Adaptive multiplexer law

Let $A \in \{-1, +1\}^k$ be k address bits and let $D_a \in \{-1, +1\}$ be one independent data bit for each address a . The target is the addressing or multiplexer function $T = D \cdot A$. Suppose h address bits are observed through independent BSC reliability α , and d data bits are then observed through BSC reliability δ .

Given address observations Y_H , let $\pi_a(Y_H) = P(A=a|Y_H)$. If a policy chooses a data-query set $Q(Y_H)$, then because the data bits are independent and unbiased, posterior chi-square information is

$$J = \delta^2 E_Y \left[\sum_{a \in Q(Y)} \pi_a(Y)^2 \right].$$

Thus, for fixed h and d , the optimal adaptive data stage queries the d largest posterior address masses. Under homogeneous address reliability and $d \leq 2^{k-h}$, this gives the closed form

$$J_{\text{ad}}(h, d) = \delta^2 d (1 + \alpha)^{2h} / 4^k.$$

For a nonadaptive policy that queries h fixed address bits and d fixed data bits,

$$J_{\text{non}}(h, d) = \delta^2 d (1 + \alpha^2)^h / 4^k.$$

The distinction is structural: nonadaptive access averages squared posterior mass at fixed data locations, whereas adaptive access follows the posterior mass to the most likely data locations.

4. Exponential adaptivity gap

At exact address reliability $\alpha=1$ and budget $m=k+1$, the adaptive policy observes all k address bits and then the selected data bit. It recovers the target exactly:

$$J_{\text{ad}}^{*(k+1)} = \delta^2.$$

The best nonadaptive policy at the same budget has

$$J_{\text{non}}^{*(k+1)} = \delta^2 2^{-k}.$$

Therefore

$$J_{\text{ad}}^{*(k+1)} / J_{\text{non}}^{*(k+1)} = 2^k.$$

Full nonadaptive certainty requires querying the address and all 2^k data coordinates, while adaptive certainty requires only $k+1$ queries. This creates an exponential separation between coordinate information content and sequentially steerable information content.

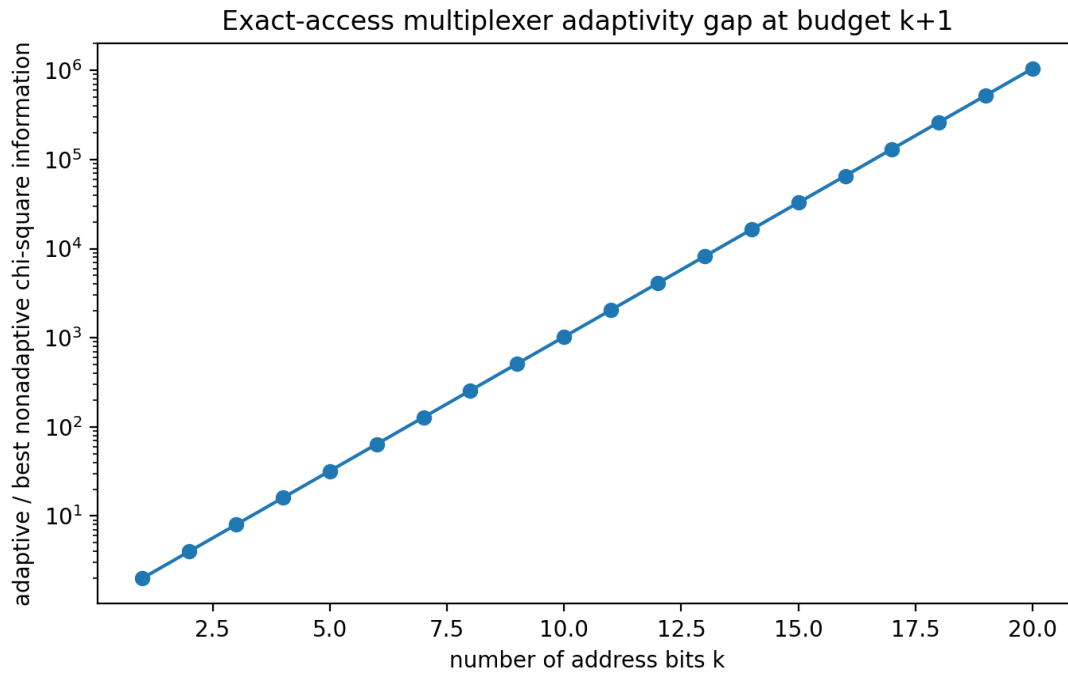


Figure v0.16.2. Exact-access adaptive/nonadaptive chi-square information ratio at budget $k+1$.

5. Noisy address/data phase curve

For budget $m=k+1$, the adaptive allocation objective is

$$\text{maximize over } h: (k+1-h)(1+\alpha)^{2h},$$

while the nonadaptive allocation objective is

$$\text{maximize over } h: (k+1-h)(1+\alpha^2)^h.$$

The full-address adaptive policy $h=k$ becomes better than $h=k-1$ exactly when

$$(1+\alpha)^2 \geq 2, \text{ i.e. } \alpha \geq \sqrt{2} - 1 \approx 0.41421356.$$

By contrast, the corresponding nonadaptive full-address comparison requires $\alpha^2 \geq 1$, so full nonadaptive address acquisition is not strictly favored until the address channel is perfect. This identifies a sharp reliability threshold where adaptivity begins to prefer completing the selector before sampling data.

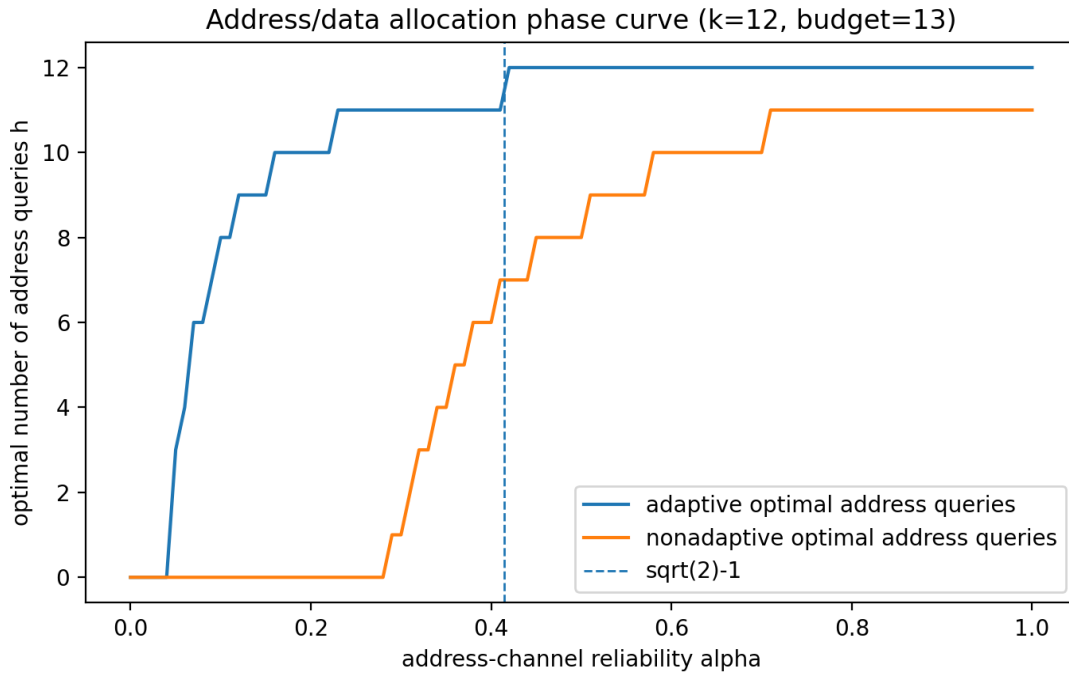


Figure v0.16.3. Optimal number of address queries at budget $k+1$ for $k=12$. Adaptive acquisition commits to the full address above $\alpha \approx 0.414$; nonadaptive acquisition remains more data-heavy until near-perfect address reliability.

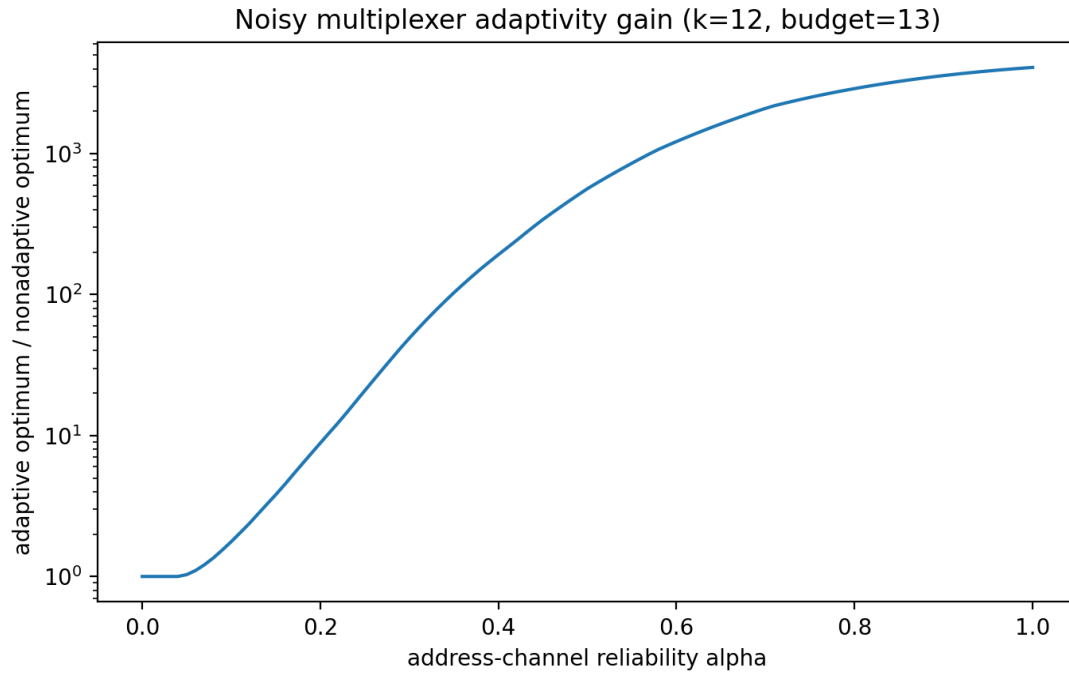


Figure v0.16.4. Noisy adaptive gain for $k=12$ and budget 13. The gain rises from 1 at $\alpha=0$ to 4096 at $\alpha=1$.

6. Exact experiments

The posterior-energy identity was tested on a random three-state stochastic target with four Boolean sources. Along the reveal order (2,0,3,1), expected posterior chi-square values were:

0.000000000000, 0.008950951165, 0.018058501005, 0.175374713047, 0.307701699166

The one-step Pythagorean increment error was 0.

Direct enumeration of the adaptive and nonadaptive multiplexer transcript distributions matched the formulas below; the maximum absolute error across these checks was approximately 1.0×10^{-15} .

k	α	δ	h,d	J_ad direct	J_ad formula	J_non direct	J_non formula
2	0.35	0.80	2,1	0.13286025	0.13286025	0.05040025	0.05040025
2	0.70	0.60	1,2	0.13005	0.13005	0.06705	0.06705
3	0.50	0.75	3,1	0.10011291503 9	0.10011291503 9	0.01716613769 53	0.01716613769 53
3	0.80	0.90	2,2	0.2657205	0.2657205	0.0680805	0.0680805

7. Interpretation

The multiplexer demonstrates that information order is not enough to characterize acquisition difficulty. In isotropic nonadaptive NRID, addressing information is distributed over many address-data hyperedges. A sequential policy can use early observations to choose which hyperedge to complete, converting a dispersed spectral object into a narrow query path. This suggests a new distinction:

static information = energy present under a fixed observation design; navigable information = energy reachable by policy-conditioned observation.

The gap can be exponential because the policy transcript need not acquire all possible data coordinates; it only has to acquire the data coordinate made relevant by the observed address context.

8. Possible applications

- Active sensing: adaptive policies should search for selector/context variables whose values concentrate posterior mass onto a small number of high-value downstream sensors.
- Privacy auditing: a release that looks safe under fixed subsets may leak strongly to an adaptive querier who can use early answers to choose later probes.
- Feature acquisition: models with gating, routing, mixture-of-experts, attention heads, or decision-tree structure may have large navigable information not visible in nonadaptive feature rankings.
- Fault diagnosis: early low-cost tests can identify which expensive component test is relevant, producing exponential savings in structured diagnostic systems.
- Experimental design: NRID hyperedge activation provides a way to quantify whether an experiment is gathering broad low-order information or steering toward completion of high-order information.

9. Targets for further study

- 1 Develop a full adaptive NRID policy theory: characterize optimal policies for general information hypergraphs and identify approximation guarantees or hardness barriers.
- 2 Define navigable NRID as the value function of a sequential observation process and compare it with static NRID, Shapley values, and adaptive submodularity.
- 3 Extend the adaptive multiplexer laws to repeated noisy measurements, heterogeneous address reliability, and nonbinary address alphabets.
- 4 Study whether high-order spectral mass can be efficiently discovered by policies using only local posterior-energy gradients, or whether gradient blindness forces group exploration.
- 5 Construct privacy mechanisms whose static and adaptive NRID leakage are both bounded under finite query budgets.
- 6 Connect activation clocks with stopping-time versions of directed information and with causal information flow in feedback systems.

10. Research ledger

- New exact theorem: posterior chi-square information is a submartingale with a Pythagorean squared-jump increment.
- New sequential law: an order- r hyperedge activates at the maximum reveal time of its r coordinates, with mean $r(n+1)/(r+1)$ under random order.
- New adaptive multiplexer law: $J_{\text{ad}}(h,d)=\delta^2 d(1+\alpha)^{\{2h\}/4^k}$ for d within the top posterior address bucket.
- New nonadaptive comparison law: $J_{\text{non}}(h,d)=\delta^2 d(1+\alpha^2)^{h/4^k}$.
- New exponential gap: exact adaptive/nonadaptive information ratio 2^k at budget $k+1$.
- New phase threshold: adaptive full-address acquisition becomes locally optimal at $\alpha=\sqrt{2}-1$, while nonadaptive full-address acquisition requires $\alpha=1$.
- New conceptual distinction: static information versus navigable information.

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v0.17 addendum: Symmetry quotients and anonymous-observation NRID

Motivation

Many observation systems do not reveal labeled source coordinates. They reveal a quotient: a histogram, a count, an unordered multiset of sensor readings, an exchangeable population summary, or a representation deliberately made invariant to relabeling. This addendum studies how NRID behaves when the observer sees not Y_ρ itself but only its orbit under a coordinate-permutation group. The resulting theory separates *amount of information*, *interaction order*, and *label geometry*. The key discovery is that anonymization by symmetry is an orthogonal projection on Fourier information: it preserves only the invariant component of each information orbit and annihilates phase-balanced, label-specific information.

Setup

Let $X \in \{-1, +1\}^n$ be uniform, let Y_ρ be the independent BSC-degraded source with reliability ρ , and let $G \leq S_n$ act on coordinates. Instead of observing Y_ρ , observe only the orbit

$$Z = [Y_\rho]_G.$$

For a finite target T , write

$$q_t(x) = \mathbb{P}(T = t \mid X = x), \quad p_t = \mathbb{P}(T = t), \quad m_t(x) = q_t(x) - p_t.$$

Let

$$m_t(x) = \sum_{S \subseteq [n]} \hat{m}_t(S) \chi_S(x), \quad \chi_S(x) = \prod_{i \in S} x_i.$$

It is useful to package the target-state coefficients into weighted vectors

$$a_S = \left(\frac{\hat{m}_t(S)}{\sqrt{p_t}} \right)_t.$$

The ordinary NRID hyperedge coefficient is $\omega_S = \|a_S\|_2^2$. Let $\mathcal{O}_G$ be the set of orbits of nonempty subsets under G .

Definition 1 (Quotient information orbit coefficient). *For an orbit $O \in \mathcal{O}_G$, define*

$$\omega_O^G = \frac{1}{|O|} \left\| \sum_{S \in O} a_S \right\|_2^2.$$

This is the squared orbit-mean coefficient. It is not the same as the total energy $\sum_{S \in O} \omega_S$ unless all coefficient vectors in the orbit are aligned.

Theorem 1 (Group-quotient NRID theorem). *For quotient observation $Z = [Y_\rho]_G$,*

$$\mathcal{J}_{G,\rho}(T; Z) = \sum_{O \in \mathcal{O}_G} \rho^{2r(O)} \omega_O^G,$$

where $r(O) = |S|$ for any $S \in O$.

Proof. For each target state, the centered posterior score after observing $Y_\rho = y$ is

$$P(T = t \mid Y_\rho = y) - p_t = (T_\rho m_t)(y).$$

Observing only $[y]_G$ replaces this score by its conditional mean over the orbit. Since Y_ρ is uniform, this is the group projection

$$P_G f(y) = \frac{1}{|G|} \sum_{g \in G} f(gy).$$

The noise operator T_ρ commutes with P_G . On Fourier supports inside one orbit O , P_G replaces all coefficients by their orbit mean. The squared L^2 norm contributed by that orbit is therefore

$$|O| \left\| \rho^{r(O)} \frac{1}{|O|} \sum_{S \in O} a_S \right\|_2^2 = \rho^{2r(O)} \frac{1}{|O|} \left\| \sum_{S \in O} a_S \right\|_2^2.$$

Summing over target states and support orbits gives the identity. □

Immediate consequences

Corollary 1 (Orbit-energy contraction). *For every orbit O ,*

$$\omega_O^G \leq \sum_{S \in O} \omega_S.$$

Equality holds exactly when the coefficient vectors a_S are identical across the orbit, up to the degenerate zero cases.

Corollary 2 (Anonymous dark information). *If*

$$\sum_{S \in O} a_S = 0$$

for every nonempty support orbit O , then the quotient observation contains zero posterior chi-square information about the target, even if the full labeled observation has positive NRID. Label anonymity kills every non-invariant component of the posterior score field.

Corollary 3 (Single-hyperedge anonymity law). *For a pure parity target supported on S ,*

$$\boxed{\mathcal{J}_{G,\rho} = \frac{\rho^{2|S|}}{|\text{Orb}_G(S)|}.$$

Thus the privacy factor of label anonymity for a single hyperedge is exactly the orbit size.

Corollary 4 (Symmetric survival). *If each posterior score $T_\rho m_t$ is G -invariant, quotienting by G loses no chi-square information. Hence identity-stripping can strongly suppress label-specific interactions while preserving all G -symmetric information.*

Full anonymity and the Krawtchouk projection

For the full symmetric group S_n , the quotient is the Hamming weight of Y_ρ . If $T = \chi_S(X)$ is a fixed r -way parity, then

$$\mathcal{J}_{S_n, \rho} = \frac{\rho^{2r}}{\binom{n}{r}}.$$

Equivalently, for $W = \#\{i : Y_i = -1\}$,

$$\mathbb{E}[\chi_S(Y) \mid W = w] = \frac{K_r(w)}{\binom{n}{r}},$$

where

$$K_r(w) = \sum_j (-1)^j \binom{w}{j} \binom{n-w}{r-j}$$

is the degree- r Krawtchouk polynomial. The identity

$$\mathbb{E}_W[K_r(W)^2] = \binom{n}{r}$$

then gives the retention factor $1/\binom{n}{r}$.

This produces an anonymity band-pass law: middle-order labeled parities are maximally hidden, but the global parity $r = n$ is not hidden at all because it is itself a function of the count.

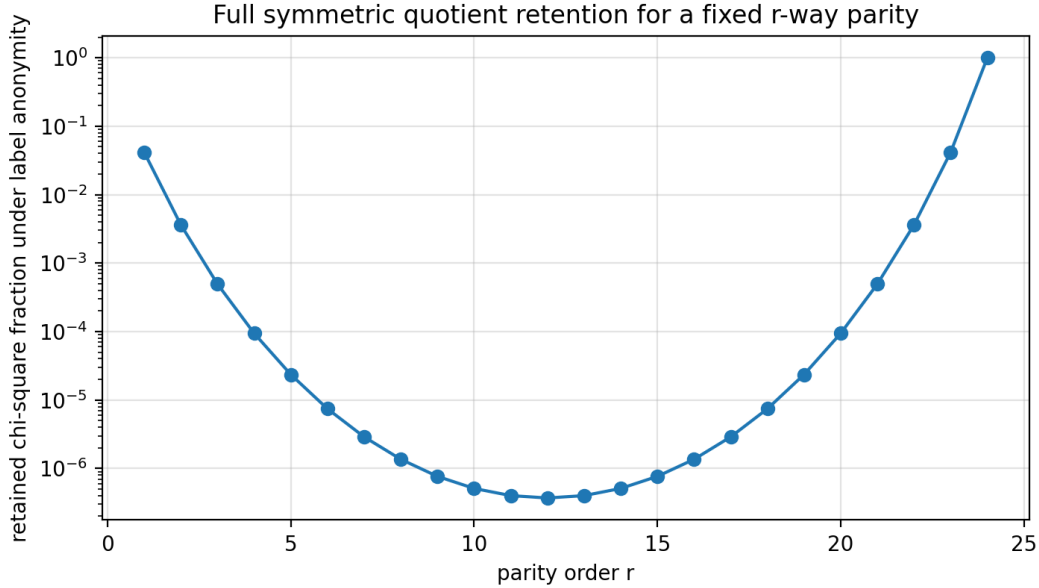


Figure 1: For a fixed r -way parity under full label anonymity, retained chi-square information is $1/\binom{n}{r}$ at $\rho = 1$. For $n = 24$, the strongest suppression occurs at $r = 12$, where the retained fraction is 3.698×10^{-7} , a suppression factor of 2,704,156. The total parity $r = 24$ is retained completely.

Burnside law for random balanced targets

For a uniformly random balanced Boolean target, the nonconstant Fourier coefficients have covariance

$$\mathbb{E}[\widehat{f}(S)\widehat{f}(A)] = \frac{\mathbf{1}\{S = A\}}{2^n - 1}, \quad S, A \neq \emptyset.$$

Therefore each nonempty support orbit contributes one invariant Fourier degree in expectation:

$$\mathbb{E}[\omega_O^G] = \frac{1}{2^n - 1}.$$

This gives a closed form by Burnside's lemma.

Theorem 2 (Burnside quotient law for random balanced targets). *Let f be a uniformly random balanced Boolean target and let $G \leq S_n$. Then*

$$\mathbb{E}_f \mathcal{J}_{G,\rho} = \frac{1}{2^n - 1} \left(\frac{1}{|G|} \sum_{\pi \in G} \prod_{C \in \text{cyc}(\pi)} (1 + \rho^{2|C|}) - 1 \right).$$

At $\rho = 1$ this reduces to

$$\mathbb{E}_f \mathcal{J}_{G,1} = \frac{N_G - 1}{2^n - 1},$$

where N_G is the number of subset orbits under G , including the empty subset.

For the trivial group this recovers the ordinary random-target response

$$\frac{(1 + \rho^2)^n - 1}{2^n - 1}.$$

For the full symmetric group,

$$\mathbb{E}_f \mathcal{J}_{S_n,\rho} = \frac{\rho^2(1 - \rho^{2n})}{(1 - \rho^2)(2^n - 1)},$$

with the $\rho = 1$ limit $n/(2^n - 1)$.

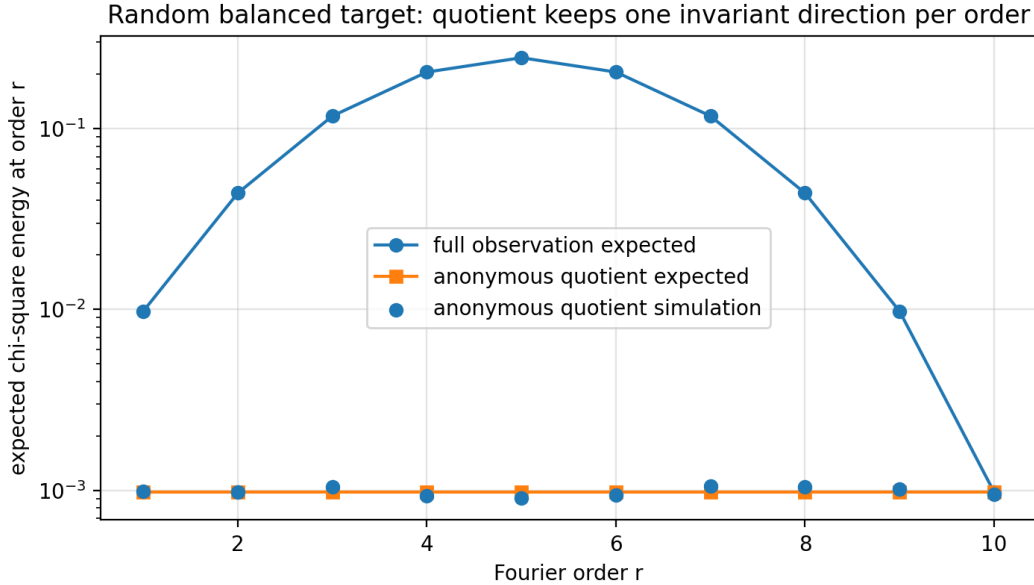


Figure 2: For $n = 10$ random balanced Boolean targets, full observation has expected order spectrum $\binom{n}{r}/(2^n - 1)$. Full label anonymity keeps only one invariant direction per order, so the expected quotient contribution is $1/(2^n - 1)$ at every nonzero order. Simulation over 1200 random balanced targets matches the prediction.

Phase cancellation inside an orbit

The quotient spectrum depends on orbit means, not orbit energies. Thus two targets can have the same full NRID spectrum but opposite anonymous visibility. Under a group that swaps coordinates 2 and 3, define score fields

$$g_+(x) = \chi_{\{1,2\}}(x) + \chi_{\{1,3\}}(x),$$

$$g_-(x) = \chi_{\{1,2\}}(x) - \chi_{\{1,3\}}(x).$$

Both have the same full second-order energy. But g_+ is invariant under the swap, while g_- is anti-invariant and has zero orbit mean. Hence

$$\mathcal{J}_G(g_+) = 2\rho^4, \quad \mathcal{J}_G(g_-) = 0.$$

At $\rho = 0.8$, both have full labeled energy 0.8192, but the anonymous quotient retains 0.8192 for g_+ and 0 for g_- . Such score fields can be embedded in stochastic binary targets by multiplying by a sufficiently small contrast parameter.

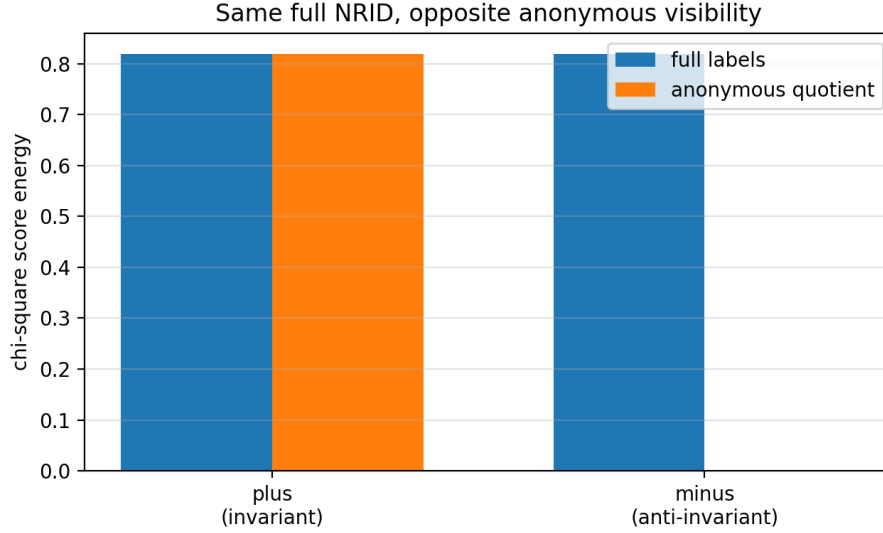


Figure 3: Two score fields with identical full NRID spectrum behave maximally differently after a symmetry quotient. The invariant combination survives; the anti-invariant combination is completely dark.

Exact-enumeration experiments

A random stochastic three-state target with $n = 5$ Boolean sources was observed through a cyclic quotient. Direct enumeration over all noisy observations gave

$$\mathcal{J}_{\text{direct}} = 0.011122310906614618.$$

The orbit-coefficient formula gave

$$\mathcal{J}_{\text{formula}} = 0.011122310906614620,$$

with absolute error

$$1.73 \times 10^{-18}.$$

For $n = 10$, the Burnside random-balanced law gives the following expected retained chi-square information.

G	nonempty subset orbits	$\mathbb{E} \mathcal{I}_{G,1}$	$\mathbb{E} \mathcal{I}_{G,0.65}$
trivial	1023	1.000000	0.03218497
cyclic	107	0.104594	0.00335364
dihedral	77	0.075269	0.00269750
S_{10}	10	0.009775	0.00071502

A simulation of 700 random balanced targets under the cyclic quotient at $\rho = 0.65$ gave empirical mean

$$0.00330516,$$

compared with the Burnside prediction

$$0.00335364.$$

The sampling discrepancy was 4.85×10^{-5} .

Interpretation

This addendum introduces a new distinction:

$$\text{order-resolved information} \neq \text{identity-resolved information}.$$

Independent noise measures how many coordinates an informational feature needs. Symmetry quotienting measures whether the feature has a stable identity under relabeling. These axes are independent: a high-order global parity is fully anonymous-invariant, while a low-order pair interaction can be hidden if many label-equivalent pairs are averaged together.

The results also clarify a limitation of anonymization. Label removal protects target-specific facts by orbit dilution, but it preserves all symmetric facts. A count release can hide which r individuals form a parity while still revealing any property that is itself a function of the count. This gives a spectrum-level view of why aggregate releases can be useful and risky at the same time.

Possible applications

1. **Privacy and anonymized release.** Quantify which labeled interactions are suppressed by identity stripping and which symmetric statistics remain exposed.
2. **Aggregate-data auditing.** Decompose released histograms into invariant NRID components and estimate leakage for protected target classes.
3. **Sensor anonymity.** Analyze systems where sensor identities are unavailable, shuffled, or deliberately randomized.
4. **Invariant machine learning.** Diagnose what target information survives permutation-invariant architectures such as Deep Sets, set transformers, and graph-level pooling.
5. **Representation design.** Choose quotient groups to preserve desired invariant information while maximizing orbit dilution of protected label-specific hyperedges.

Targets for further study

1. Develop a full representation-theoretic NRID that decomposes information into irreducible G -modules, not only the trivial invariant module.
2. Add noise to quotient statistics themselves, such as randomized histograms, and derive Krawtchouk-plus-count-noise filters.
3. Extend quotient NRID to exchangeable but nonuniform sources, where quotient projection and source-dependence heterodyning interact.
4. Study auxiliary-information attacks: side information can break the group symmetry and turn anonymous dark information back into labeled information.
5. Optimize anonymization groups for a protected family of hyperedges using orbit-size, orbit-phase, and utility-retention objectives.

Research ledger

- New theorem: quotient observation by a coordinate-permutation group keeps exactly the orbit-mean Fourier information vectors.
- New coefficient: $\omega_O^G = |O|^{-1} \|\sum_{S \in O} a_S\|^2$.
- New law: a pure hyperedge under anonymity is suppressed by its orbit size, $\mathcal{J}_{G,\rho} = \rho^{2|S|}/|\text{Orb}_G(S)|$.
- New full-symmetry result: a fixed r -way parity observed only through Hamming weight retains $1/\binom{n}{r}$ of its chi-square information.
- New Burnside law: random balanced quotient information equals a weighted count of subset orbits divided by $2^n - 1$.
- New counterexample: identical full NRID spectra can produce full versus zero anonymous visibility through invariant versus anti-invariant orbit phases.
- New conceptual distinction: noise fragility measures interaction order; quotient fragility measures label identity.

References noted for v0.17

Boolean Fourier analysis and the diagonal noise-operator view are standard in R. O’Donnell, *Analysis of Boolean Functions*, Cambridge University Press, 2014, and R. de Wolf, *A Brief Introduction to Fourier Analysis on the Boolean Cube*, Theory of Computing, 2008. Krawtchouk polynomials arise naturally in symmetric Boolean and Hamming-scheme analysis. Aggregate-data privacy risks and reconstruction/tracing attacks are surveyed in C. Dwork, A. Smith, T. Steinke, J. Ullman, and S. Vadhan, *Exposed! A Survey of Attacks on Private Data*, Annual Review of Statistics and Its Application, 2017, and in recent reviews of anonymization and privacy attacks.

Addendum v0.18: Resolution-Filtered NRID for Coarsened Anonymous Counts

Noise-Resolved Information Decomposition Working Paper

Summary

The preceding quotient addendum showed that removing labels projects a Boolean information field onto orbit means. Under the full symmetric group, the orbit statistic is the Hamming weight W of the noisy observation. This addendum studies the next blind spot: real aggregate releases often do not expose the exact count. They expose a bin, a threshold, a parity bit, a censored value, or another coarse count sketch

$$B = \beta(W).$$

The main discovery is that such sketches act as *Krawtchouk resolution filters*. Exact count keeps one symmetric degree of freedom at every Fourier order. A sketch with only L occupied output values has exactly $L - 1$ nonconstant symmetric degrees of freedom to spend across all orders. Where that small resolution budget is placed can change noise leakage by many orders of magnitude.

Setup and notation

Let $X \in \{-1, +1\}^n$ be uniform, and let Y_ρ be an independent BSC degradation of X with reliability ρ . Write

$$W(Y) = |\{i : Y_i = -1\}|.$$

For $r = 0, \dots, n$, define the binary Krawtchouk polynomial

$$K_r(w) = \sum_{j=0}^r (-1)^j \binom{w}{j} \binom{n-w}{r-j}.$$

Equivalently, if y has w negative entries, then

$$K_r(w) = \sum_{|S|=r} \chi_S(y), \quad \chi_S(y) = \prod_{i \in S} y_i.$$

The orthogonality relation is

$$\mathbb{E}[K_r(W)K_s(W)] = \binom{n}{r} \mathbf{1}_{r=s}, \quad W \sim \text{Bin}(n, 1/2).$$

For a count sketch $B = \beta(W)$, define

$$\mu_r(b) = \mathbb{E}[K_r(W) \mid B = b],$$

and the sketch Gram matrix

$$G_{rs}^\beta = \mathbb{E}[\mu_r(B)\mu_s(B)].$$

Theorem 0.18.1: count-sketch NRID filter law

Let T be a finite target generated from X , with

$$m_t(x) = \mathbb{P}(T = t \mid X = x) - p_t, \quad p_t = \mathbb{P}(T = t),$$

and Fourier expansion

$$m_t(x) = \sum_S \hat{m}_t(S) \chi_S(x).$$

Define the orbit-sum coefficient

$$A_{t,r} = \sum_{|S|=r} \hat{m}_t(S).$$

Then the posterior chi-square information in a coarsened anonymous count observation $B = \beta(W(Y_\rho))$ is exactly

$$\mathcal{J}_{\beta,\rho}(T; B) = \sum_t \frac{1}{p_t} \sum_{r=1}^n \sum_{s=1}^n \rho^{r+s} \frac{A_{t,r} A_{t,s}}{\binom{n}{r} \binom{n}{s}} G_{rs}^\beta.$$

Thus a finite-resolution count sketch does two things. First, it discards all orbit-internal label geometry, keeping only the orbit sums $A_{t,r}$. Second, it mixes symmetric orders through the matrix G^β .

Proof sketch. The noisy posterior score is

$$\mathbb{E}[m_t(X) \mid Y_\rho = y] = \sum_S \rho^{|S|} \hat{m}_t(S) \chi_S(y).$$

Conditioning further on $B = \beta(W(y))$ gives

$$\mathbb{E}[m_t(X) \mid B] = \sum_{r=1}^n \rho^r \frac{A_{t,r}}{\binom{n}{r}} \mu_r(B),$$

because $\mathbb{E}[\chi_S(Y) \mid W] = K_r(W) / \binom{n}{r}$ for $|S| = r$. Squaring, averaging over B , dividing by p_t , and summing over target states gives the formula.

Corollaries

Exact count as the diagonal case

If $B = W$ is the exact anonymous count, then

$$G_{rs}^W = \binom{n}{r} \mathbf{1}_{r=s},$$

so

$$\mathcal{J}_{W,\rho}(T; W) = \sum_t \frac{1}{p_t} \sum_{r=1}^n \rho^{2r} \frac{A_{t,r}^2}{\binom{n}{r}}.$$

This is the full-symmetric-group quotient law from v0.17.

Resolution dimension spectrum

Let

$$\phi_r(W) = \frac{K_r(W)}{\sqrt{\binom{n}{r}}},$$

so that $\{\phi_0, \dots, \phi_n\}$ is an orthonormal basis for symmetric functions on the cube. Define

$$d_r(\beta) = \|\mathbb{E}[\phi_r(W) \mid B]\|_2^2 = \frac{G_{rr}^\beta}{\binom{n}{r}}.$$

Then $d_r(\beta)$ is the fraction of the one-dimensional symmetric order- r channel retained by the sketch.

Resolution conservation law

If the count sketch has L occupied output bins, then

$$\sum_{r=1}^n d_r(\beta) = L - 1.$$

Indeed, conditional expectation onto B is an orthogonal projection of rank L . The constant function consumes one dimension, leaving exactly $L - 1$ nonconstant symmetric dimensions. A count sketch therefore has a fixed resolution budget. It can place that budget at low orders, high orders, or a mixture of orders, but it cannot exceed the budget.

Random balanced target leakage

For a uniformly random balanced Boolean target $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$,

$$\mathbb{E}_f[\widehat{f}(S)\widehat{f}(R)] = \frac{\mathbf{1}_{S=R}}{2^n - 1}, \quad S, R \neq \emptyset.$$

Therefore

$$\mathbb{E}_f \mathcal{J}_{\beta, \rho} = \frac{1}{2^n - 1} \sum_{r=1}^n d_r(\beta) \rho^{2r}.$$

At $\rho = 1$, every L -bin count sketch has the same expected random-target leakage,

$$\mathbb{E}_f \mathcal{J}_{\beta, 1} = \frac{L - 1}{2^n - 1}.$$

For $\rho < 1$, the placement of the same $L - 1$ dimensions becomes decisive.

One-bit count sketches

For a balanced one-bit symmetric sketch $h(W) \in \{-1, +1\}$, the retained dimensions are exactly the Fourier level weights of h :

$$d_r(h) = \sum_{|S|=r} \widehat{h}(S)^2.$$

Count parity $h(W) = (-1)^W$ spends its whole dimension at degree n :

$$d_n = 1, \quad d_r = 0 \quad (r < n).$$

Majority spends its dimension over odd degrees, heavily favoring low odd orders. Thus two one-bit aggregate releases can have equal output cardinality and equal total symmetric resolution, yet radically different noise response.

Resolution-induced order aliasing

Under exact count, different symmetric orders are orthogonal and the response is a polynomial in ρ^2 . Coarse count sketches can make $G_{rs}^\beta \neq 0$ for $r \neq s$. The response then contains cross-order terms ρ^{r+s} and depends on relative signs of orbit-sum coefficients.

This is a new phase sensitivity at the level of quadratic information. Full NRID is phase-blind within an order. Exact count is phase-blind across symmetric orders. A coarsened count can become phase-sensitive across orders because it maps several formerly orthogonal Krawtchouk modes into the same small output space.

Experiments

All experiments were exact enumerations or exact finite sums; no Monte Carlo estimates were used unless explicitly noted.

Exact formula check

A random three-state stochastic target with $n = 6$ was observed through the three-bin count sketch

$$\beta(w) = 0 \ (w \leq 1), \quad \beta(w) = 1 \ (2 \leq w \leq 3), \quad \beta(w) = 2 \ (w \geq 4),$$

at $\rho = 0.57$. Direct enumeration of the joint distribution gave

$$\mathcal{J}_{\text{direct}} = 0.0025305703647542322,$$

while Theorem 0.18.1 gave

$$\mathcal{J}_{\text{formula}} = 0.0025305703647542704.$$

The absolute error was

$$3.82 \times 10^{-17}.$$

Dimension conservation

For $n = 31$, the following count sketches satisfy the resolution conservation law numerically.

sketch	L	$\sum_{r \geq 1} d_r$	$L - 1$
majority bit	2	1.000000	1
count parity	2	1.000000	1
4 quantile bins	4	3.000000	3
8 equal-width bins	8	7.000000	7
exact count	32	31.000000	31

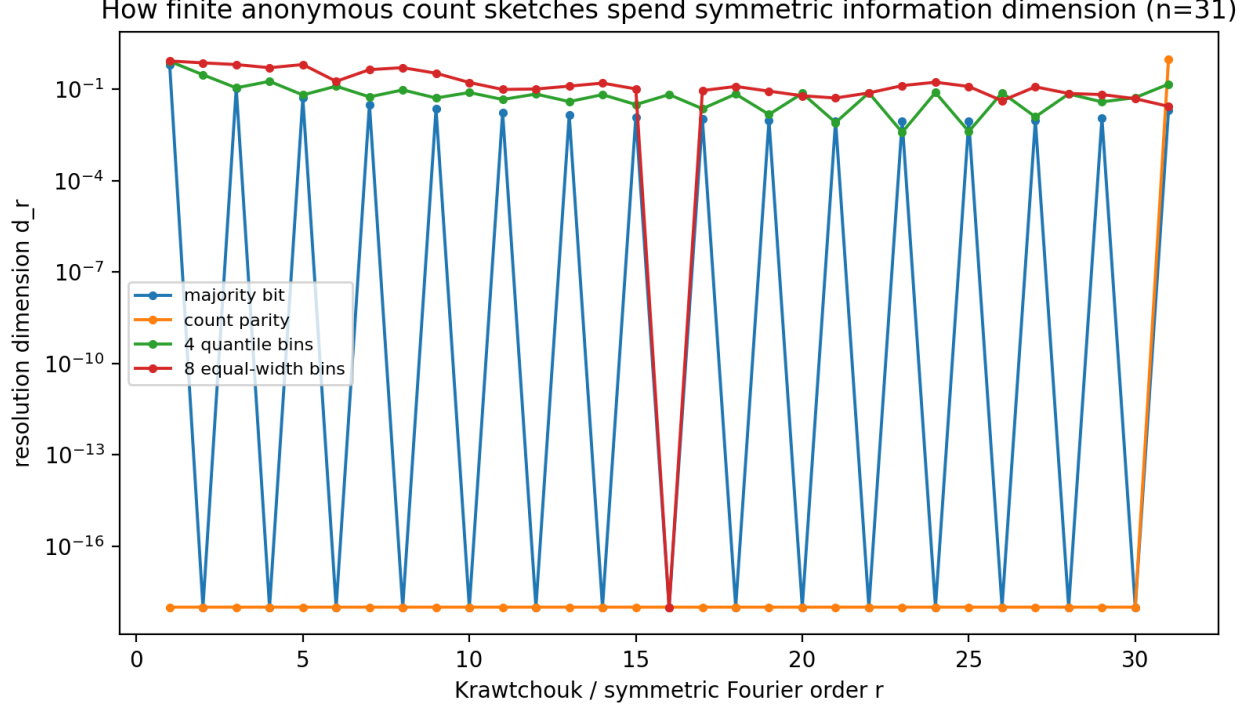


Figure 1: Resolution dimension spectra $d_r(\beta)$ for several anonymous count sketches at $n = 31$. Majority and parity are both one-bit releases, but majority spends dimension at low odd orders while count parity spends it at the top order.

Same one-bit cardinality, eleven orders of magnitude apart

At $n = 31$ and $\rho = 0.65$, the random-balanced-target expectation formula gives

$$\mathbb{E}_f \mathcal{J}_{\text{majority}} = 1.3157114131 \times 10^{-10},$$

while count parity gives

$$\mathbb{E}_f \mathcal{J}_{\text{parity}} = 1.1713811917 \times 10^{-21}.$$

The majority bit therefore leaks about

$$1.123 \times 10^{11}$$

times more expected random-target chi-square information than the count-parity bit, despite both having exactly two output values and exactly one nonconstant symmetric dimension.

Fixed hyperedge retention

For a fixed r -way parity observed only through exact count, the retained fraction of labeled chi-square information is

$$\frac{1}{\binom{n}{r}}.$$

For $n = 24$, a fixed 12-way parity retains only

$$3.6980 \times 10^{-7}$$

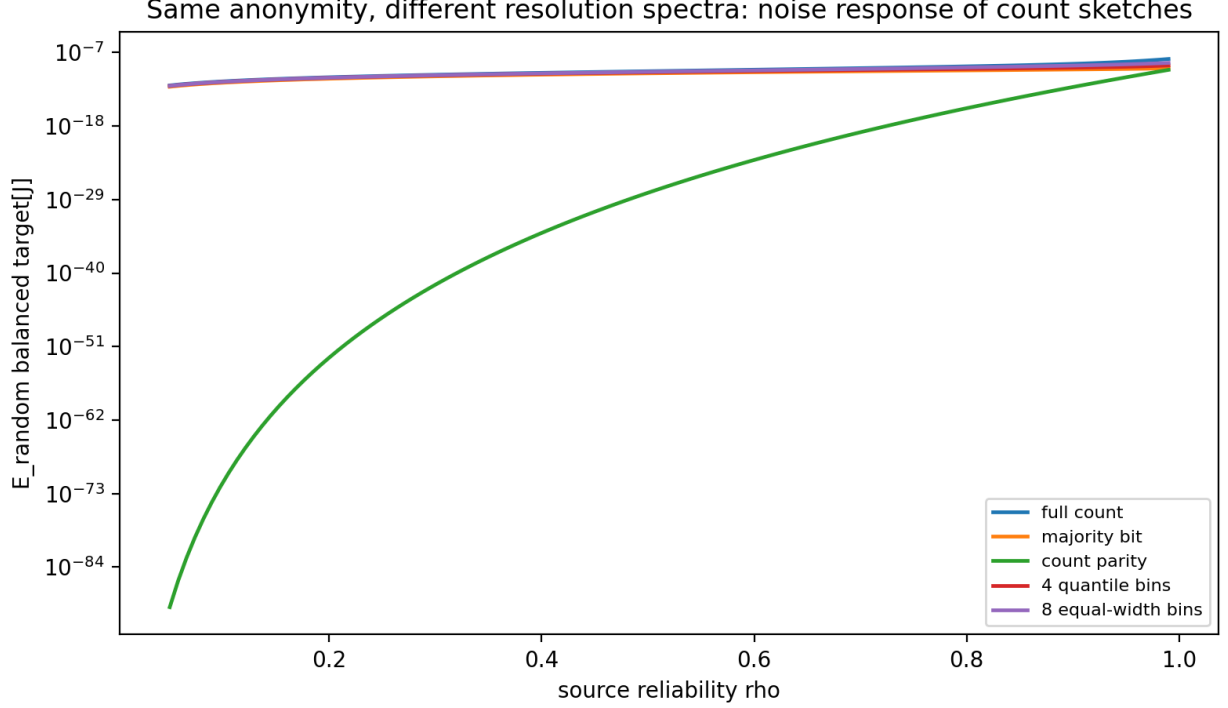


Figure 2: Expected random-balanced-target leakage for several count sketches. Output cardinality alone is not a leakage measure; the order placement of the same resolution dimension controls the noise response.

through exact count. Count parity reveals none of the proper fixed parities but reveals the global parity perfectly:

$$q_r = 0 \quad (1 \leq r < 24), \quad q_{24} = 1.$$

Thus count parity is an extreme high-order pass filter.

Coarse bins mix orders

Phase interference between orders 1 and 3

Let

$$f_{\pm} = \frac{1}{\sqrt{2}}\phi_1 \pm \frac{1}{\sqrt{2}}\phi_3.$$

The two score fields have the same exact-count energy

$$\frac{1}{2}\rho^2 + \frac{1}{2}\rho^6.$$

At $\rho = 0.82$, this is

$$0.488203335712$$

for both. But after coarsening to the majority bit, the two energies become

$$0.113008795490 \quad \text{and} \quad 0.355924206295,$$

a factor of about 3.15. The difference is not caused by different spectra; it is caused by cross-order phase after finite-resolution projection.

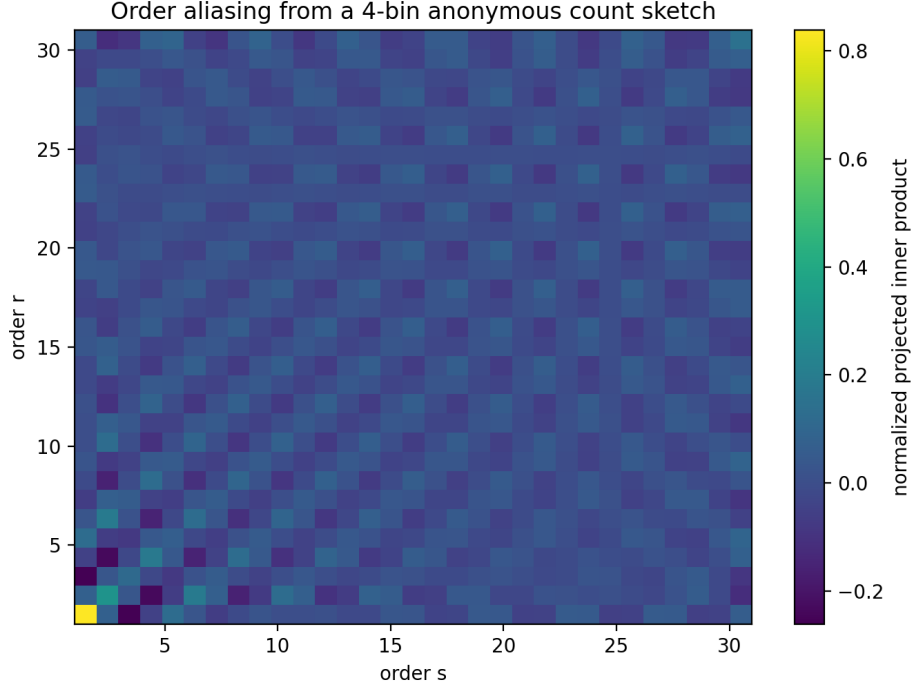


Figure 3: Normalized matrix $G_{rs}^\beta / \sqrt{\binom{n}{r} \binom{n}{s}}$ for a four-bin anonymous count sketch. Exact count would be diagonal. Coarse counts create order aliasing.

Interpretation

This addendum separates three layers of aggregate information:

$$\begin{aligned}
 \text{label removal} &\rightarrow \text{orbit means,} \\
 \text{count release} &\rightarrow \text{one symmetric dimension per order,} \\
 \text{coarse count sketch} &\rightarrow \text{a finite Krawtchouk filter bank.}
 \end{aligned}$$

A one-bit release does not have a single leakage level. A majority bit leaks low-order symmetric information and remains visible under noise. A parity-of-count bit leaks only top-order symmetric information and becomes nearly invisible under independent noise unless the protected target is itself global parity. The hidden risk is therefore not just how many aggregate values are released, but where the aggregate statistic places its resolution dimension in the Krawtchouk order spectrum.

Possible applications

1. **Aggregate privacy auditing.** Estimate which target interaction orders are exposed by released histograms, threshold counts, count parity, or bucketed counts.
2. **Census and telemetry design.** Choose binning functions that preserve low-order utility while pushing protected interactions into high-order or null Krawtchouk bands.

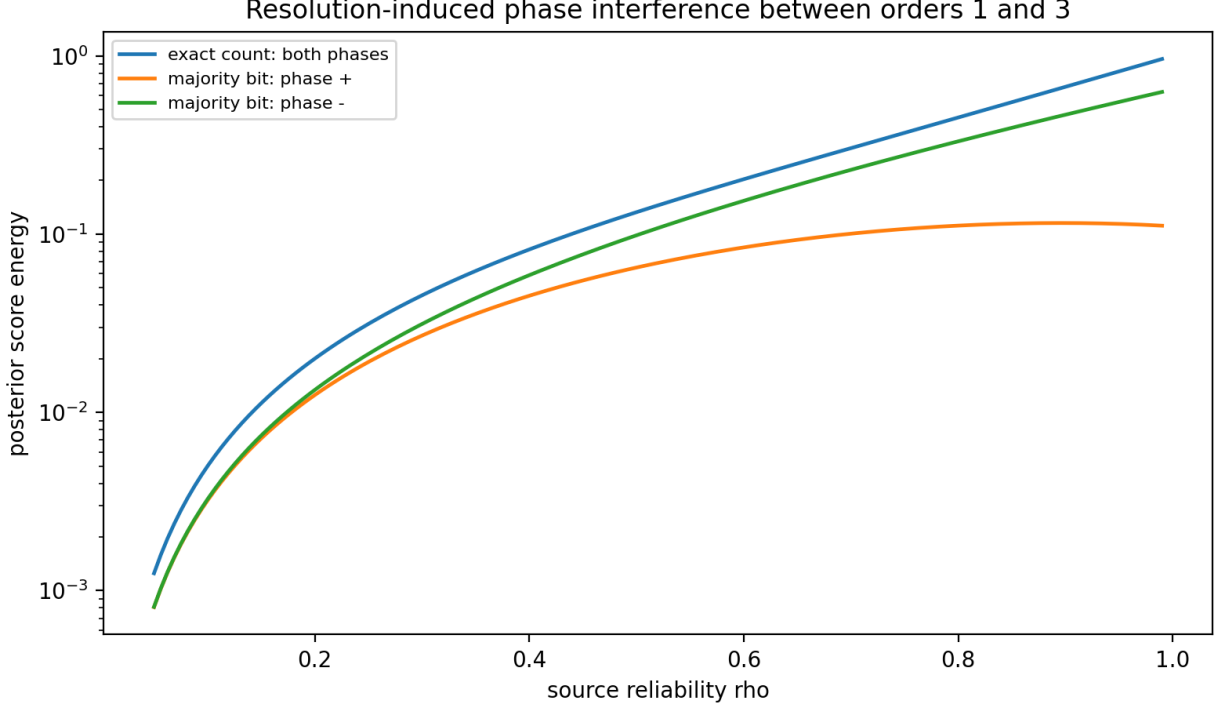


Figure 4: Two symmetric score fields with identical exact-count energy can become distinguishable after a one-bit majority sketch because the sketch aliases orders 1 and 3 into the same output direction.

3. **Permutation-invariant learning.** Diagnose which symmetric degrees are available to models that consume only count summaries or pooled set statistics.
4. **Robust aggregate sensing.** Use majority-like sketches when low-order population signals must survive noise; use parity-like sketches when exposure should be high-order and noise-fragile.
5. **Information-optical filter design.** Treat count sketches as finite-dimensional filters and optimize $d_r(\beta)$ for desired passbands and notches.

Targets for further study

1. Solve the optimal L -bin count-sketch design problem: maximize or minimize $\sum_r d_r \rho^{2r}$ subject to bin contiguity, balance, or utility constraints.
2. Extend the Krawtchouk filter law to noisy count releases, including randomized response on bins and additive noise on counts.
3. Develop asymptotic Hermite limits for $d_r(\beta)$ when $n \rightarrow \infty$ and β is a threshold, interval, or quantile map.
4. Combine source-dependence heterodyning with count-sketch filters to study aggregate releases under nonexchangeable populations.
5. Investigate whether adversaries with side information can rotate orbit-internal information back into the retained symmetric filter bank.

Research ledger

- New theorem: arbitrary coarsened anonymous count observations have an exact Krawtchouk Gram-matrix NRID response.
- New coefficient: $d_r(\beta) = G_{rr}^\beta / \binom{n}{r}$, the resolution dimension retained at symmetric order r .
- New conservation law: an L -bin count sketch has exactly $L - 1$ nonconstant symmetric information dimensions.
- New random-target law: expected leakage is $(2^n - 1)^{-1} \sum_r d_r \rho^{2r}$.
- New one-bit contrast: majority and count parity have the same output cardinality but can differ by more than 10^{11} in expected noisy leakage at $n = 31, \rho = 0.65$.
- New phenomenon: coarse count sketches can make quadratic information phase-sensitive across different symmetric orders.
- New design principle: aggregate release risk is controlled by the Krawtchouk placement of finite resolution, not only by the number of released bins.

References noted for v0.18

Boolean Fourier analysis, the hypercube noise operator, and the Fourier spectrum of symmetric Boolean functions are standard background in R. O’Donnell, *Analysis of Boolean Functions*, Cambridge University Press, 2014. Krawtchouk polynomials and the Hamming association scheme are classical tools in coding theory and algebraic combinatorics; see P. Delsarte, *Association schemes and coding theory*, Philips Research Reports Supplements, 1973, and modern expositions of Hamming-scheme MacWilliams/Krawtchouk identities. Recent work on pseudorandomness, symmetry, and smoothing also uses Krawtchouk expansions of symmetric Boolean tests.